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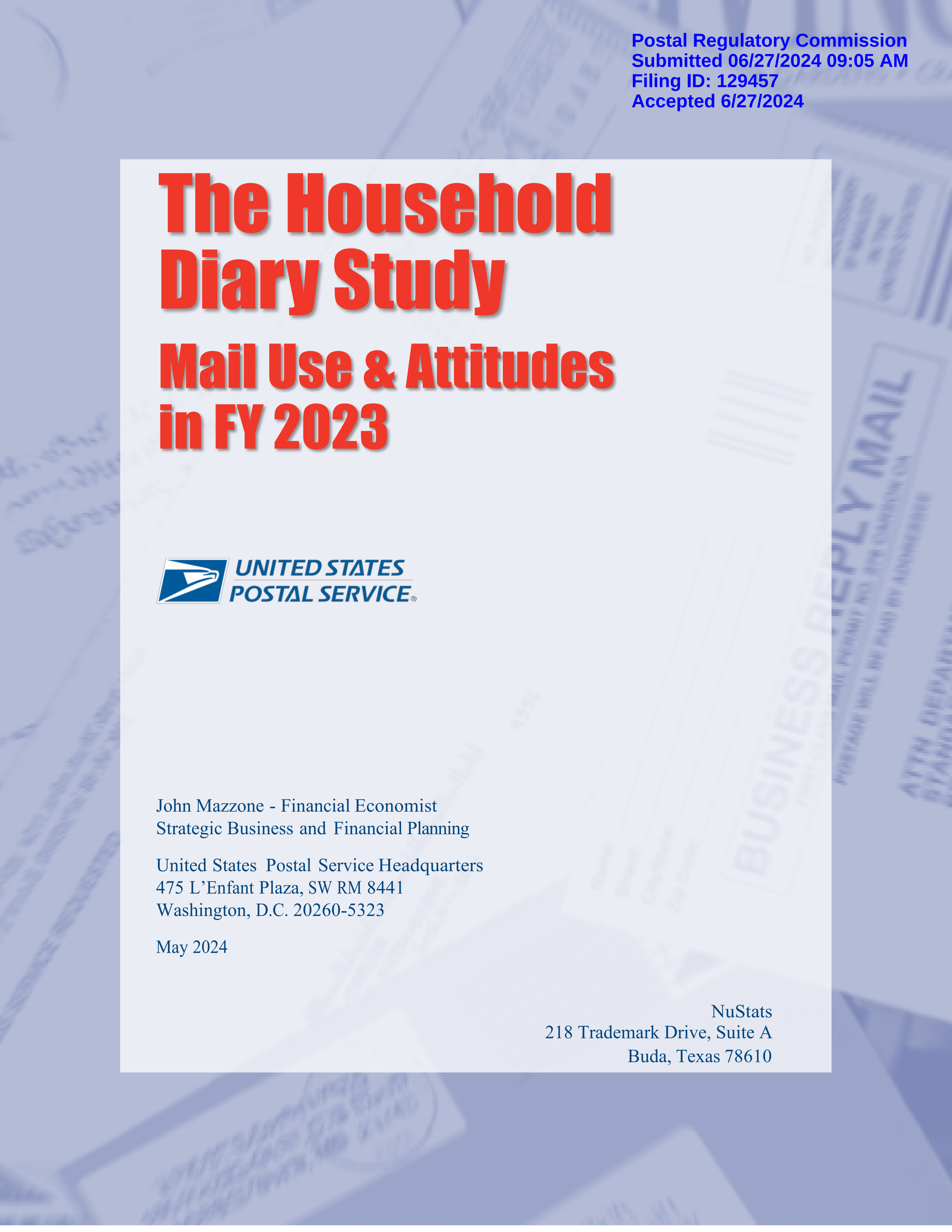

The Household Diary Study

Mail Use & Attitudes in FY 2023

UNITED STATES
POSTAL SERVICE

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Executive Summary

This report documents the findings of the United States Postal Service's Household Diary Study (HDS) for Fiscal Year (FY) 2023. The main objectives of the study are to:

- Measure the types and volumes of mail sent and received by U.S. households,
- Track trends in mail usage over time, and
- Compare mail usage by household demographic characteristics.

The report examines these trends in the context of changes and developments in the wider markets for communications and package delivery.

Background

The Household Diary Study survey, fielded continuously since 1987, aims to collect information about households' use of the mail and how that use changes over time. The information collected includes household demographics, lifestyle, attitudes toward mail advertising, bill paying practices, and use of the Internet and other information technologies.

The FY 2023 report covers Government Fiscal Year 2023, with comparisons to 2022, 2021, and other years, as appropriate.

Overview

As seen in Table E.1, in 2023, U.S. households received 94.0 billion pieces of mail and sent 5.2 billion pieces. After excluding the double-count of 1.8 billion pieces of mail that households sent to each other, households sent and received a total of 97.4 billion pieces, accounting for 84% of total U.S. mail. The remaining 16% of U.S. mail (18.8 billion pieces was sent from non-households to non-households (mostly businesses and government mail).

Table E.1:
Mail Received and Sent by Households
(Millions of Pieces)

Mail Classification	Received	Sent
First-Class Mail	32.2	4.6
Marketing Mail-Regular	41.1	—
Marketing Mail Nonprofit	10.8	—
Periodicals	3.4	—
Package & Shipping Services	6.4	0.6
Total	94.0	5.2
Household to Household	1.8	
Total Mail Received and Sent by Households *	97.4	
FY 2022 RPW Total	116.2	
Non-household to Non-household	18.8	
Unaddressed	0.9	—

Source: HDS Diary Sample, FY 2023.
Note: Totals may not sum due to rounding.
* Total (97.4) is less than received (94.0) plus sent (5.2) because household to household mail (1.8) is included in both categories.

Households sent and received 84% of all U.S. mail. B-to-B mail accounted for most of the remaining 16% of the volume.

Mail Markets

The Household Diary Study examines mail by the markets it serves. This design cuts across classes to provide a foundation for understanding mail flows and the marketplace changes that affect them. Table E.2 shows household mail volumes by market for 2023 and for the preceding two years.

Table E.2:
Household Mail Volume Sent and Received
By Market Served
(Billions of Pieces)

Market	2021	2022	2023
Correspondence	14.8	14.5	13.4
Transactions	18.5	17.2	16.9
Advertising	65.2	65.3	57.3
Periodicals	3.7	3.3	3.4
Packages	6.8	7.3	6.7
Unclassified	1.5	1.1	1.5
Total	**107.7**	**105.9**	**97.4**

Source: HDS Diary Sample, FY 2021, 2022, and 2023.
Notes:
1. Correspondence and packages exclude double-counted pieces reported as both sent and received by households.
2. Advertising includes 2.1 billion pieces of First-Class advertising-enclosed mail but excluded from total.
3. Package volumes include 0.1 billion pieces of CD/DVD rental reported in First-Class Mail letters in Tables E.1, 1.5, and 1.6.

In 2023, correspondence mail totaled 13.4 billion pieces, accounting for 14% of all mail sent and received by households. Historically, volumes declined continuously since the early 2000's, driven by a rapidly growing migration of mail correspondence to the various electronic alternatives available on the Internet, including emails, text, and social media. More recently, from 2021 to 2023, correspondence mail volume fell almost 10%, driven by sharp declines in personal (-16%) and social (-23%) correspondence and a more gradual decline in business and government correspondence (-6%). Businesses and government are often limited in their ability to communicate electronically due to legal or regulatory protocols requiring the exclusive use of hard-copy correspondence.

In 2023, households paid 82% of their bills electronically and only 15% by mail

As in the case of correspondence, transaction mail was also influenced by the rapid growth of electronic transaction alternatives, such as online payments and online presentments of bills and statements. Payments started migrating online in 2004 and, within ten years, one-third of the volume had already moved online.

The speed of the electronic migration of payments was in large part fueled by various promotional campaigns by banks and businesses to quickly broaden their customers' use of online bill payments. While most promotions offered incentives for switching online, some companies chose to impose fees on customers who wanted to continue paying by mail. Measured on how quickly payments migrated to the Internet, it seems clear that those promotional efforts proved to be very successful. For instance, as seen in Table E-3, the share payments by mail, which by 2013 had already fallen to 37%, continued to decline steadily over the next ten years, falling to just 15% by 2023. In contrast, the rapid migration of mail payments to the internet increased the share of electronic payments from 59% in 2013 to 82% in 2023.

Table E.3:
Bill Payment Methods

Shares	2013	2023
By Mail	37%	15%
Electronically	59%	82%
In Person	5%	2%
Total Bills Paid	**100%**	**100%**

Bills and statements (also referred to as presentments) started migrating online a few years later and more gradually than payments. Initially, customers were reluctant to switch to electronic presentments because of data security, privacy concerns, and the lack of hard-copy records. However, as access to electronic copies improved and data security concerns eased, migration began accelerating around 2013 and continued to grow steadily through 2023. As seen in Table E.3b, from 2013 to 2023, the share of online presentments more than doubled, increasing from 20% to 43%, respectively. In contrast, the share of mail presentments declined from 80% in 2013 to just 57% in 2023.

Table E.3 b:
Bills and Statements Received by Method

Shares	2013	2023
By Mail	80%	57%
Electronically	20%	43%
Total Bills Received	**100%**	**100%**

Advertising mail has represented the largest share of household mail since 2005, when it surpassed First-Class Mail for the first time. In 2023, it accounted for 59% of all mail sent and received by households, with 57.3 billion mail pieces. As Table E.4 illustrates, 91% of total advertising mail consisted of Marketing Mail (51.9 billion commercial and non-profit pieces). An additional 9% (5.3 billion pieces) represented First-Class Mail advertising, either stand-alone advertising (3.2 billion pieces) or advertising-enclosed (2.1 billion pieces). The latter is not included in the total volume count of advertising mail received because the primary piece (be it a bill, statement, or correspondence) is already accounted for in the corresponding First-Class Mail category. The large decline in advertising from 2022 to 2023 (shown in Table E.2) reflects the impact of the continuing migration of ad mail to digital and mobile media.

Table E.4:

Advertising by Mail Class

Mail Classification	Volume *(Billions)*	Percent of Total Advertising
First-Class Advertising	5.3	9%
Marketing Mail-Commercial	41.1	72%
Marketing Mail Nonprofit	10.8	19%
Total Advertising Mail	57.3	100%

Source: HDS Diary Sample, FY 2023.

Note: First-Class Advertising includes 2.1 billion pieces of ads enclosed.

Periodical mail volumes have declined for a two or more decades, reflecting dwindling levels of readership and circulation. Deliveries by mail have also faced strong competition from online publications, as they provide cheaper (and often free) access to hard-copy content. More recently, from 2021 to 2023, periodicals received by households declined 6%, falling from 3.7 billion pieces to 3.4 billion, respectively (see Table E-2).

Table E.5 shows, in 2023, magazines accounted for the majority of periodical mail received by households, representing 63% of the volume, with 2.1 billion magazines. Newspapers were the second largest component, with 1.0 billion pieces and accounting for a 29% of the volume. The volume of newsletters was minimal, with just 200 million pieces

Table E.5:

Periodical Type Received

Mail Classification	Volume *(Billions)*	Percent of Total Periodicals
Magazines	2.1	63%
Newspaper	1.0	29%
Newsletters	0.2	6%
Unclassified	0.1	2%
Total Periodicals	3.4	100%

Source: Household Diary Study, FY 2023.

Note: Totals may not sum due to rounding.

Households sent and received a total of 6.7 billion packages in 2023, a decline of 8% from 7.3 billion pieces received in 2022 (see Table E-2). They received 6.4 billion packages and sent 0.6 billion (Table E.6). Compared to 2022, households received and sent 5% and 19% fewer packages, respectively.

Table E.6

Packages Received and Sent via the U.S. Postal Service

(Millions of Pieces)

Mail Classification	2023			
	Received		Sent	
	Number	Percent	Number	Percent
Parcel Select	2803	44%	--	--
Ground Advantage	2,330	37%	320	53%
Expedited	782	12%	133	22%
Other	474	7%	150	25%
Total Packages	**6,372**	**100%**	**603**	**100%**

Source: HDS Diary Sample, FY 2023.

Notes: Totals may not sum due to rounding. Expedited includes Priority Mail and Express Mail.

Chapter 1: Introduction – Volumes & Trends

The United States Postal Service Household Diary Study (HDS) Report documents the findings of Fiscal Year (FY) 2023 study. The HDS measures the mail sent and received by U.S. households, tracks volume trends over time, and compares mail usage by households' demographic characteristics.

The Survey

The Household Diary Study survey, fielded continuously since 1987, gathers information on household's use of the mail and how usage changes over time. The types of information covered by the survey include:

- Volumes and types of mail sent and received,
- Demographics of mail users,
- Attitudes toward mail and advertising,
- Bill payment behavior, and
- Use of the Internet and other information technologies and its impact on mail usage.

Findings are used to assist in market research, forecasting, and strategic planning within the Postal Service and can be viewed by the public on the Postal Rate Commission website.

The Survey Consists of Two Parts:

1) An **entry, or recruitment interview**, conducted by phone or Web, collects demographic and attitudinal information from about 8,500 households.
2) These households then receive a **mail diary**, which collects information on the mail the household sends and receives in a one-week period. Annually, about 5,200 households complete the diary.

The data generated by these two instruments are the basis of the analysis in this report. The FY 2023 HDS report covers the period from October 2022 through September 2023, equivalent to the Government Fiscal Year (GFY) used by the Postal Service. Data from FY 2021 and FY 2022 are also reported on a GFY basis.

U.S. Postal Service Volumes

(This section reviews all mail volumes processed by the Postal Service, as reported in RPW reports. They include mail to/from households as well as business-to-business mail. All other sections of the report will focus exclusively on household mail.)

The Household Diary Study examines the types, volumes, and uses of mail sent and received by households.

In 2023, the Postal Service delivered 116.2 billion pieces of mail, a decrease of 8.9% from 127.5 billion pieces delivered in 2022 (see Table 1.1). The decrease was largely due to lower volumes of Marketing Mail and First-Class Mail, which fell 11.5% and 6.1%, respectively.

The steep decline in Marketing Mail (consisting mostly of advertising materials) reflected a rapid increase in commercial mailers' use of digital and mobile advertising as alternatives to direct mail -- a trend that was greatly accelerated by the pandemic.

The decline in FCM was also driven by the migration of mail to the Internet, with the latter providing a convenient and lower costing alternative to correspondence and transaction mail, such as payments and the presentment of bills and statements.

Competitive mail volume (mostly package products) fell 5.6% in 2023, partly reflecting a lingering impact of post-pandemic deceleration in e-commerce activity.

The Postal Service publishes the revenues, volumes, and weight of mail pieces going through the postal network in the Revenue, Pieces, and Weight (RPW) Reports. Table 1.1 presents the RPW volumes for FY 2021, 2022 and 2023.

Letters, flats, and parcels are defined as follows:

- **Letters** refer to pieces that are less than 11.5 inches wide by 6.125 inches tall and less than .25 inches thick.
- **Flats** consist of pieces that are greater than 11.5 inches wide, 6.125 inches tall, or .25 inches thick, but less than 12 by 15 by .75 inches.
- **Parcels** are pieces that are larger than 12 by 15 inches, or thicker than 0.75 inches.

Because of the difficulty involved in recording mailpiece characteristics in the Household Diary, these categories do not correspond precisely to the shape categories used in RPW reports.

Table 1.1:

Total Mail Volume: FY 2021, 2022, and 2023

(Billions of Pieces)

	2021	2022	2023
Market Dominant Mail:			
First-Class Mail:			
Single-Piece	13.9	12.9	11.8
Presort	35.6	35.0	33.2
Other	1.4	1.3	1.1
Total First-Class Mail	**50.9**	**49.2**	**46.2**
Marketing Mail:			
Letters	47.8	49.7	43.7
Flats	17.9	16.8	15.1
Parcels & Other	0.6	0.5	0.6
Total Marketing Mail	**66.2**	**67.1**	**59.4**
Periodicals	**3.7**	**3.4**	**3.0**
USPS and Free Mail	0.4	0.5	0.4
Package Services Mail	0.5	0.5	0.5
Tot Market Dominant Mail	**121.6**	**120.6**	**109.5**
Total Competitive Mail	**7.3**	**7.2**	6.8
Total All Mail	**128.8**	**127.5**	**116.2**

Source: RPW Reports.

Note: Totals may not sum due to rounding.

Mail Flows

Mail volume can be broken into four basic flows, based on origin and destination. These flows are:

1) Household to household,
2) Household to non-household,
3) Non-household to household, and
4) Non-household to non-household, which is calculated as the residual of total RPW volumes less mail sent and received by households, as determined from HDS surveys.

Tables 1.2 and 1.3 measure volumes by type of mail flow. Table 1.4 shows pieces per household per week for each flow.

Table 1.2:

Total Domestic Mail Flows

(Billions of Pieces)

Sent By:	Received By:		
	Household	Non-household	Total Originating
Household	1.8	3.4	5.2
Non-household	92.1	18.8	111.0
Total Destinating	94.0	22.2	116.2

Source: HDS Diary Sample, FY 2023.

Note: Totals may not sum due to rounding.

Table 1.3:

Total Domestic Mail Flows

(Billions of Pieces)

Mail Flows	Billions of Pieces	Percent of Total Mail
Sent by Household	5.2	5%
Received by Household	94.0	81%
Total Household Mail*	97.4	84%
Non-Household to Non-Household	18.8	16%
Total Mail	**116.2**	**100%**

**Excludes 1.8 billion pieces sent from household to household*

Table 1.4:

Domestic Mail Flows per Household per Week

Sent By:	Received By:	
	Household	Non-household
Household	0.3	0.6
Non-household	15.2	N/A

Source: Household Diary Study, FY 2023.

Household Mail

As shown in Table 1.3, in 2023, domestic mail to and from households accounted for 84% of total USPS mail volume, equating to 97.4 billion pieces sent and received by U.S. households. As seen in Table 1.5, households received 94.0 billion pieces of mail and sent 5.2 billion. Both totals include 1.8 billion pieces of mail that households sent to each other. After subtracting the 1.8-billion-piece double-count (counted as both sent and received), total mail sent and received by households was 97.4 billion pieces. Since RPW reports encompass all USPS mail, including business to business and business to/from government mail, subtracting 105.9 billion pieces of household mail from a total of 116.2 billion pieces reported in RPW, leaves a residual of 18.8 billion pieces of non-household to non-household mail.

Table 1.5:

Mail Received and Sent by Households

(Billions of Pieces)

Mail Classification	Received	Sent
First-Class Mail	32.2	4.6
Marketing Mail Regular	41.1	—
Marketing Mail Nonprofit	10.8	—
Periodicals	3.4	—
Packages & Shipping Services	6.4	0.6
Total	94.0	5.2
Household to Household	1.8	
Total Mail Received and Sent by Households	97.4	
FY 2023 RPW Total	116.2	
Non-household to Non-household (Residual)	18.8	
Unaddressed	0.9	—

Source: HDS Diary Sample, FY 2023.

Note: Totals may not sum due to rounding.

Table 1.6 presents these data in two other forms, annual volumes per household and number of pieces per household per week. Many of the subsequent results in this report are presented in terms of pieces per household per week. As the table shows, in 2023, households received almost 16 pieces of mail per week and sent slightly less than one piece.

Table 1.6:

Pieces Received and Sent per Household

Classification	Annual Pieces per Household	Pieces per Household per Week
Mail Received		
First-Class Mail	277	5.3
Marketing Mail Regular	354	6.8
Marketing Mail Nonprofit	93	1.8
Periodicals	29	0.6
Packages & Shipping Services	55	1.1
Total Mail Received	**809**	**15.6**
Mail Sent		
First-Class Mail:	40	0.8
Packages & Shipping Services	5	0.1
Total Mail Sent	**45**	**0.9**
Unaddressed	8	0.2

Source: HDS Diary Sample, FY 2023.

Note: Totals may not sum due to rounding.

Classes and Markets

- **First-Class Mail** is used to send transactional mail, correspondence, and advertising. Because it is limited to pieces weighing thirteen ounces or less, it primarily includes letters, cards, and large envelopes.
- **Marketing Mail** includes advertisements and marketing packages weighing less than 16 ounces. For the most part, Marketing Mail comprises letters and flats, although it contains a few postcards and packages as well.
- **Periodicals** are magazines, newspapers, and newsletters and are predominantly flat-shaped.
- **Package and Shipping Services** is used to deliver merchandise, books, catalogs, and media. Most of this mail is parcel shaped.

In 2023, the average households received a total of 809 pieces of mail and sent 45 pieces.

Table 1.7 Crosswalks between classes of mail and the markets they serve.

Table 1.7:

Mail Received and Sent by Households

Class	Market (Billions of Pieces)						
	Correspondence	Transactions	Advertising	Periodicals	Packages	Unclassified	Total
First-Class Mail	13.6	16.9	5.3	—	0.1	1.4	**35.2**
Marketing Mail	—	—	51.9	—	0.0		**52.0**
Periodicals	—	—	—	3.4	—		**3.4**
Packages & Shipping Services	—	—	—	—	6.6	0.1	**6.8**
Total Mail Received and Sent by Households	**13.6**	**16.9**	**57.3**	**3.4**	**6.7**	**1.5**	**97.4**

Source: Household Diary Study, FY 2023

1. Correspondence and Packages exclude double-counted pieces reported separately as both sent and received by households.

2. Advertising includes 2.1 billion pieces of First-Class advertising-enclosed mail but excluded from total

Note: Totals may not sum due to rounding

Report Organization

The rest of the Household Diary Study report is organized around the markets the mail serves. Each chapter contains an analysis of the trends in the HDS data, as well as a discussion of how those trends affect and are affected by changes in the broader market. The following provides an overview of each chapter.

Chapter 2: Profile of Mail Usage gives an analysis of household demographics. This chapter examines demographic trends over time and their impact on the mail, and discusses attributing factors, such as access to technology and changing attitudes.

Chapter 3: Correspondence examines mail that is used solely or primarily to deliver (non-sales-related) communications, such as letters and greeting cards. This chapter includes analysis of both personal and business correspondence.

Chapter 4: Transactions reviews financial transactions in the mail and the impact of new technologies on that market. It analyzes household bill payment trends with a focus on technological and demographic change.

Chapter 5: Advertising Mail presents the trends in mail used to deliver sales-related messages. It contains information on household attitudes towards advertising by various media, treatment of advertising mail, and demographic determinants of advertising mail receipt.

Chapter 6: Periodicals examines magazines and newspapers delivered in the mail. It looks at how changing demographics are affecting the market for periodicals, and what the implications are for future volume.

Chapter 7: Packages analyzes household use of various types of packages.

In addition, there are two appendices to the report:

Appendix A contains a set of trend tables for major classes and types of mail

Appendix B documents the study methodology and discusses how the data were collected, weighted, and adjusted, and compares demographic data in the sample to that of the population as a whole.

A copy of the instruments used to administer the survey is available upon request.

Chapter 2: Profile of Mail Usage

Introduction

This chapter provides an overview of demographic trends and other factors affecting mail flows and usage, as they are helpful tools for assessing the causes of change in mail volumes. The breakouts introduced provide the framework for much of the analyses in subsequent chapters.

The first section compares and analyzes long-term changes in mail volume relative to relevant national demographic changes, such as the U.S. population and household formation, and delivery points. The next section examines the demographic characteristics of mail users, contrasting higher-mail-volume households with lower-volume households. The third section details emerging demographic and technological trends that will affect the future of mail.

Mail Volume and Demographics

Table 2.1 illustrates the progression of mail growth over the last four decades (with 13 years for the period 2011-2023) and compares it to changes in national demographics.

Over the entire period, from 1981 to 2023, total mail volume grew from 110 billion pieces to 116.2 billion, or 6%, in part reflecting the long-term growth in population and household formation. The last two, however, increased at much faster rates than mail volumes, with adult population and the number of households surging 48% and 60%, respectively. As household formation increased, the number of places to which the Postal Service delivered increased even faster, surging 70% between 1981 and 2023. Such growth disparities led the Postal Service to incur significant financial losses, by delivering less mail (lowering revenues) to more delivery points (increasing costs).

As Table 2.1 indicates, total mail started declining in the 2001-2010 decade, falling at an average annual rate of 2.1%. The volume continued declining through the next period of 2011 to 2023 at an average annual rate of 3.0%. In contrast, all key demographic factors affecting delivery costs (population, household formation, and delivery points) continued to rise over both periods, each growing about 1% annually.

The last two decades saw significant declines in mail volumes as household formation and delivery points continued to rise

The 1980s were a time of extraordinary growth in mail volume, reflecting strong population growth and an increasing number of delivery points. Importantly, there were also significant technological innovations that helped raise revenue and reduce processing costs. Advances in computerized data management and techniques for sorting large amounts of data created a fertile climate for Marketing Mail growth. Additionally, computerization of financial systems encouraged a greater flow of bills and payments through First-Class Mail. These innovations, in turn, led the Postal Service to introduce work-sharing discounts, encouraging mailers to prepare the mail in ways that reduced total system costs for creating and delivering mail. Mailers could take advantage of these discounts by sorting their mail by Zip code in advance.

The 1990's also saw significant growth in mail volume (especially in the second half of the decade), spurred by a strong economy and postal rates that increased by less than inflation. Additionally, in this decade the U.S. economy rapidly embraced information technology and integrated the Internet into its business processes. Those events, however, also led to a significant migration of mail volumes to the Internet which took place in the decades that followed.

Table 2.1:
Mail Volume and Demographics
Annual Growth, 1981-2023

	1981–1990	1991–2000	2001–2010	2011–2023
Total Mail Volume	4.6%	2.3%	-2.1%	-3.0%
Delivery Points	1.7%	1.5%	1.0%	0.8%
Adult Population	1.5%	1.3%	1.2%	0.9%
Households	1.4%	0.9%	1.2%	1.0%

Source: U.S. Postal Service, U.S. Census Bureau.

The 2001-2010 decade saw the start of deteriorating mail volumes brought by the rapid rise and ubiquity of the Internet as well as significant economic downturns. After the 2001 economic recession and the terrorist attacks on the World Trade Center led to significant volume losses for the USPS, a strong economic recovery spurred a rebound in Marketing Mail volume that reached new highs, surpassing First-Class Mail volume for the first time in 2005. First-Class Mail, however, continued to decline, as large volumes of transaction and correspondence mail started migrating to the Internet. Later, the 2008 economic recession proved to have a severe impact on all types of mail, resulting in a 12.8 percent drop in total mail, the largest volume decrease since the Great Depression. When the 2008 recession was over, it was followed by a slow but long recovery that culminated in 2020.

Between 2011 to 2023, as U.S. population, household formation, and delivery points continued to rise, each growing about to 1% annually, mail volumes fell at an average annual rate of 3.0%. The decline in mail volumes was mostly due to the persistent outflow of First-Class Mail and (to a lesser extent) Marketing Mail to the Internet. Importantly, the 2020 COVID-19 pandemic returned the economy into a recession, further contributing to the downturn in both First-Class and Marketing Mail volumes.

Characteristics of Higher- and Lower-Volume Households

Tables 2.2 and 2.3 illustrate how certain households' demographic characteristics can affect mail usage more significantly than others. For example, while the volume of mail received by households was found to be strongly correlated to income, it did not seem to be much influenced by educational attainment.

More specifically, Table 2.2 shows that, while households earning the lowest annual income ($54,637) received less than 12 pieces of mail per week, those earning almost twice as much ($95,407) received 45 or more pieces. Table 2.3, however, shows that, while college graduates received a much larger share of mail at every level of volume, this relationship was not consistent at lower levels of education.

Table 2.2:
Characteristics of Higher- and Lower-Mail-Volume Households

Mail Received *(Pieces per Household per week)*	**Households** *(Millions)*	**Median Annual Household Income**	**Households w/ Internet Access** *(Percent)*	**Total Bills Paid** *(Pieces per Household per week)*	**Bills Paid by Internet** *(Pieces per Household per week)*	**Mail Sent** *(Pieces per Household per week)*
45 or more	2.4	$95,407	99%	3.4	1.1	4.7
36-44	3.2	$135,201	100%	3.2	1.5	2.8
30-35	5.0	$100,556	100%	2.8	1.3	2.0
24-29	10.4	$98,745	100%	3.2	1.7	1.1
18-23	20.7	$93,568	99%	3.0	1.5	1.0
12-17	29.1	$80,249	98%	3.0	1.6	0.9
Less than 12	60.7	$54,637	96%	2.3	1.2	0.3
Total	**131.4**	**$71,893**	**97%**	**2.7**	**1.4**	**0.8**

Source: HDS Diary Sample, FY 2023.
Note: Mail received includes USPS and Non-USPS mail.

Table 2.3:
Education of Higher- and Lower-Mail-Volume Households

Mail Received *(Pieces per Household per week)*	**Households** *(Millions)*	**Educational Attainment of Head of Household**			
		Less than High School	**High School Graduate**	**Some College or Technical School**	**College Graduate**
45 or more	2.4	0%	27%	16%	58%
36-44	3.2	4%	23%	18%	54%
30-35	5.0	5%	21%	20%	53%
24-29	10.4	7%	19%	20%	53%
18-23	20.7	5%	27%	18%	49%
12-17	29.1	6%	22%	23%	48%
Less than 12	60.7	9%	29%	20%	41%
Total	**131.4**	**7%**	**26%**	**20%**	**46%**

Source: HDS Diary Sample, FY 2023.
Note: Percentages may not total 100 percent due to heads of households who did not answer the educational attainment question.
Percentages in this table are row percentages.
Excludes households not receiving any mail delivery at their home address (using mailbox only).

Demographic Characteristics of U.S. Households

This section presents breakouts of households by the demographic characteristics that typically influence the volume of mail they send and receive. It looks at both traditional and newly emerging factors. The following chapters will show how mail volume varies with these household characteristics.

Income, Education, and Age

Historically, mail usage has shown a strong correlation with household income, age, and (to a lesser extent) education. As Table 2.4 illustrates, except for households earning less than $35K, the relationship between income and education was fairly consistent. At the lower end of the income scale (under $35K), only 22% of households were college graduates, while the largest percentage (32%) were HS graduates. At the highest level of income (over $100K) the vast majority (68%) of households had college degrees and only 14% had HS degrees.

While the relationship between income and age can be somewhat more complicated, overall, household income and age were found to be closely related. More specifically, in Table 2.5 we see that, except for incomes over $100K, at every other level of income the share of households increased as the age of the head of household increased. However, at incomes over $100K, the majority of household were between 35 and 54 years old, while the share of households over 55 was slightly lower instead of increasing. In general, this result reflects the different composition of members of the over 55 cohort. The latter typically includes a large share of households who are either close to or of retirement age and usually earn less than $100K. However, it is also true that, by that point, mail behavior is pretty well set and older households continue to receive similar amounts of advertising materials and periodicals. They also continue to receive and pay a similar number of bills, even though their incomes decline.

Table 2.4:

Households by Income and Education

(Percent of Households)

Household Income *(Thousands)*	**Educational Attainment of Head of Household**				**Total**
	Less than High School	**High School Graduate**	**Some College or Technical School**	**College Graduate**	
Under $35	18%	32%	27%	22%	19%
$35 to $65	4%	37%	24%	35%	18%
$65 to $100	7%	27%	18%	49%	19%
Over $100	3%	14%	14%	68%	28%
Don't know/Refused	6%	26%	20%	43%	16%
Total	**7%**	**26%**	**20%**	**46%**	**100%**

Source: HDS Diary Sample, FY 2023.
Notes: Totals may not sum due to rounding.
Percentages may not total 100 percent due to heads of households who did not answer the educational attainment question.

Table 2.5:

Households by Income and Age

(Percent of Households)

Household Income *(Thousands)*	**Age of Head of Household**				**Total**
	Under 35	**35 to 54**	**Over 55**	**Don't Know/ Refused**	
Under $35	24%	30%	46%	0%	19%
$35 to $65	27%	27%	47%	0%	18%
$65 to $100	23%	29%	48%	0%	19%
Over $100	15%	43%	42%	0%	28%
Don't know/Refused	14%	30%	49%	7%	16%
Total	**20%**	**33%**	**46%**	**1%**	**100%**

Source: HDS Diary Sample, FY 2023.
Note: Totals may not sum due to rounding.

Household Size

The changing composition of households is another important determinant of the volumes and types of mail sent and received. As an example, and as discussed in later chapters, family structure was found to influence the number of bills and the types of advertising materials households receive in the mail.

As highlighted in Table 2.6, in 2023, the majority of U.S. households included either one or two adults. Households with three or more adults represented only 16% of households. However, as noted in Table 2.7, when all household members are included (adults and children), the share of households with three or more members rose to 34%.

Table 2.6:

Households by Number of Adults

(Millions of Households)

Number of Adults	
One	34.3
Two	75.7
Three or more	21.5
Total Households	**131.4**

Source: HDS Diary Sample, FY 2023.
Note: Totals may not sum due to rounding.

Table 2.7:

Households by Size

(Millions of Households)

Household Size	
One person	30.1
Two	56.6
Three	19.1
Four	15.4
Five or more	10.2
Total Households	**131.4**

Source: HDS Diary Sample, FY 2023.
Note: Total may not sum due to rounding.

Internet Access

Since its outset, the Internet has shown to have a strong influence on mail usage. While bills, statements and, to a lesser extent, bill payments are major components of mail sent and received by households, online activity in this area has diverted large volumes of mail once used for these purposes. However, the rapid growth in eCommerce activity and online shopping activities that emerged with the Internet proved to have a very positive influence on the volume of package deliveries by the Postal Service.

Table 2.8 shows that, in 2023, 97% of US households had access to the Internet and 95% of those had Broadband access. As seen in Fig 2.1a, households with incomes over $100,000 had the highest levels of Internet and Broadband access, as the share of households reached 99% and a 97%, respectively. Access decreased only slightly at lower levels of income but, when income fell below $35K, the shares of households with access fell to 89% and 82%.

Table 2.8:

Households by Type of Internet Access

(Millions of Households)

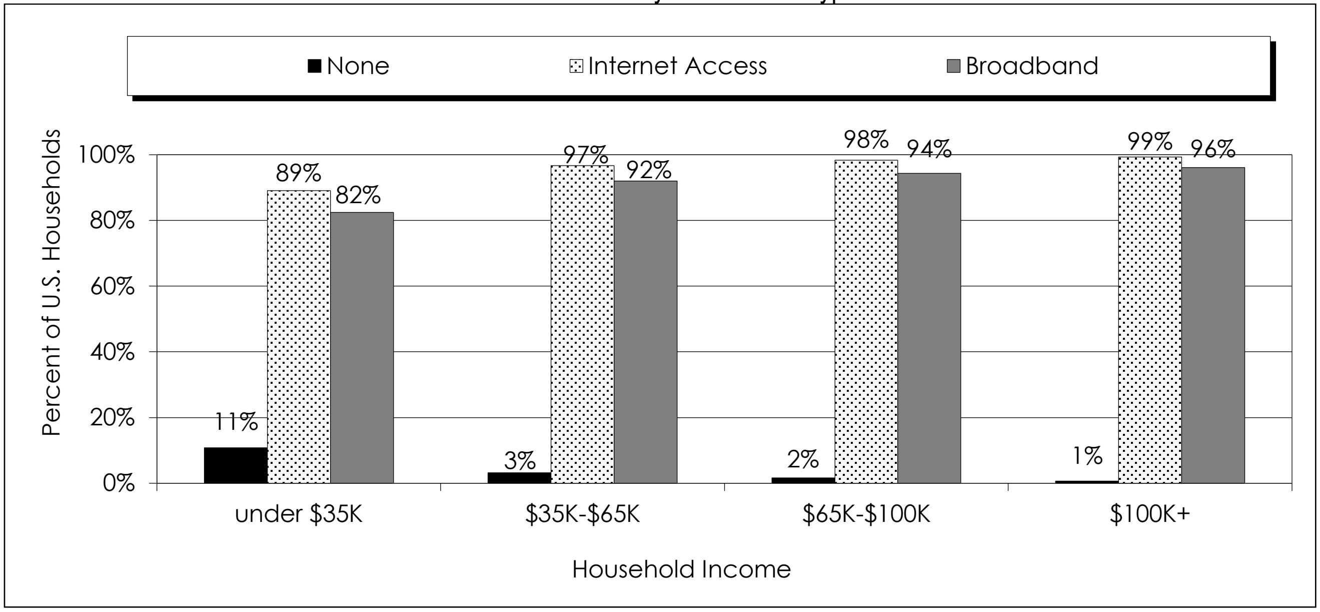
Type of Internet Access	
Broadband	121.7
None	3.3
Others	6.4
Total Households	**131.4**

Source: HDS Diary Sample, FY 2023.
Note: Totals may not sum due to rounding.

While almost all households had access to the Internet in 2023, Figure 2.1b indicates that the age of heads of household continued to have a substantial influence on the level of access. Heads of household in the 35-55 age cohort had the most access to both the Internet and Broadband, with shares of 99% and 95% of households, respectively. They were followed by heads of household under-35, who showed the second highest level of access. Heads of household over-55 had the lowest access to the Internet and Broadband, with shares of 89% and 82% of households, respectively.

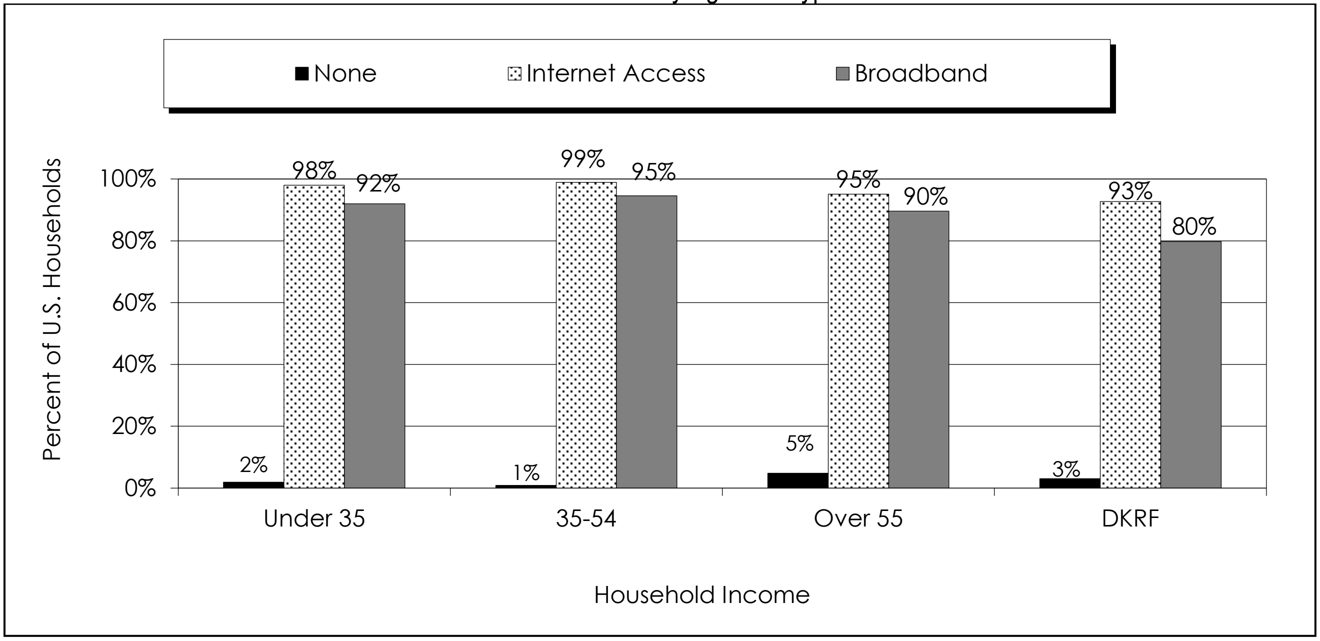
In Figure 2.2 we see that, after a sharp rise in Broadband subscribers from 1998 through the 2008 recession, the rate of growth slowed significantly. Nonetheless, since 2008, the number of subscribers increased by another 44 million, reaching a total of 111 million subscribers by 2023 and accounting for almost half (46%) the U.S. adult population. The strong growth in high-speed connections was a key factor driving the rapid rise of the migration of mail to the Internet and the overall decline in mail usage.

Figure 2.1a:
Internet Access by Income and Type



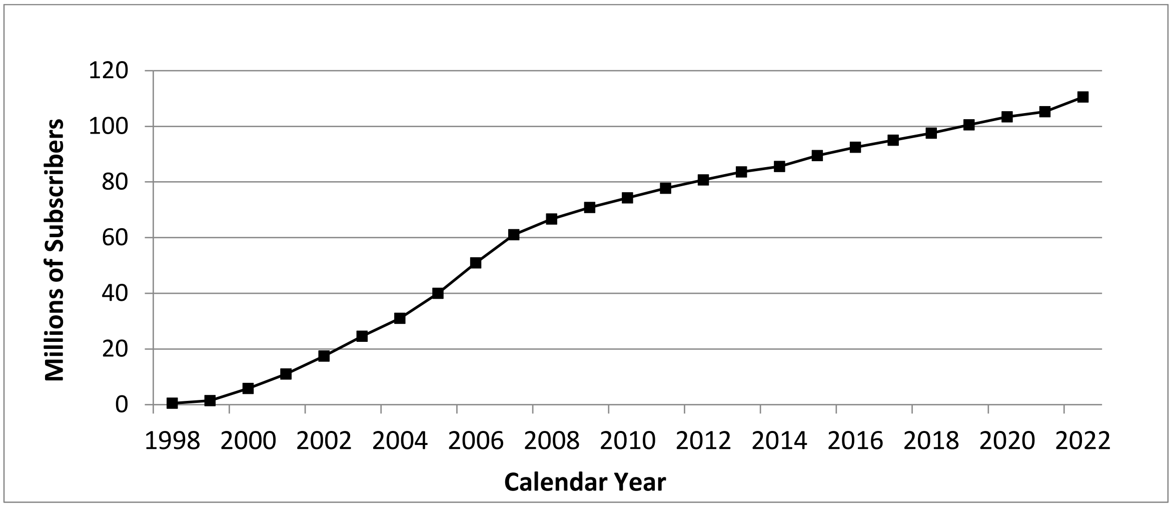
Source: HDS Recruitment Data, FY 2023.
Note: Sum of Internet Access and None does not equal 100 percent due to missing responses and access outside the home only.

Figure 2.1b:
Internet Access by Age and Type



Source: HDS Recruitment Data, FY 2023.
Note: Sum of Internet Access and None does not equal 100 percent due to missing responses and access outside the home only.

Figure 2.2:
Broadband Subscribers



Source: Leichtman Research Group.

Chapter 3: Correspondence

Introduction

This chapter examines the flow of correspondence mail from household to household and between households and businesses, including letters, greeting cards, invitations, and announcements. In several cases, this chapter, and several following it, examines comparisons in data between 2021, 2022, and 2023, providing an illustration of mail trends over time.

Correspondence Mail Volume

Historically, correspondence mail has declined steadily since the early 2000's, when both personal and business communications began migrating to the Internet. Electronic communications and other technological advances have enabled both individuals and businesses to exchange information at faster speeds and at substantially lower costs - key features that persuaded a large portion of the population to move to the Internet. Such developments completely changed the marketplace and continue to have a strong influence on the use of correspondence mail today. Between 2001 and 2023, personal correspondence mail volumes (mostly letters and greeting cards) plummeted 70%, falling from 7.5 billion pieces to 2.3 billion, respectively. However, business and government correspondence volumes sent to households declined more slowly, falling only 20% over the same period.

As seen in Table 3.1, the migration of correspondence remained strong and steady through 2023. For example, in just one-year, from 2022 to 2023, total correspondence mail fell almost 9%. Analyzing volumes by the sectors outlined in the table, we see that the main cause of the 9% decline was a significant reduction in household-to-household, or personal, correspondence. The latter fell almost 15%, three times faster than correspondence from non-households to households (representing mostly business and government mail sent to households), which declined close to 5%. Mail sent by households to non-households represented the smallest component of correspondence but fell the most in terms of percentages (-20%).

The rapid decline in personal correspondence over time was fueled primarily by the availability of a large variety of electronic alternatives (e.g.: emails, social media, and texts) which, at first, were particularly popular with the younger generation but later became ubiquitous across all age groups. The slower decline in businesses and government correspondence sent to households, however, was likely influenced by legal, regulatory, or business protocols which often require the use of hard-copy documents.

According to Table 3.1, in 2023, the vast majority of correspondence mail (78.6%) flowed from non-households to households, with only a small portion of correspondence (17.4%) still flowing from household to households.

Table 3.1:

First-Class Mail Correspondence Sent and Received by Sector

Sector	Volume *(Millions of Pieces)*			Change, 2022–2023
	2021	2022	2023	
Household to household	**2,780**	**2,698**	**2,298**	**-14.8%**
Non-household to household	11,075	10,914	10,388	-4.8%
Household to non-household	980	901	720	-20.1%
Total	**14,835**	**14,512**	**13,214**	**-8.9**
Sector	**Pieces per Household per Week**			**Share of 2023 Total**
	2021	2022	2023	
Household to household	0.4	0.4	0.4	17.4%
Non-household to household	1.6	1.6	1.5	78.6%
Household to non-household	0.1	0.1	0.1	5.4%
Total	**2.2**	**2.1**	**2.0**	**100%**

Source: HDS Diary Sample, FY 2021, 2022 and 2023.
Notes: Totals may not sum due to rounding.

Correspondence Mail and Household Characteristics

The following tables categorize correspondence mail sent and received by households according to the demographic characteristics developed in Chapter 2.

Income, Education, and Age

Tables 3.2 and 3.3 show that household income and, to a lesser extent, educational attainment influenced the volume of correspondence sent and received by households. In most cases, both correspondence sent and received increased as household income increased. For instance, households with income greater than $100K received almost two times as much correspondence as those earning less than $35K. They also received 28% more correspondence than households earning between $35K and $65K. Notably, these relationships were more evident when measured against household income than against educational levels.

Table 3.2:
Correspondence Mail Received by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	**Educational Attainment of Head of Household**				**Average**
	Less than High School	**High School Graduate**	**Some College or Technical School**	**College Graduate**	
Under $35	1.2	1.6	1.5	1.4	**1.4**
$35 to $65	3.2	1.5	1.5	1.6	**1.6**
$65 to $100	2.3	2.4	2.1	2.0	**2.1**
Over $100	2.7	2.6	2.2	2.1	**2.2**
Average	**1.9**	**1.9**	**1.8**	**1.9**	**1.9**

Table 3.3:
Correspondence Mail Sent by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	**Educational Attainment of Head of Household**				**Average**
	Less than High School	**High School Graduate**	**Some College or Technical School**	**College Graduate**	
Under $35	0.3	0.3	0.3	0.2	**0.3**
$35 to $65	0.6	0.3	0.3	0.4	**0.3**
$65 to $100	0.1	0.7	0.4	0.5	**0.5**
Over $100	0.0	1.1	0.4	0.6	**0.6**
Average	**0.2**	**0.5**	**0.4**	**0.5**	**0.5**

Source: HDS Diary Sample, FY 2023.
Note: Excludes Don't Know/Refused.

As Tables 3.4 and 3.5 illustrate, age was another important determinant of household correspondence mail. Regardless of income, younger heads of household received and sent significantly fewer pieces of correspondence than their older counterparts. While young adults have traditionally been less inclined to correspond by mail, the emergence of emails, social media and texting have widened the gap between age groups even further. According to the survey, heads of household over 55 received almost twice as many pieces of correspondence as heads of households under 35. They also sent three times more piece than those under 35.

Table 3.4:
Correspondence Mail Received by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	**Age of Head of Household**			**Average**
	Under 34	**35 to 54**	**Over 55**	
Under $35	1.0	1.5	1.6	**1.4**
$35 to $65	.6	1.7	2.1	**1.6**
$65 to $100	1.8	1.5	2.7	**2.1**
Over $100	1.3	2.1	2.6	**2.2**
Average	**1.2**	**1.8**	**2.3**	**1.9**

Source: HDS Diary Sample, FY 2023.

Table 3.5:
Correspondence Mail Sent by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	**Age of Head of Household**			**Average**
	Under 34	**35 to 54**	**Over 55**	
Under $35	0.1	0.3	0.3	**0.3**
$35 to $65	0.1	0.3	0.5	**0.3**
$65 to $100	0.4	0.3	0.7	**0.5**
Over $100	0.2	0.7	0.8	**0.6**
Average	**0.2**	**0.4**	**0.6**	**0.5**

Source: HDS Diary Sample, FY 2023.

Household Size

As would be expected, household size was strongly correlated to the volume of correspondence sent and received by households. In Tables 3.6 and 3.7, the steps from one- to two-person households were associated with substantial increases in correspondence mail use. Further increases, however, had no discernible (and at times opposite) impact on volume. In Table 3.7, however, we do see that for each additional adult present in the household, the volume of correspondence received increased in step. On the other hand, correspondence sent increased only when the number of adults rose from one to two.

Table 3.6:
Correspondence Mail Received and Sent by Household Size
(Pieces per Household per Week)

Household Size	Received	Sent
One person	1.2	0.3
Two	2.0	0.5
Three	2.2	0.6
Four	1.9	0.4
Five or more	2.3	0.4
Average	**1.9**	**0.5**

Source: HDS Diary Sample, FY 2023.

Table 3.7:
Correspondence Mail Received and Sent by Number of Adults in Household
(Pieces per Household per Week)

Number of Adults	Received	Sent
One	1.3	0.3
Two	2.0	0.5
Three or more	2.4	0.5
Average	**1.9**	**0.5**

Source: HDS Diary Sample, FY 2023.

Internet Access

As illustrated in Table 3.8, households with Internet access received significantly more correspondence than households without the service. While counter-intuitive, the explanation for this result lies in the strong relationship that both income and education have with the presence of an Internet connection in the home. As shown in Table 3.9, households with Internet access on average earned substantially more income than those without a connection and, as shown in tables 3.2 and 3.3 above, the volume of correspondence mail received and sent was highly correlated to income.

Table 3.8:
Correspondence Mail Received and Sent by Type of Internet Access
(Pieces per Household per Week)

Type of Internet Access	Received	Sent
Broadband	1.9	0.5
None	0.9	0.2
Others	2.3	0.4
Average	**1.9**	**0.5**

Source: HDS Diary Sample, FY 2023.

Table 3.9:
Income and Education by Type of Internet Access

Type of Internet Access	Median Income	% w/ College Degree
Broadband	73,321	45%
None	27,712	24%
Others	44,811	30%

Source: HDS Diary Sample, FY 2023.

Personal Correspondence

In 2023. households sent and received 2.3 billion pieces of personal correspondence, a significant decline of 14.8% from 2022. Table 3.10 details volumes by type (or purpose) of correspondence. As shown, except for "Other Personal", all types of personal correspondence contributed to the decline in total volume. Holiday greeting cards, the largest category, fell 3.5% in 2023, continuing a steady decline that has persisted for several years. Non-Holiday greeting cards, the second largest component of personal correspondence, saw the sharpest decline, both in terms of percentages (-24.4%) and volume (-211 million pieces). In terms of percentage, all correspondence categories fell more rapidly in 2023 than in 2022. In general, all volume decreases between 2021 and 2023 reflected the impacts of the continuing migration of personal correspondence to the Internet, be it in the form of emails, texts, or social media.

Table 3.10:
Personal Correspondence Sent and Received

Correspondence Type	**Volume *(Millions of Pieces)***			**Change: 2022–2023**
	2021	2022	2023	
Personal Letters	470	427	323	-24.4%
Holiday Greeting Cards	1116	1,101	1,062	-3.5%
Non-Holiday Greeting Cards	856	864	653	-24.4%
Invitations/Announcements	281	274	256	-6.5%
Other Personal	58	32	34	6.6%
Total	**2,780**	**2,698**	**2,298**	**-14.8%**
Correspondence Type	**Pieces per Household per Week**			**Share of 2023 Total**
	2021	2022	2023	
Personal Letters	0.1	0.1	0.0	14.1%
Holiday Greeting Cards	0.2	0.2	0.2	46.2%
Non-Holiday Greeting Cards	0.1	0.1	0.1	28.4%
Invitations / Announcements	0.0	0.0	0.0	11.1%
Other Personal	0.0	0.0	0.0	1.5%
Total	**0.4**	**0.4**	**0.4**	**100.0%**

Source: HDS Diary Sample, FY 2021, 2022 and 2023.
Note: Totals may not sum due to rounding

Figure 3.1 illustrates the relationship between household income and the main types of personal correspondence mailed in 2023. The chart indicates that holiday greeting cards sent by households experienced a much stronger correlation with income that other types of correspondence. The disparity between the number of holiday greeting cards sent by households with incomes lower than $35K (3 annual pieces) and those earning over $100K (17 annual pieces) was the most significant, with the latter sending almost six times more cards than the former. There was also a substantial increase in holiday cards sent when income increased from $35K-$64K (3 pieces) to $65K-$100K (11 piece), almost a four-fold increase.

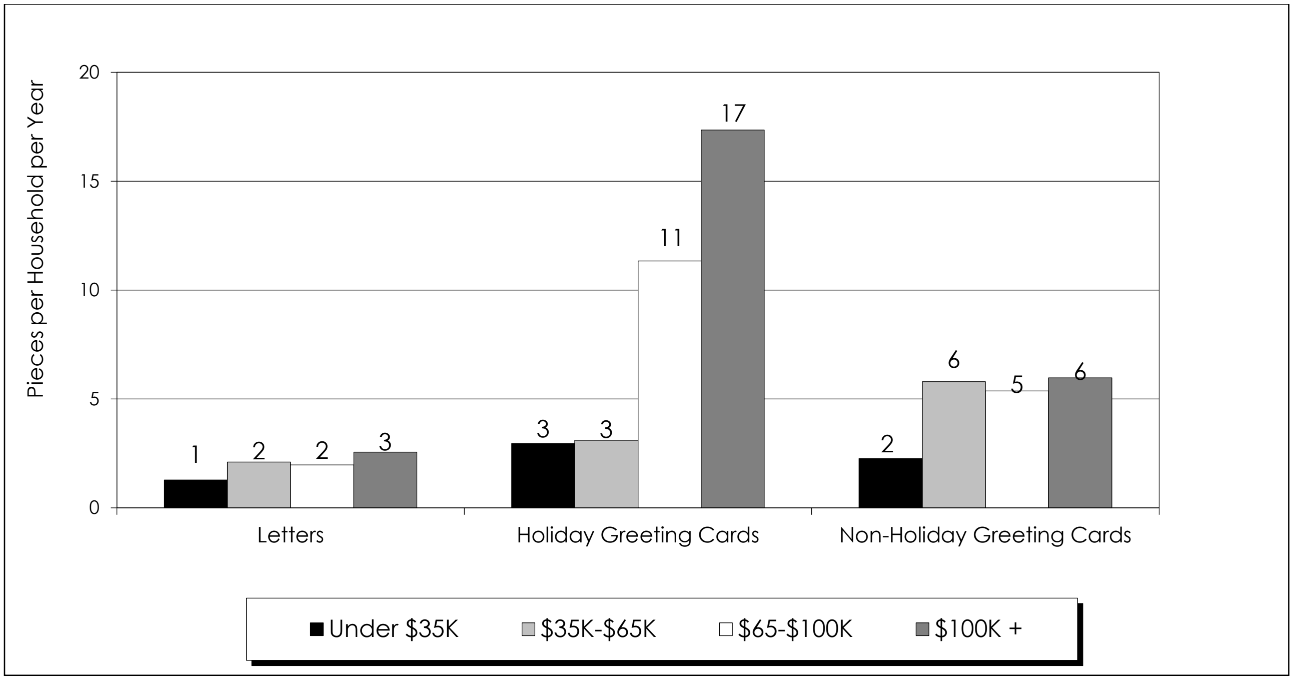
The income relationships for letters and non-holiday greeting cards were both very weak, except when incomes increased from under $35K to $35-$65K.

Figure 3.1:

Personal Correspondence Sent by Income Group

(Pieces per Household per Year)

FY 2023



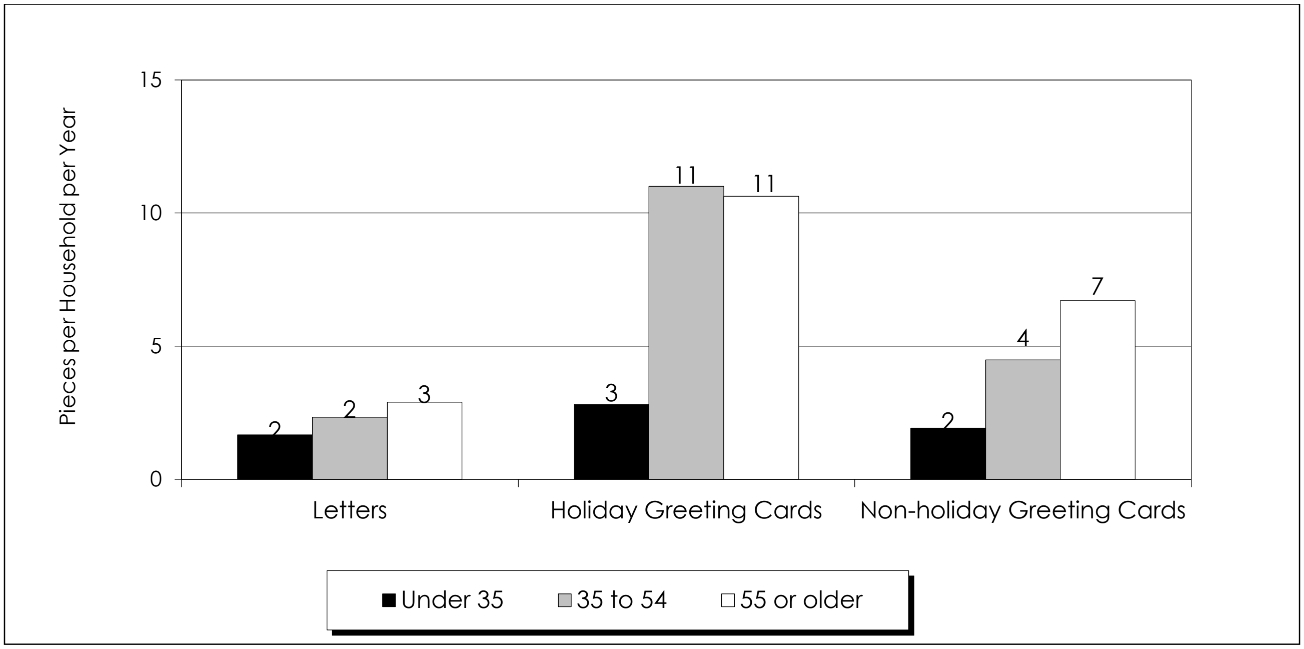
Source: HDS Diary Sample, FY 2023.

Figure 3.2 illustrates the relationship between the age of heads of households and the number of personal correspondence pieces sent. Here we see that the number of letters and non-holiday greeting cards sent was more strongly correlated to age than holiday cards. The former categories increased proportionally with each step increase in the age group, both cases confirming that younger households are less engaged with written correspondence than their older counterparts. Holiday greeting cards, however, showed no distinction in the number of cards mailed by heads of household in the 35-54 and over-55 groups. Heads of household younger than 35, however, sent significantly fewer holiday cards than the older age groups. They mailed just 3 holiday cards in 2023 compared to 11 cards sent by the older groups, probably relying mostly on eCards or social media to send their greetings.

Figure 3.2:

Personal Correspondence Sent by Age Cohort

(Pieces per Household per Year)

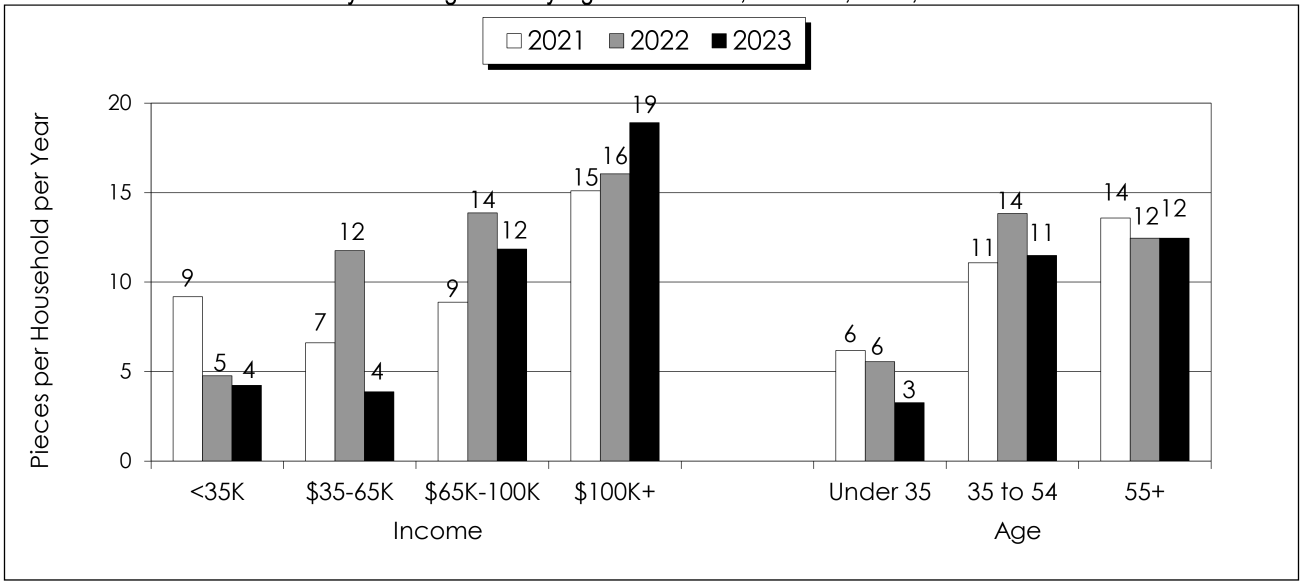


Source: HDS Diary Sample, FY 2023.

Figure 3.3 shows three-year trends in the number of holiday greeting cards sent by households at different levels of income and age. The over $100K group was the only one experiencing a constant increase in the number of holiday cards sent each year, with growth of 7% in 2022 and 16% in 2023. Except for incomes under $35K, all other income groups saw growth in holiday cards only from 2021 to 2022. All income groups other than those earning over $100K mailed fewer holiday cards in 2023 than in 2022. The $35K-$65K income group saw the sharpest decline, with the number of holiday cards falling 67% from 2022 to 2023 from 2022. In contrast, households earning less than $35K sent fewer greeting cards each year, as the number of cards fell 44% in 2022 and 20% in 2023.

When examined by age, in 2023, heads of household over 55 sent the same number of holiday cards as they sent in 2022 but more than the other age groups. Conversely, the number of holiday cards sent in 2023 by the two younger cohorts -- those under 35 and 35-54 years-old -- fell 50% and 21%, respectively.

Figure 3.3:
Holiday Greetings Sent by Age and Income, FY 2021, 2022, and 2023



Source: HDS Diary data, Diary Sample only, FY 2021, 2022, and 2023.

Table 3.11 shows the number of personal correspondence pieces sent and received weekly by households with and without Internet access. While households with Internet access are usually more likely to communicate with friends and family via email or social media, in 2023 they also mailed more greeting cards than households without the Internet. As shown, they mailed about twice as many letters and holiday cards and five times as many non-holiday cards as households without Internet access.

Table 3.11:
Personal Correspondence by Type of Internet Access
(Pieces per Household per Week)

Correspondence Type	No Internet Access	Internet Access
Personal Letters	0.02	0.05
Holiday Greeting Cards	0.07	0.18
Non-Holiday Greeting Cards	0.02	0.10
Total	**0.11**	**0.33**

Source: HDS Diary Sample FY 2023

Business Correspondence

This section examines the types and uses of correspondence mailed between households and non-households. Non-households comprise primarily businesses but also include government and social organizations, such as charities and political groups. In addition to correspondence, households and businesses exchange bills, payments, and advertising materials. These topics are discussed in detail in Chapters 4 (for transactions mail) and Chapter 5 (for advertising mail).

As shown earlier in Table 3.1, in 2023, correspondence received from the non-household sector was the by far the largest component of correspondence mail received by households, accounting for over 79% of the volume.

Table 3.12 presents business correspondence volumes by type and sector for the years 2021, 2022, and 2023. As shown, business, government, and social correspondence received by households fell 4.8% from 2022 to 2023. The largest decline was in the insurance correspondence category, which fell 16.5%. Social correspondence (mostly non-profit and political mail) fell 2.6%, driven entirely by a decline in announcements.

Business correspondence sent by households fell 20.1% from 2022 to 2023, with the business and government category experiencing the largest decline in terms of volume. However, since the volume sent by households was much smaller than volume received, the impact of the decline on total business correspondence mail was also relatively small.

Table 3.12:

Business Correspondence Sent and Received by type and Sector

Business Correspondence Type (millions of pieces)	2021	2022	2023	Change, 2022-2023
Business/Government/Social Received by Households				
Invitation/Announcement	4,309	4,259	4,498	5.6%
Insurance Related (EOBs, etc.)	2,286	2,204	1,841	-16.5%
Tax Related	1,314	1,247	1,301	4.3%
Other Business/Government	2,328	2,445	2,010	-17.8%
Total Business/Government Received	**10,236**	**10,156**	**9,649**	**-5.0%**
Announcement	363	364	313	-14.0%
Other Social	475	394	425	7.9%
Total Social Received	**839**	**758**	**738**	**-2.6%**
Total Business Correspondence Received	**11,075**	**10,914**	**10,388**	**-4.8%**
Business/Government/Social Sent by Households				
Tax-Related	184	119	103	-13.4%
Surveys/Sweepstakes	161	188	130	-30.8%
Greeting Cards/ /Invitations	146	172	112	-34.7%
Other Business/Government	293	306	314	2.6%
Total Business/Government Sent	**784**	**785**	**660**	**-16.0%**
Letters/Surveys/Election Materials	76	80	27	-66.7%
Other Social	120	36	33	-6.6%
Total Social Sent	**196**	**115**	**60**	**-48.1%**
Total Business Correspondence Sent	**980**	**901**	**720**	**-20.1%**

Source: HDS Diary Sample, FY 2021, 2022 and 2023.
Notes: Totals may not sum due to rounding.
Social includes non-profit and political.

Chapter 4: Transactions

Introduction

This chapter examines the volumes and types of transaction mail sent and received by households. Transaction categories include bills, statements, payments, donations, rebates, and other categories.

Transaction Mail

Transaction mail sent and received by households is a major and important component of the mail-stream. In 2023, its volume accounted for 48% of all First-Class Mail sent and received by households and 17% of total household mail (see Table E.2).

As Table 4.1 illustrates, total household transaction mail fell by almost 300 million pieces, or 1.7%, from 2022 to 2023. The main contributors to the decline were bill payments (-196 million pieces), requests for donations (-169 million pieces), and donations (-31 million pieces). Offsetting some of the lower volumes were minor increases in bills and statements. The steep decline in payments was mostly driven by the continuing migration of those transactions to the Internet.

Prior to 2008, bills and statements (also referred to as presentments) were presented almost entirely by mail. However, around 2009, volumes began a slow downtrend which coincided with a gradual increase in the migration of presentments to the Internet. Starting 2018, the pace of migration accelerated significantly and by 2022 the respective volumes of bills and statements had declined 6% and 3% annually. In 2023, both bills and statements saw a slight trend reversal, as volumes rose 0.4% and 0.9%, respectively. The increases were likely related to a strong 2023 economy and significant growth in (presentments-generating) consumer spending.

The long-term rise in online presentments was largely spurred by business promotions offering incentives (and occasionally imposing penalty fees) to customers who were willing to discontinue presentments by mail and agree to receive them only online. The resulting migration of bills and statements to the Internet allowed businesses to significantly reduce mail processing costs, particularly those associated with labor, printing, and postage expenses.

According to the 2023 household survey, respondents who chose to continue using mail presentments explained that their main concern with migrating online was the inability to maintain physical records of their activities. They also expressed concerns about the security and privacy of electronic transactions. However, judging by the strong rise in online presentment over the years, some of those concerns appear to have largely subsided. The survey also found that respondents who chose to switch to online presentments were mostly motivated by the convenience of electronic transactions and by the absence of clutter, which they associated with mail presentments.

In the early 2000's, payments were the second largest component of transaction mail after bill presentments, accounting for approximately 25% of total transaction mail. However, starting in 2004, shortly after the development of online bill payment services, mail volumes started to rapidly migrate to the Internet. By 2010, payments had already plummeted 40%, falling from 10 billion pieces in 2004 to 6 billion in 2010. After 2010, mail payments continued to decline steadily through 2023, at which time volumes fell another 70%, totaling less than 2 billion pieces and accounting for only 10% of total transaction mail.

As was the case for presentments, businesses promoted vigorously the use of electronic payments to reduce the costs associated with processing mail remittances. For their part, many customers moved their payments to the internet very quickly, mostly driven by the lower the cost of paying bills online.

Table 4.1 provides a comparison between the total number of bills received and the total number of bills paid in 2023. As shown, only 20% of all bill received by households were paid by mail (9.1 billion bills received vs. 1.8 billion paid). The majority of the remaining 80% were paid electronically (approximately 2% of bills were paid in person).

Table 4.1:
Transactions Mail Sent and Received

Transaction Type	Volume (Millions of Pieces)			Change, 2022–2023
	2021	2022	2023	
Business				
Bills	9,658	9,095	9,130	0.4%
Bill Payments	2,415	1,946	1,750	-10.1%
Statements	3,577	3,445	3,477	0.9%
Payments (to households)	881	725	778	7.3%
Orders	134	145	112	-22.9%
Rebates	103	98	90	-8.5%
Other Transactions	699	656	664	1.3%
Total Business	**17,468**	**16,109**	**16,000**	**-0.7%**
Social/Charitable				
Request for Donation	536	632	463	-26.7%
Donations	272	267	236	-11.6%
Bills	42	58	33	-43.5%
Confirmations	212	197	244	24.0%
Total Social/Charitable	**1,061**	**1,155**	**976**	**-15.5%**
Total Transactions	**18,530**	**17,263**	**16,976**	**-1.7%**

Table 4.1:
Transactions Mail Sent and Received (cont.)

Transaction Type	Pieces per Household per Week			Share 2023
	2021	2022	2023	
Business				
Bills	1.4	1.3	1.3	53.8%
Bill Payments	0.4	0.3	0.3	10.3%
Statements	0.5	0.5	0.5	20.5%
Payments (to HH)	0.1	0.1	0.1	4.6%
Orders	0.02	0.02	0.02	0.7%
Rebates	0.02	0.01	0.01	0.5%
Other Transactions	0.1	0.1	0.1	3.9%
Total Business	**2.6**	**2.4**	**2.3**	**94.3%**
Social/Charitable				
Request for Donation	0.1	0.1	0.1	2.7%
Donations	0.0	0.04	0.03	1.4%
Bills	0.01	0.01	0.00	0.2%
Confirmations	0.03	0.03	0.04	1.4%
Total Social/Charitable	**0.2**	**0.2**	**0.1**	**5.7%**
Total Transactions	**2.8**	**2.5**	**2.5**	**100.0%**

Source: HDS Diary Sample, FY 2021, 2022 and 2023.

Transaction Mail and Household Characteristics

The following tables examine transaction mail sent and received by households based on the demographic characteristics outlined in Chapter 2.

Income, Education, and Age

Tables 4.2 and 4.3 show that, while transaction mail sent and received by households was positively correlated to income, educational attainment, had a less discernible effect on volumes received.

In general, the reason why household income impacts the volume of transaction mail received is that, even though higher-income households are heavy users of the Internet and more likely to receive bills electronically, they also have more financial accounts. More accounts generate more bills and statements and create more transaction mail.

As seen in Table 4.3, neither income nor education had measurable impacts on the volume of transaction mail sent by households. While higher income households are inclined to have more financial accounts and more bills to pay, they are also likely to have greater access to the Internet and pay at least a portion of their bills online. In net, the positive impact of higher income and the negative impact of electronic payments tend to offset each other, leaving the volume of payments by mail relatively unaffected.

Heads of household older than 55 received two and half times more transaction mail than heads under 35

Table 4.2:
Transactions Mail Received by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	Educational Attainment of Head of Household				Average
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	1.0	1.4	1.7	1.4	**1.4**
$35 to $65	2.6	2.3	2.1	1.6	**2.0**
$65 to $100	2.6	3.5	2.6	1.9	**2.5**
Over $100	2.2	2.9	2.8	2.1	**2.3**
Average	**1.9**	**2.4**	**2.3**	**2.0**	**2.2**

Source: HDS Diary Sample, FY 2023.

Table 4.3:
Transactions Mail Sent by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	**Educational Attainment of Head of Household**				**Average**
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	0.1	0.1	0.3	0.3	**0.2**
$35 to $65	0.1	0.3	0.4	0.2	**0.3**
$65 to $100	0.3	0.8	0.3	0.2	**0.4**
Over $100	0.1	0.4	0.3	0.3	**0.3**
Average	**0.2**	**0.4**	**0.3**	**0.3**	**0.3**

Source: HDS Diary Sample, FY 2023.

As illustrated in Tables 4.4 and 4.5, age had a strong influence on the volume of transaction mail sent and received by households. Across all income categories, younger heads of household (under 35) sent and received the lowest number of transaction mail pieces, averaging 1.1 piece per week. In contrast, the oldest cohort (over 55) received the most transaction pieces (across all income categories), averaging 2.8 pieces per week, or about 2.5 times more than was received by the youngest cohort. In part, this can be explained by the relationship between age and other household characteristics, such as home ownership and marriage. Younger heads of households are less likely to own a home or be married, thus they are likely to have fewer bills to pay, fewer insurance policies, and the like. Furthermore, it is also the case that younger heads of household are more active users of the Internet and more likely to conduct transaction electronically instead of using the mail. This is particularly evident in Table 4.5, which shows that in 2023 heads of household under 35 did not send any transactions by mail, whereas those over 55 sent 0.5 piece per week.

Table 4.4:
Transactions Mail Received by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	**Age of Head of Household**			**Average**
	Under 35	35 to 54	Over 55	
Under $35	1.0	1.2	1.8	**1.4**
$35 to $65	1.2	1.8	2.6	**2.0**
$65 to $100	1.2	2.1	3.4	**2.5**
Over $100	1.0	2.1	3.0	**2.3**
Average	**1.1**	**2.0**	**2.8**	**2.2**

Source: HDS Diary Sample, FY 2023.

Table 4.5:
Transactions Mail Sent by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	Age of Head of Household			Average
	Under 35	35 to 54	Over 55	
Under $35	0.0	0.1	0.3	**0.2**
$35 to $65	0.1	0.1	0.6	**0.3**
$65 to $100	0.0	0.2	0.7	**0.4**
Over $100	0.0	0.2	0.5	**0.3**
Average	**0.0**	**0.1**	**0.5**	**0.3**

Source: HDS Diary Sample, FY 2023.

Household Size

In terms of household size, Table 4.6 shows that an increase from a one-person household to two-person household added 44% more transaction mail pieces received per week and 33% more transaction mail pieces sent. However, larger household had no clear impact on the number of pieces received or sent.

Looking at the influence that the number of adults in the households had on transaction mail, Table 4.7 shows that an increase from one to two adults added 47% more transaction mail received. Moving from two to three adults, transaction mail received increased an additional 36%. However, the number of adults was found to have no impact on the number of transaction pieces sent.

Table 4.6:
Transactions Mail Received and Sent by Household Size
(Pieces per Household per Week)

Household Size	Received	Sent
One person	1.6	0.3
Two	2.3	0.4
Three	2.5	0.3
Four	2.4	0.1
Five or more	2.6	0.2
Average	**2.2**	**0.3**

Source: HDS Diary Sample, FY 2023.

Table 4.7:
Transactions Mail Received and Sent by Number of Adults in Household
(Pieces per Household per Week)

Number of Adults in Household	Received	Sent
One	1.5	0.3
Two	2.2	0.3
Three or more	3.0	0.3
Average	**2.2**	**0.3**

Source: HDS Diary Sample, FY 2023.

Internet Access

As seen in Table 4.8, households with Internet access received significantly more transaction mail than those without a connection. As Table 4.9 illustrates, households with access earned significantly higher incomes and, as a result, were more likely to own a home and have more bills to pay. In contrast, household with Internet access sent approximately the same number of transaction mail pieces as households with no access, which was probably because they paid a portion of their bills online.

Table 4.8:

Transactions Mail Received and Sent by Internet Access

(Pieces per Household per Week)

Type of Internet Access	Received	Sent
Broadband	2.2	0.3
None	1.4	0.2
Others	2.2	0.5
Average	**2.2**	**0.3**

Source: HDS Diary Sample, FY 2023

Table 4.9:

Income and Education by Type of Internet Access

Type of Internet Access	Median Income	% w/ College Degree
Broadband	73,321	47%
None	27,712	20%
Others	44,811	36%

Source: HDS Diary Sample, FY 2023.

Bill Payments

The emergence of the Internet and other technologies has spurred the development of several alternative methods for paying bills. The last two decades saw the use of electronic payments increase exponentially and become the predominant method used by households.

The household diary survey collected the volume of payments made for each payment methods. For purposes of the survey, electronic payments included those made by personal computers, mobile phones, automatic deductions from bank and credit card accounts, and landline phones.

The share of payments by mail fell from 27% in 2018 to 15% in 2023

Table 4.10 presents the average number of bills paid monthly in 2023 for each payment method. It also provides a comparison to 2018 to show changes in payment methods used over five years. As illustrated, in 2023, the share of payments by mail fell to just 15% (2.1 payments by mail out of a total of 13.4 payments), its lowest point to-date and a significant decline from 27% in 2018. Conversely, the share of payments by Internet increased from 44% in 2018 to 51% in 2023, representing (for the first time) more than half of all household payments.

It is also notable that, since 2018, there were substantial increases in the use of automated deductions and credit card payments, with their share rising to 19% and 10% of payments, respectively. Total payments increased 2.4 pieces per month (or 22%) from 2018, mostly driven by a 2-piece increase in online payments per month. In general, the increase in total payments was likely fueled by the growth in the number of bills generated by long-term growth in the economy and consumer spending.

Looking at shares by major category, Table 4.10 shows that 82% of bills were paid electronically, 15% by mail and 2% in person.

Table 4.10 also shows that the share of households who made at least one payment by mail fell from 59% in 2018 to less than half (47%) for the first time in 2023. This result implies that, for the first time in 2023, more than half of US households did not make a single payment by mail. In contrast, almost all households (95%) reported making at least one payment electronically (up from 86% in 2018).

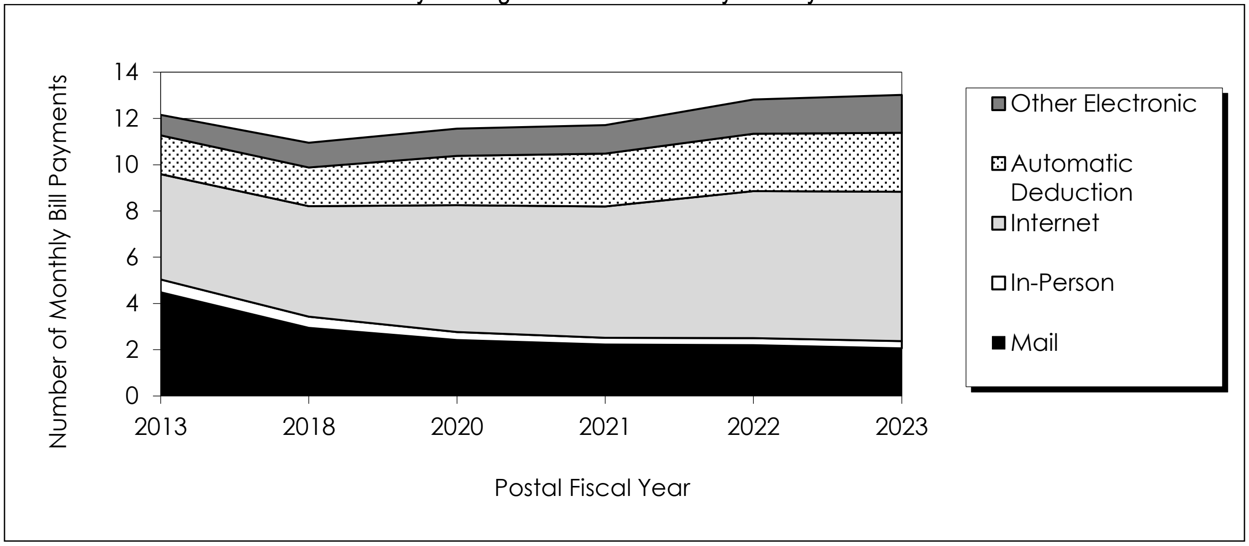
Table 4.10:
Bill Payment by Method, FY 2018 vs. 2023

Bill Payment Method	2018			2023		
	Average Number of Bills Paid per Month	Share of Bills Paid	Percent of Households Making at Least One Payment Using Method	Average Number of Bills Paid per Month	Share of Bills Paid	Percent of Households Making at Least One Payment Using Method
Mail	2.9	27%	59%	2.1	15%	47%
Using Internet	4.8	44%	68%	6.8	51%	83%
Automatic Deduction	1.7	15%	54%	2.6	19%	67%
Credit Card	0.6	6%	25%	1.3	10%	42%
Telephone	0.5	4%	18%	0.3	3%	16%
In-person	0.5	4%	23%	0.3	2%	17%
Total	**11.0**	**100%**	—	**13.4**	**100.0%**	—
Total Electronic	7.6	69%	86%	11.0	82%	95%

Source: HDS Recruitment Sample, FY 2018 and 2023.

Figure 4.1 shows long-term trends (2013-2023) in the number of monthly payments made by each method. As illustrated, the number of payments made by mail declined continuously over the ten-year period, falling 54%, from 4.5 payments per month to 2.1 per month. In contrast, online payments increased continuously, rising 42%, from 4.5 payments per month in 2013 to 6.5 in 2023. Automatic deductions also increased during that period, rising 52%, from 1.7 per month in 2013 to 2.6 in 2023.

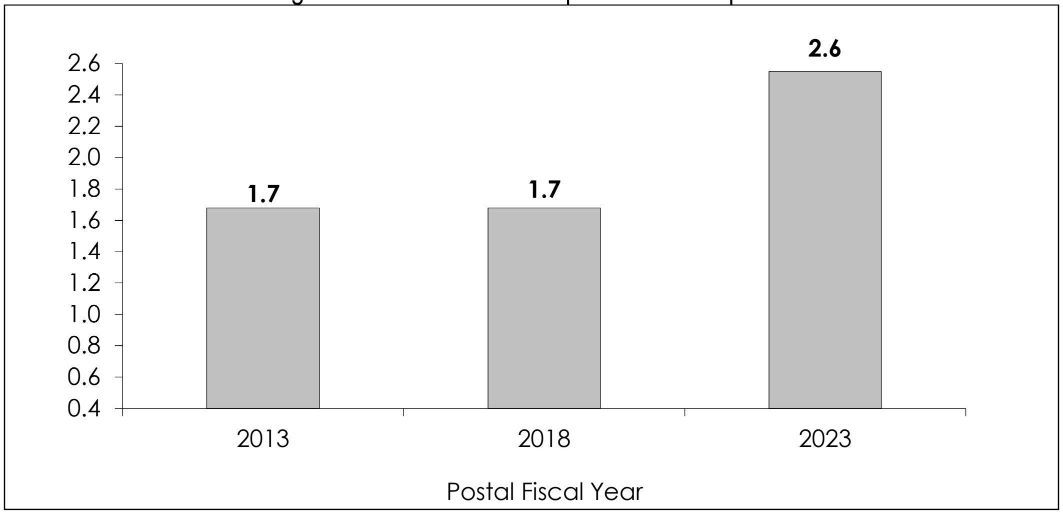
Figure 4.1:
Monthly Average Household Bill Payment by Method



Source: HDS Recruitment Sample, FY 2013-2023.
Note: Other Electronic includes telephone.

Figure 4.2 shows that, between 2013 and 2018, automatic deductions from bank accounts reached a plateau, averaging 1.7 payments per month each year. However, after 2018, deductions increased sharply, rising to 2.6 payments per month by 2023. It is notable that, despite the growing popularity and usage of online bill-pay, the number of bills paid by automatic deductions continued to rise. In fact, as seen in Table 4.10, automatic deductions in 2023 ranked second compared to other electronic payments, and their volume was higher than payment by mail, which they overtook for the first time in 2021.

Figure 4.2:
Average Automatic Deductions per Household per Month



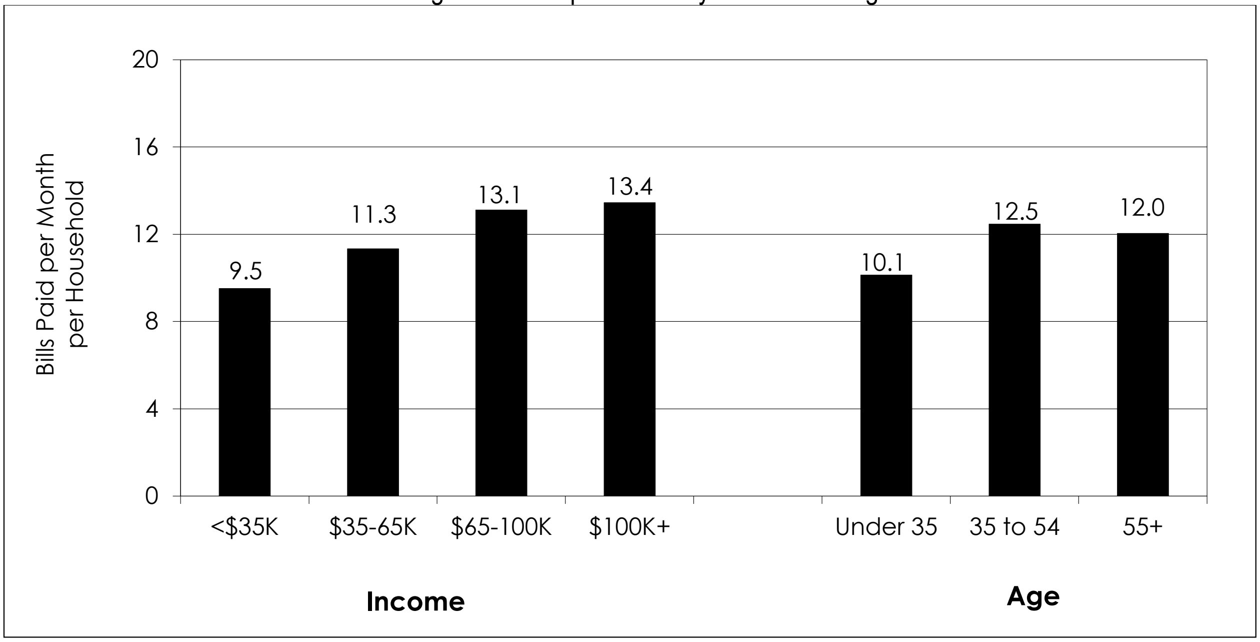
Source: HDS Recruitment Sample, FY 2013-2023.

As noted earlier in the report, the household diary survey found that the number of bills paid by mail was correlated to both the age and income of the head of household. Figure 4.3 illustrates those relationships for 2023 in terms of monthly payments.

The left side of the chart shows the number of bills paid by major income categories and illustrates the strong relationship that exists between the number of payments and household income. Comparing the two extreme income points, we see that households earning over $100K paid an average of 13.4 bills per month. Those payments were 41% greater than the number paid by households earning less than $35,000, which averaged 9.6 per month.

The link between bills paid by mail and age was only partial. Heads of household between 35 and 54 years of age made 12.5 monthly payments in 2023, 24% more than heads of household younger than 35, who paid 10.1 bills. However, they also made 4% more payments than heads of household over 55, who paid 12.0 bills per month. This finding is likely due to the high proportion of people over 55 who are either approaching or at retirement age. In general, retirees tend to moderate their spending habits and are usually left with fewer bills to pay.

Figure 4.3:
Average Bills Paid per Month by Income and Age

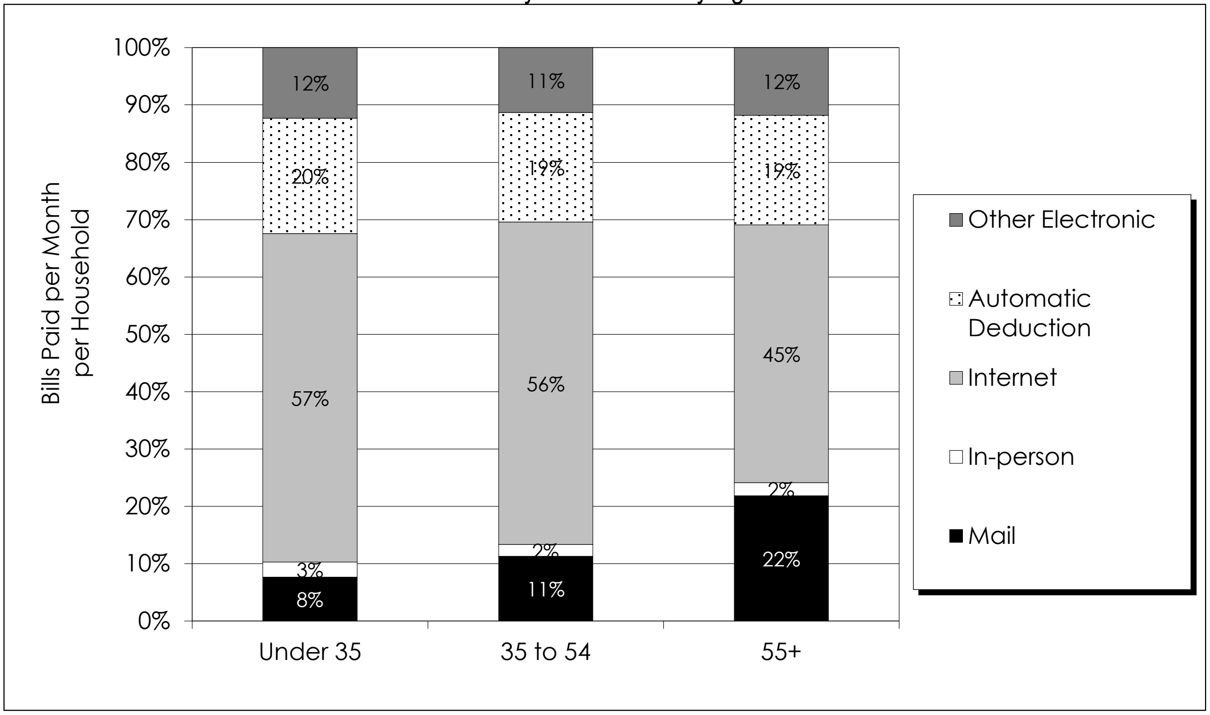


Source: HDS Recruitment Sample, FY 2023.

Figure 4.4 shows how the age of heads of household influenced the share of payments made for each method. Unsurprisingly, heads of household younger than 35 paid only 8% of their bills by mail. Instead, they paid the majority of their bills (57%) online and another 20% by automatic deductions from bank accounts.

Heads of household over 55 paid a substantially greater share of bills by mail than the younger cohorts, with almost one-quarter (22%) sent by mail. They also showed a significant rise in the use of online payments compared to prior years, with the share approaching half of their payments (45%). The share of bills paid by automatic deductions (19%) was similar across all age groups.

Figure 4.4:
Bill Payment Method by Age



Source: HDS Recruitment Sample, FY 2023.
Note: Other Electronic includes telephone.

Bills and Statements Received by Households

Table 4.11 shows an industry breakdown of bills and statements received by households. In 2023, households received a total of 12.6 billion bills and statements by mail, which accounted for 39% of all First-Class Mail received by households. Bills totaled 9.2 billion pieces, originating primarily from the financial and Service industries, which represented 48% and 46% of total household bills, respectively. Credit card companies were the largest mailers of financial bills, with 59% of industry volume. Utility companies and medical/other professional businesses were the two largest mailers of bills in the service industry, sending 43% and 33% of the volume, respectively.

The number of statements received by households totaled 3.5 billion pieces. They originated almost entirely from the financial industry, which mailed 91% of all statements. Banks, S&Ls, and credit unions combined were the largest mailers of financial statements, accounting for 53% of industry volume.

Table 4.11:
Bill and Statement Volumes by Industry

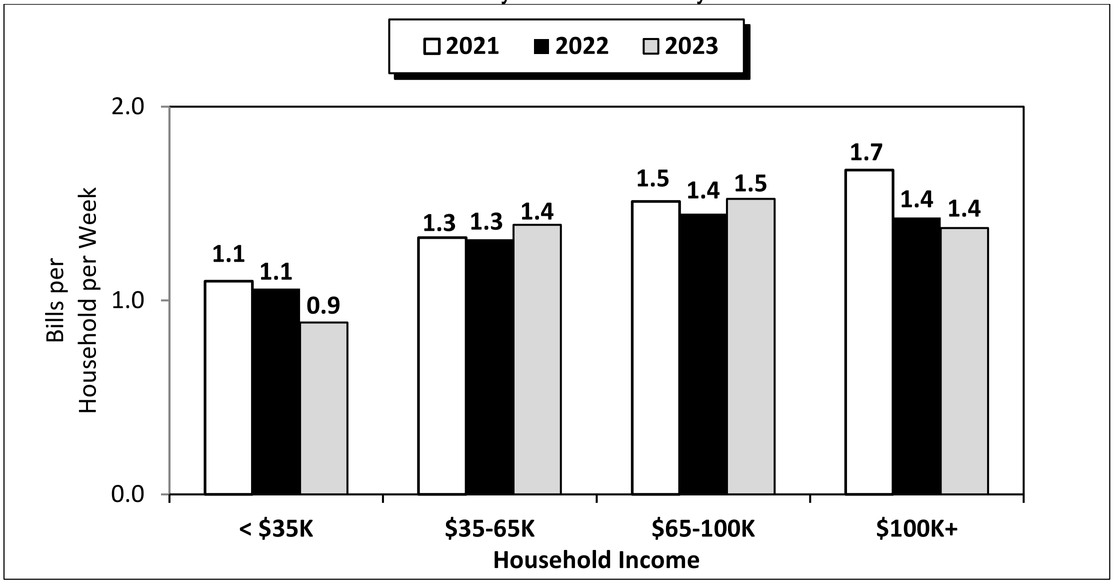
Industry	Volumes	
	Bills *(Millions)*	Statements *(Millions)*
Financial		
Bank, S&L, Credit Union	498	1,673
Credit Card	2,621	7
Insurance Company	1,024	214
Real Estate/Mortgage	157	59
Other Financial	52	1,254
Total Financial	4,352	3,207
Merchants		
Department Store	15	3
Publisher	51	1
Mail Order Company	9	0
Other Merchants	50	10
Total Merchants	125	14
Service		
Telephone/Cable Company	666	16
Utility Company	1,846	6
Medical and Other Professional	1,428	101
Other Service	258	20
Total Service	4,199	143
Manufacturers	60	10
Government	383	78
Social/Nonprofit	33	24
Other/Don't Know/Refused	12	1
Total – All Industries	9,163	3,477

Source: HDS Diary Sample, FY 2023.
Note: Social/Nonprofit bill and statement volumes were not collected separately.

Figures 4.5a and 4.5b, show three-year trends for the number of bills and statements received by households, by major income category. In Fig. 4.5a, we see that, except for households earning more than $100K, higher income households received increasingly more bills than those with lower incomes. This result is mostly explained by the relationship that typically exists between household income and the number of financial accounts. That is: higher incomes usually bring more financial accounts, and more accounts generate a greater number of bills received.

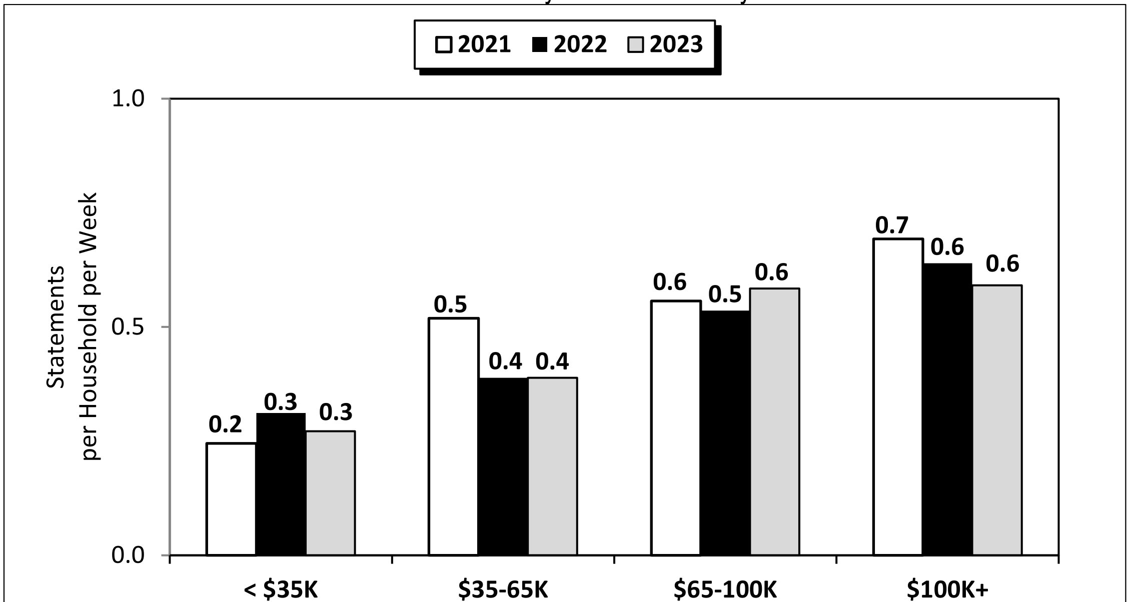
It is also notable that, for both households earning over $100K and those earning less than $35K, the number of bills received in 2023 was significantly lower than in 2021, each falling 18%. This likely reflected the stronger propensity for those two groups to use electronic billing, though most likely for different reasons. Similar trends were found for statements received, shown in Fig. 4.5b

Figure 4.5a:
Bills Received by Mail Per Week by Income



Source: HDS Diary Sample, FY 2021, 2022 and 2023. Amounts are rounded.

Figure 4.5b:
Statements Received by Mail Per Week by Income



Source: HDS Diary Sample, FY 2021, 2022 and 2023. Amounts are rounded.

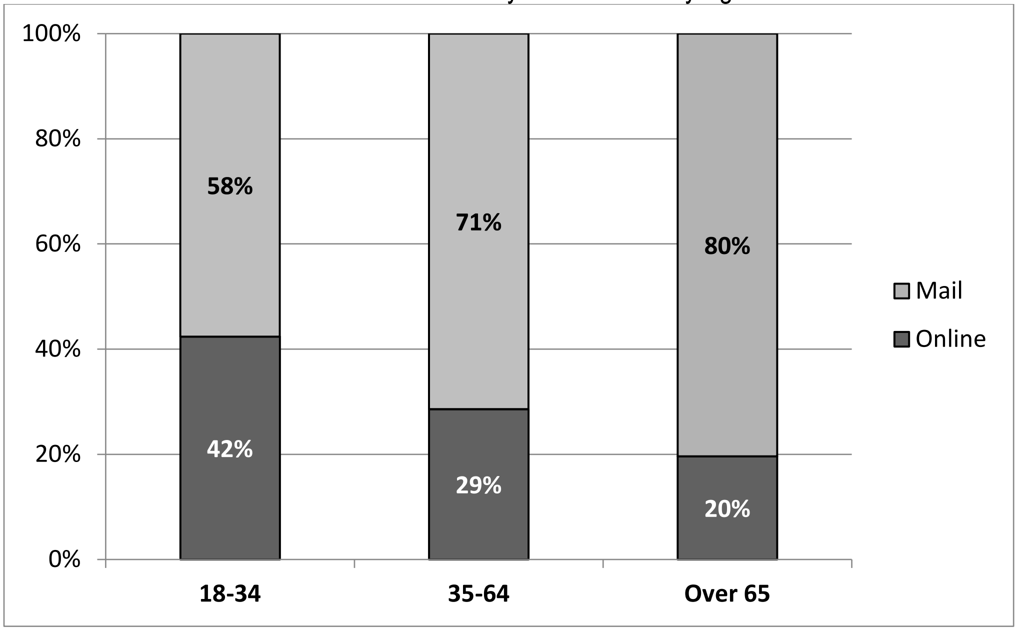
Table 4.12 compares the number of bills received by mail and online from 2021 to 2023. Though most bills received in 2023 were still presented by mail (61%), the volume was 8% lower compared to 2021. Conversely, bills received online increased 8% from 2021 and their share of total bills increased from 36% in 2021 to 39% in 2023.

Table 4.12:
Average Monthly Bills Received by Method

Method	2021	2022	2023
Mail	**6.3**	**5.8**	**5.8**
Internet	**3.4**	**3.7**	**3.7**
Total	**9.7**	**9.5**	**9.5**
Share Received Online	**36%**	**39%**	**39%**

Figure 4.6 illustrates the percentage of bills received by presentment method and by age cohort. Because younger heads of household (18-34 years old) are more active users of the Internet, they received almost half (42%) of their bills online. By comparison, heads of household over 65 continued to rely mostly on paper billing, receiving 80% of their bills by mail and only 20% online.

Figure 4.6:
Shares of Bills Received by Methods and by Age



Source: HDS Recruitment Sample, FY 2023

Chapter 5: Advertising Mail

Introduction

This chapter examines advertising mail, which consists of any advertising, promotional, or sales material sent via the Postal Service. Advertising mail can be sent as First-Class or Marketing Mail.

The Advertising Market

In 2023, American businesses spent approximately $364 billion advertising their products and services, an increase of 5% from 2022, following another 5% increase from 2021 (Table 5.1). Of this total spending, 4% was spent on direct mail advertising, the third largest spending medium after the Internet and TV.

As Table 5.1 illustrates, direct mail spending fell 7% in 2023, after increasing 2% in 2022. Spending on most other media types also fell in 2023, after rising in 2022. However, Internet ad spending rose both years, growing 11% in 2023 and 7% in 2022. Over the past several years, the rapid growth in Internet advertising absorbed funds otherwise available for spending on direct mail and other types of advertising media, thus reducing their share of the market.

Table 5.1:

U.S. Advertising Spending Growth by Medium, 2021-2023

(Percent Growth from Prior Year)

Medium	2021	2022	2023
Direct Mail	15%	2%	-7%
TV	14%	3%	-5%
Radio	21%	1%	-7%
Newspapers	6%	-6%	-3%
Magazines	6%	-8%	-9%
Internet	43%	7%	11%
Other	12%	12%	0%
Total	**30%**	**5%**	**5%**

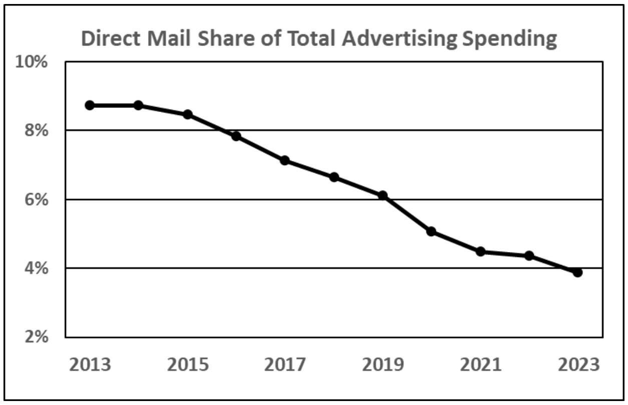
Source: GroupM—Excludes Political Advertising

*Other fluctuations are due to political/Olympic advertising

As seen in Figure 5.1, GroupM's estimates that direct mail's share of total advertising spending took a downturn in 2015 which continued through 2023, falling from 8.5% to 3.9%, respectively.

Figure 5.1:

Direct Mail as a Share of Total Advertising, 2013-2023



Source: U.S. Postal Service calculations based on GroupM data.

Despite the decline in market share, direct mail continued to be a popular choice for businesses because of its effectiveness and versatility. Direct mail can be targeted to the interests of individual customers and used to both locate new customers and maintain relationships with existing customers. In addition, direct mail allows for a variety of different types of advertising, such as letters, postcards, catalogs, and free samples. It can be sent as First-Class or Marketing Mail, allowing advertisers to trade off expeditious, personalized First-Class mailings against cost-savings from Marketing Mail.

Importantly, the effectiveness of direct mail is readily measurable, more so than most other media. Businesses can track the response rate to a mailing far more precisely than to television, magazine, or online ads. This feature alone gives advertising mail a key advantage over other media.

Advertising Mail Volume

As mentioned, businesses can send direct mail advertising as either First-Class Mail or Marketing Mail. In 2023, households received a total of 57.3 billion pieces of advertising mail (Table 5.2), which accounted for 59% of total household mail. From 2022 to 2023 total ad volume fell 12.3%, reflecting the growing migration of direct mail to digital advertising. The decline also reflected concerns over the possibility of a 2023 economic recession, which never materialized but nonetheless constrained business spending decisions.

Marketing Mail volume totaled 51.9 billion pieces, representing 91% of total advertising mail received by households and a significant 11.3% decline from 58.6 billion pieces received in 2022. Regular and ECR Marketing Mail, the main component of total Marketing Mail, fell 12.1% and were the main drivers of the total decline.

First-Class Mail advertising totaled 5.3 billion pieces, a 20.8% decline from 2022, representing the remaining 9% of total ads received by households. Of this volume, 3.2 billion pieces contained only advertising materials (also referred to as advertising-only mail), while the remaining 2.1 billion pieces were secondary advertising. The latter (also referred to as advertising-enclosed mail), was not included in RPW and other reports of total advertising mail volume because the primary piece (be it a bill, statement, or correspondence mailing) was already accounted for in the corresponding First-Class Mail category.

Both advertising-only and ads-enclosed mail experienced significant declines from 2022 to 2023, falling 18.0% and 24.7%, respectively. Prior to 2023, ads-only volume was relatively stable but ads-enclosed has fell continuously since 2015.

Table 5.2:
Advertising Mail Received by Households by Mail Classification
(Volume in Billions of Pieces)

Mail Classification	Volume *(Billions of Pieces)*			Change 2022–2023
	2021	2022	2023	
First-Class Mail Advertising	**6.6**	**6.7**	**5.3**	**-20.8%**
Advertising-Only	3.6	3.9	3.2	-18.0%
Secondary Advertising	3.0	2.8	2.1	-24.7%
Marketing Mail Advertising	**58.5**	**58.6**	**51.9**	**-11.3%**
Regular and ECR	46.3	46.8	41.1	-12.1%
Nonprofit	12.2	11.8	10.8	-8.3%
Unsolicited Packages	0.03	0.05	0.02	-60.2%
Total Advertising	**65.2**	**65.4**	**57.3**	**-12.3%**
Unaddressed Mail	0.8	1.3	0.9	-25.6%

Source: HDS Diary Sample, FY 2021, 2022, and 2023.
Notes: Totals may not sum due to rounding. Unaddressed Mail is not included in totals.
Minor changes to historical data reflect changes to sample expansion factors related to revised Census population estimates.

Table 5.3:
Advertising Mail by Mail Classification
(Pieces per Household per Week)

Mail Classification	Pieces per HH per Week			Share of Total
	2021	2022	2023	
First-Class Advertising	**1.0**	**1.0**	**0.8**	**9.3%**
Advertising Only	0.5	0.6	0.5	5.6%
Secondary Advertising	0.4	0.4	0.3	3.7%
Marketing Mail	**8.7**	**8.6**	**7.6**	**90.6%**
Regular and ECR	6.9	6.9	6.0	71.8%
Nonprofit	1.8	1.7	1.6	18.9%
Total Advertising	**9.7**	**9.6**	**8.4**	100.0%
Unaddressed Mail	0.1	0.2	0.1	1.6%

Source: HDS Diary Sample, FY 2021, 2022 and 2023.
Note: Totals may not sum due to rounding. Unaddressed Mail is not included in totals.

Advertising Mail and Household Characteristics

Income, Education, and Age

Given that advertising mail is used to sell goods and services, it is not surprising that the volume of advertising mail received by households is closely tied to their income and education.

As Table 5.4 illustrates, in 2023, the relationship between advertising mail and household income was quite strong. As an example, heads of household with incomes of $100,000 or more received twice as many pieces per week as those earning less than $35,000 (10.6 and 4.9 pieces, respectively).

The table shows that education also played a key role in the volume of advertising mail received. Households headed by someone without a high school degree received an average of 5.8 ad pieces per week, while households headed by a college graduate received 9.3 pieces per week, or 60% more pieces. At higher income brackets, we see that household with less education received the same or more ad pieces than the those with higher education. This result probably reflected situations where successful entrepreneurs received large volumes of business-related ad mail.

In general, the role that education plays in advertising mail is two-fold. First, direct mail is a written type of communication, and education may play some role in its relative effectiveness compared to television or radio advertising. Second, education is not only tied to current household income, but also to future household income. A college graduate who currently has a relatively low income may, in a few years, earn a much higher income.

Table 5.4:
Advertising Mail Received by Income and Education
(Pieces per Household per Week)

Household Income (Thousands)	Education of Head of Household				Average
	Less than High School	High School graduate	Some College or Technical School	College graduate	
Under $35	2.6	4.3	6.3	5.8	**4.9**
$35 to $65	8.5	7.7	7.2	7.3	**7.4**
$65 to $100	9.4	8.6	10.0	8.5	**8.9**
Over $100	10.9	10.3	10.9	10.7	**10.6**
Average	**5.8**	**7.5**	**8.4**	**9.3**	**8.4**

Source: HDS Diary Sample, FY 2023.

As Table 5.5 illustrates, the volume of advertising mail received was also strongly correlated with age. At each level of income, households headed by an older person received substantially more advertising pieces than their younger counterparts.

This relationship occurs because the age of the head of households is also correlated to other demographic characteristics such as marriage, home ownership, and the presence of children in the household. Moreover, the older a person is, the longer his or her buying history and the more businesses with which the person has relationships that advertising mail can help maintain.

The number of ads received weekly was maximized when heads of households were over 55 and earned higher incomes. For example, in Table 5.5 we see that, heads of household older than 55 who earned over $100,000, received 13.9 ad pieces weekly, more than any other demographic combination.

The amount of advertising mail received is strongly correlated to income, age, and education.

Table 5.5:

Advertising Mail Received by Income and Age

(Pieces per Household per Week)

Household Income *(Thousands)*	Age of Head of Household			Average
	Under 34	35 to 54	Over 55	
Under $35	2.5	3.4	7.1	**4.9**
$35 to $65	3.0	6.2	10.7	**7.4**
$65 to $100	4.7	7.1	12.0	**8.9**
Over $100	5.3	9.4	13.9	**10.6**
Average	**3.9**	**7.3**	**11.1**	**8.4**

Source: HDS Diary Sample, FY 2023.

Household Size

Tables 5.6 and 5.7 show the relationships between the size and number of adults in the household and the volume of advertising mail received. In Table 5.6 we see that, while the relationship with the size of the household was evident when looking at changes from one to two household members (ad pieces increased 37%), additional members over two did not increase but, in fact, reduced the number of ad pieces received.

Table 5.6:

Advertising Mail Received by Size of Household

(Pieces per Household per Week)

Household Size	
One person	6.7
Two	9.2
Three	8.5
Four	8.5
Five or more	8.5
Average	**8.4**

Source: HDS Diary Sample, FY 2023.

Examining the impact that the number of adults in the household had on the volume of ads received, the survey found that the number of pieces received increased continuously with additional adults.

Table 5.7 we see that households with two adults received 38% more ad pieces than when only one adult was present. When three or more adults were present, households received 13% additional ad pieces. These results stem from the likelihood that households with more than one adult also have two or more incomes.

Table 5.7:
Advertising Mail Received by Number of Adults
(Pieces per Household per Week)

Number of Adults	
One	6.4
Two	8.8
Three or more	9.9
Average	**8.4**

Source: HDS Diary Sample, FY 2023.

Internet Access

Table 5.8 shows that households with Internet access received 8.5 ads pieces per week, or 57% more pieces than was received by households without access. This can be explained by the relationships between income, education, and access. As shown in Table 5.9, households with access tend to have more income and a higher education, traits that typically drive ad volumes higher (see Table 5.4). Still, households without access received a substantial number of ads, in part because advertising mail is often sent as a complement to Internet sales to direct potential customers to specific company websites.

Table 5.8:
Advertising Mail Received by Internet Access
(Pieces per Household per Week)

Type of Internet Access	
Broadband	8.5
None	5.4
Others	8.5
Average	**8.4**

Source: HDS Diary Sample, FY 2023.

Table 5.9:
Income and Education by Type of Internet Access

Type of Internet Access	Median Income	% w/ College Degree
Broadband	73,321	47%
None	27,712	20%
Others	44,811	36%

Source: HDS Diary Sample, FY 2023.

Senders of Advertising Mail

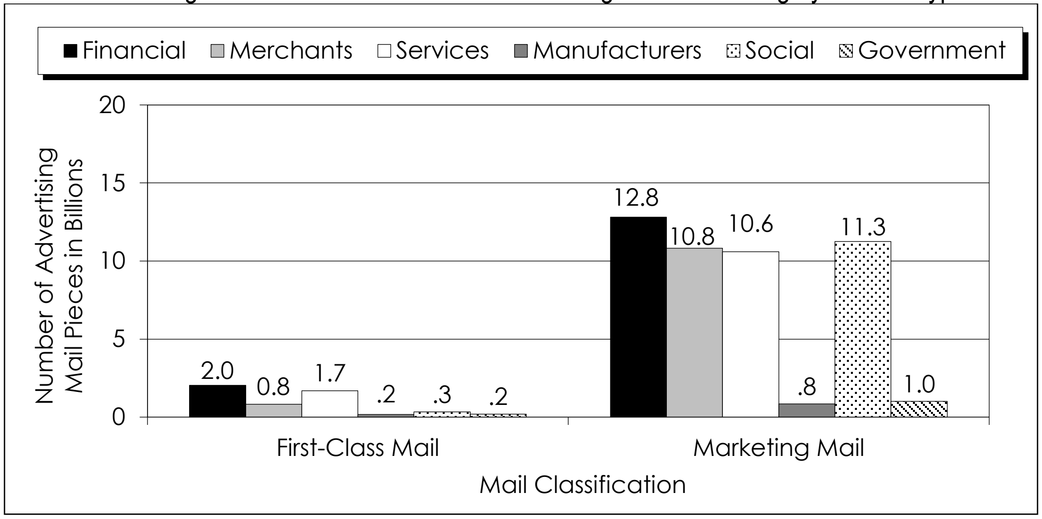
Figure 5.2 shows advertising mail volumes for key mailing industries. In prior years, merchants were consistently the largest mailers of Marketing Mail ads. However, as the rapid rise in ecommerce activity absorbed a large share of business from retailers, merchants began to make significant cuts to their advertising budgets. Lower ad spending led to a constant decline in Marketing Mail ads from merchants that started in 2007 and continued through 2023, with. volumes falling 64%, from 30 billion pieces to 10.8 billion, respectively. As a result, merchants' share of total Marketing Mail fell from 36% in 2007 to 21% in 2023, ranking third, behind the financial and service industries.

Figure 5.2 also shows that, in 2023, financial companies mailed the largest share of Marketing Mail volume (after surpassing merchants in 2021 for the first time), sending 12.8 billion pieces of advertising mail to households and 25% of total ad volume. Within the financial industry, credit card companies accounted for most of the mailings, with 34% of industry volume. The social industry (mostly non-profit and political mail) was the second largest mailer of Marketing Mail, sending 11.3 billion ad pieces and accounting for 22% of the volume.

As in case of Marketing Mail, the financial industry was also the largest mailer of First-Class Mail advertising, mailing 2.0 billion ads, representing 38% of the volume. Service companies ranked second, mailing 1.7 billion pieces of First-Class ads and 32% of the volume. Merchant ranked third, with 0.8 billion pieces and 17% of the volume.

Figure 5.2:

Advertising Volumes for First-Class and Marketing Mail Advertising by Sender Type



Source: HDS Diary Sample, FY 2023

Note: First-Class Mail advertising includes advertising-only and advertising enclosed volumes

Attitudes toward Advertising

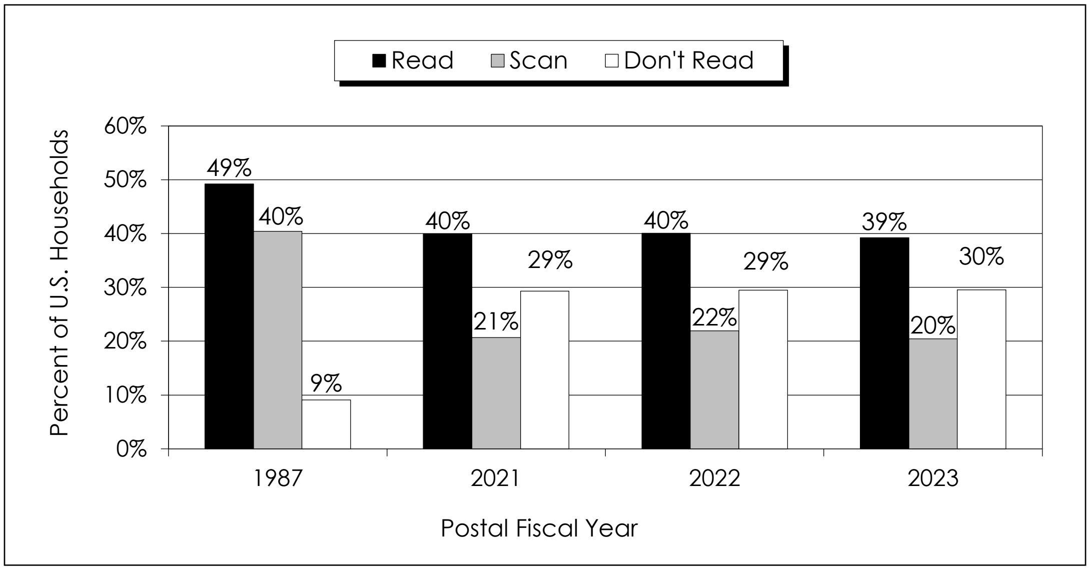
With $364 billion spent on advertising in 2023 in the US, few households would probably wish they received more.

Whether they wish to receive more or not, the household diary survey reveals that most households either read or scanned their advertising mail pieces. In Figure 5.3 we see that 39% of households reported reading their advertising mail, a slight decline from 2022 but remarkably stable over the long-term. An additional 20% of households scanned their ads, a small decline from 22% scanned in 2022 but significantly lower than the 40% reported in 1987.

Offsetting the decline in ad mail scanned was a slight increase in the share of households who did not read ads, which rose from 29% in 2022 to 30% in 2023.

Compared to 1987, Figure 5.3 shows significant declines in the shares of ads read and scanned in 2023. However, given the large increase in the volume of advertising mail received since 1987, it is clear that households read and scanned significantly more advertising mail pieces in 2023 than in 1987.

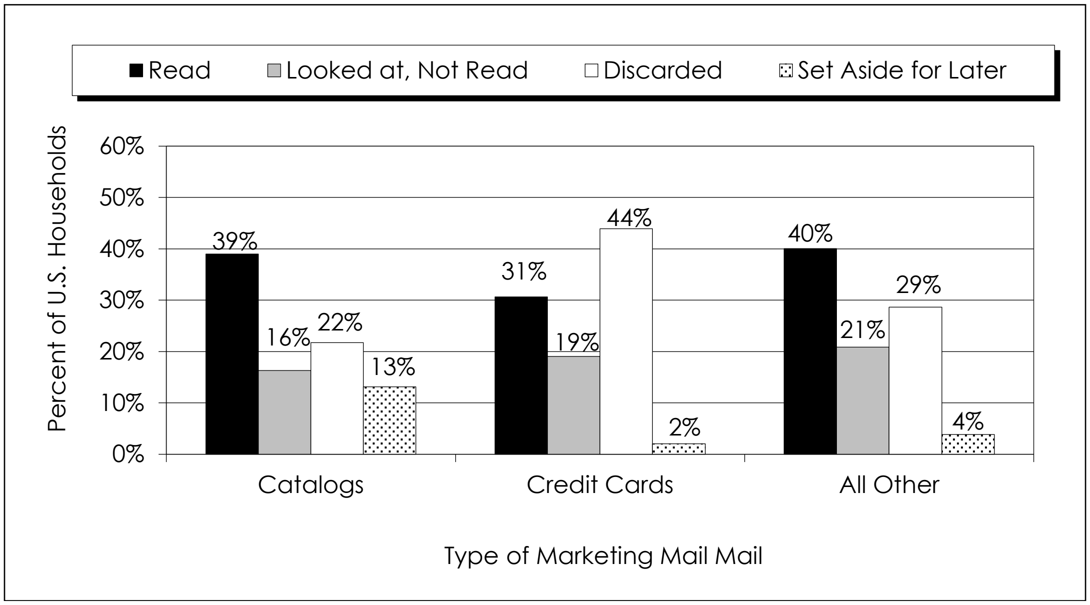
Figure 5.3:
Marketing Mail Advertising Reading Trends, FY 1987, 2021, 2022, and 2023



Source: HDS Diary Sample, FY 1987, 2021, 2022, and 2023.

As would be expected, not all advertising mail is treated equally. As Figure 5.4 illustrates, catalogs attract considerably more attention than credit card ads because they are typically more interesting to read. In 2023, 39% of households read catalogs received and only 22% discarded them. In contrast, 31% of households read credit card ads and 44% discarded them. The reading rate for all other ad types was similar to catalogs (around 40%) but the discard rate was higher, at 29%.

Figure 5.4:
Treatment of Marketing Mail Advertising by Type



Source: HDS Diary Sample, FY 2023.

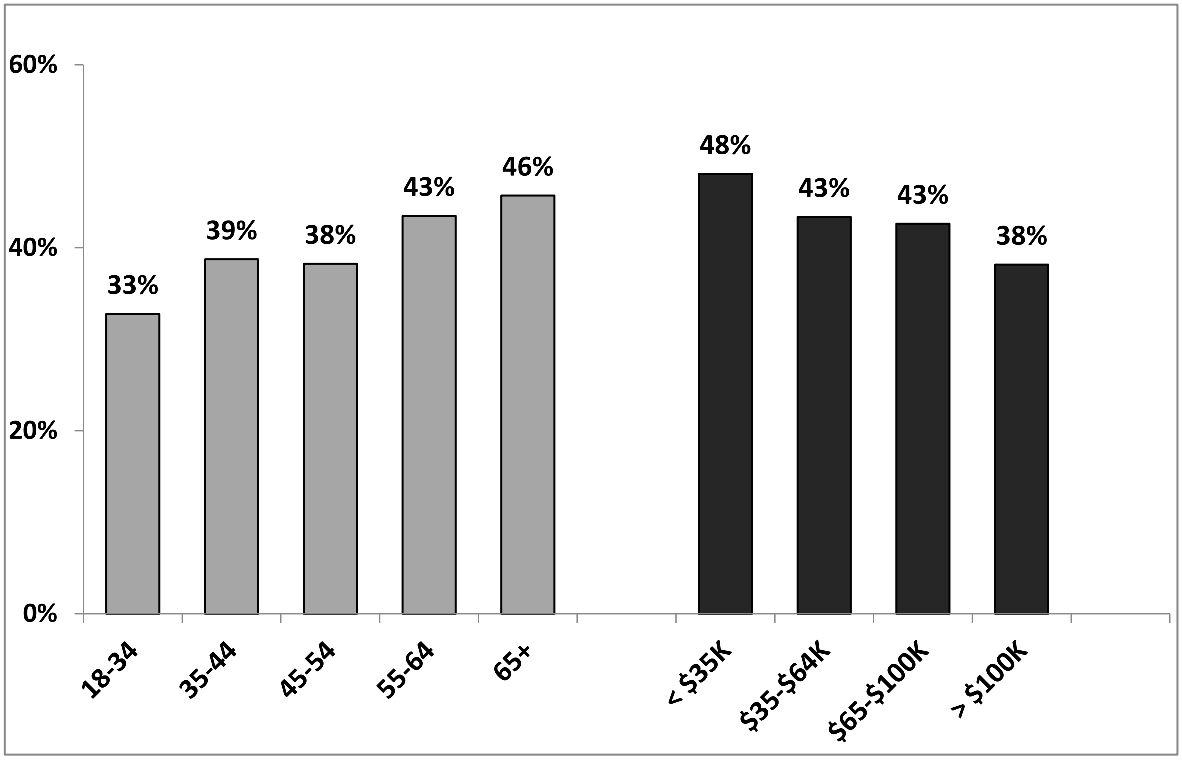
Figure 5.5 illustrates how household demographics can influence advertising reading rates.

On the right side of the chart, we see that low-income households reported reading a significantly larger share of advertising than higher income earners. This was likely related to a previous finding that lower income households tend to receive less advertising mail (see Table 5.4). With fewer ads to read, those households are likely to be less fatigued and more inclined to read more of them. According to the chart, households earning less than $35K read almost half (48%) of their ads, while those earning over $100K read only 38%.

Looking at reading rates by age cohort, we see that, just as older heads of household typically read more than younger ones, they also tend to read more advertising material. Figure 5.5 shows that heads of household over 65 read almost half (46%) of their advertising mail pieces, while the youngest cohort read only one-third.

Figure 5.5

Marketing Mail Reading Rates by Household Demographics



Source: HDS Diary Sample, FY 2023.

Effectiveness of Advertising Mail

Ultimately, advertisers send direct mail because it works—household members read and respond to it. Table 5.10 presents the intended response rates to advertising mail. Households reported that they intended to respond to 11% of First-Class and 8% of Marketing Mail ads. While intended response rates are usually higher than actual response rates, the data demonstrate that direct mail can have a significant impact on household purchasing behavior.

The table also shows that households reported that they "may" respond to an additional 10% of First-Class Mail ads and 12% of Marketing Mail ads. This is not to say that a similar mail piece may receive a higher response rate if mailed via Marketing Mail; instead, it is more probable that it is due to the difference in the mix of ads received. For example, catalogs, which typically enjoy a high response rate (linked to the high read rate shown in Figure 5.4), are routinely mailed via Marketing Mail and only rarely via First-Class Mail. Credit card ads, on the other hand, have the lowest response rate (linked to the low read rate shown in Figure 5.4) and are often mailed First-Class.

Table 5.10:
Intended Response to Advertising Mail by Class
(Percentage of Pieces)

Response	First-Class	Marketing Mail
Yes	11%	8%
Maybe	10%	12%
No	65%	72%
No Answer	14%	7%

Source: HDS Diary Sample, FY 2023.

Table 5.11 illustrates how the existence of a past business relationship between the mailer and the recipient can impact the effectiveness of advertising mail. Advertising mail sent to a recipient with whom the business had a prior relationship (e.g.: a previous or existing customer) was almost three times as likely to be read and eight times more likely to get a response than ads sent to a recipient with whom they had no prior relationship (e.g.: prospects).

Table 5.11:
Intended Response to Advertising Mail by Class
(Percentage of Pieces)

Past Business Relationship with Recipient	Reading Rate	Intended Response Rate
Existing Customers	59%	16%
Prospects	23%	2%
All Recipients	**42%**	**10%**

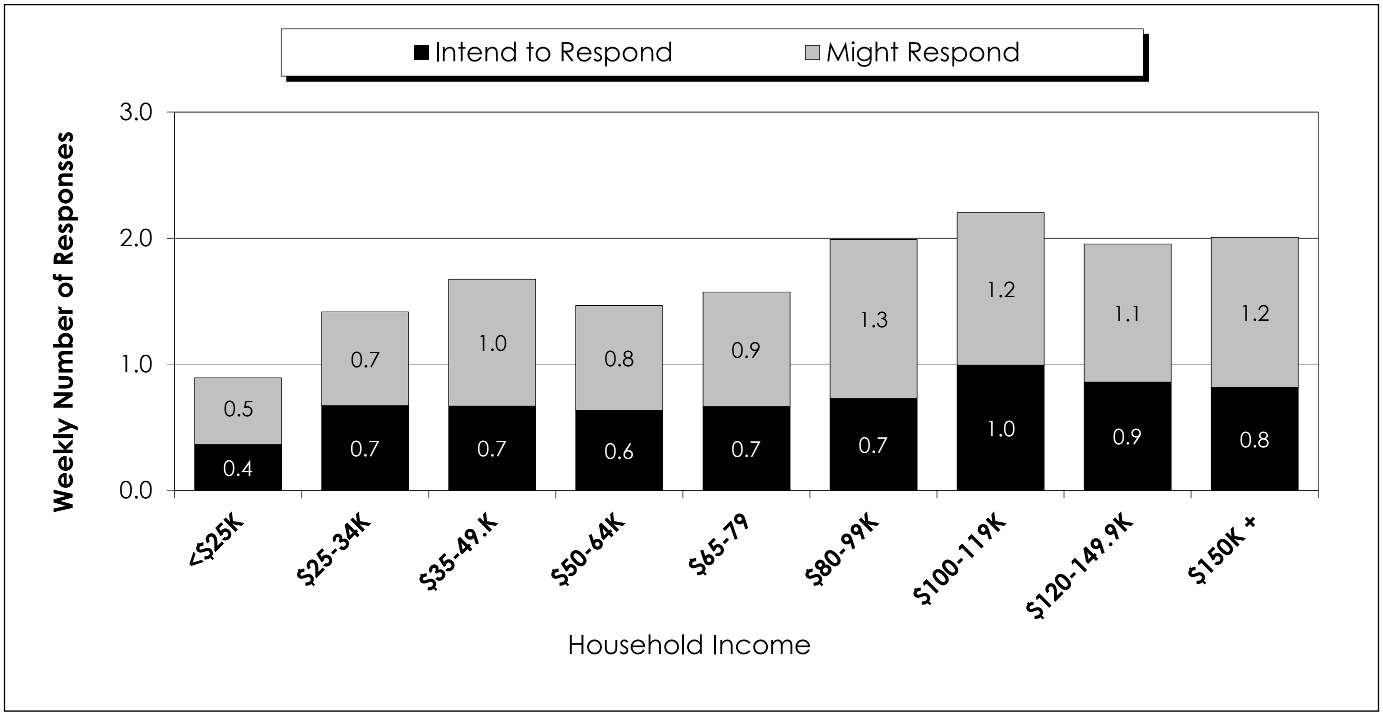
As illustrated earlier in Tables 5.4 and 5.5, higher income households received more advertising mail than low-income earners. Figure 5.6 combines the weekly volume of advertising mail received by each income group with the respective intended response rates reported in Table 5.10. The result is the average number of intended responses per week for each income level.

As the chart illustrates, households with incomes greater than $150,000 reported intending to respond to 0.8 piece of advertising per week and may respond to 1.2 pieces. Other high-income households also indicated they intended to respond to about one piece of advertising mail per week. However, starting from households earning $99,000 or less, both types of response rates started to decline.

Intended response do not always lead to actual responses, but the data presented in Table 5.10 and Figure 5.6 help explain why direct mail is a popular choice for advertisers in the US.

While there is no completely equivalent measure for intended response to Internet ads, the click-through rate (the percentage of online visitors who viewed an ad and clicked on it) is widely used as an indicator of consumers' interest in online ads and, as such, it is somewhat comparable to the read rates for mail ads. Based on recently published reports, most click-through rates for online advertising fall between 3% and 5%, depending on formats, visuals, and the type of industry. Read rates for advertising mail, on the other hand, have been significantly higher. As seen earlier in Figure 5.3 for 2023, households read 39% of Marketing Mail pieces and scanned an additional 20%.

Figure 5.6:
Weekly Number of Intended Responses by Income



Source: HDS Diary Sample, FY 2023.

Chapter 6: Periodicals

Introduction

This chapter examines the volumes and types of periodicals received by households in 2023. Periodical mail consists of newspapers, magazines, and newsletters regularly sent to households, typically as part of a subscription. This chapter analyzes only periodicals delivered by the Postal Service to households. Newspapers or magazines delivered by a local carrier or purchased at a newsstand or store are not included in Household Diary Study data. Additionally, periodicals examined here are only a portion of the total volume, as some periodicals are received by non-households, such as doctors' offices and other businesses.

The Periodicals Market

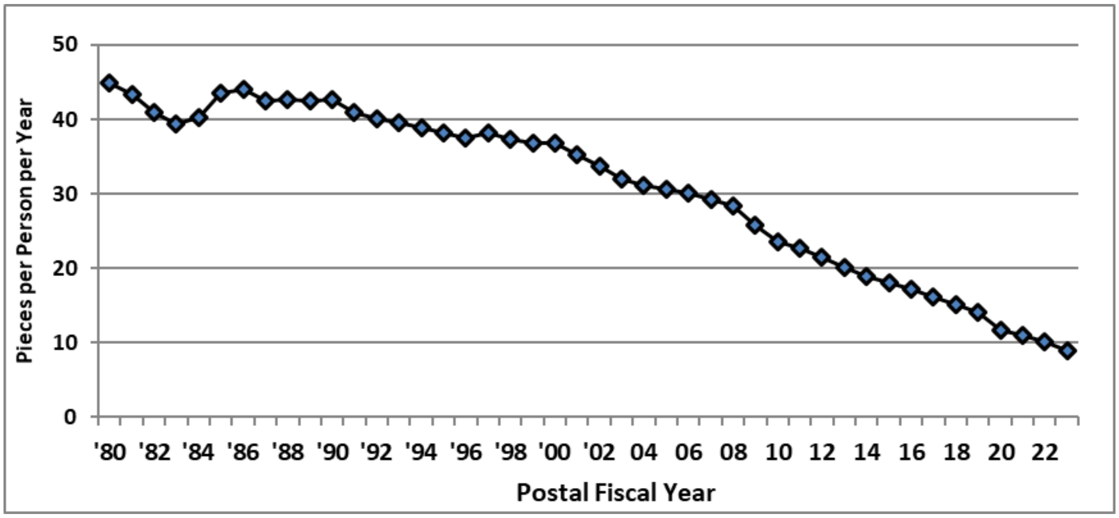
As Fig. 6.1 illustrates, the volume of periodicals has fallen out of pace with population growth since the early nineties. After reaching a peak of 42.7 pieces per person in 1990, periodicals took a downturn that by 2023 brought the annual volume down to only 8.9 pieces per person.

The emergence of the Internet catalyzed the decline in periodicals by providing a vast selection of content at lower cost and more easily accessible than hard copy publications. The recession of 2008 and, more recently, the 2020 pandemic, provided additional incentives for individuals to switch to online subscriptions, further accelerating the decline in periodicals.

Figure 6.1:

Number of Periodicals per Person, 1980-2023

(Annual Pieces per Person)



Source: U.S. Postal Service RPW Reports, U.S. Census Bureau.

Advertising's impact on Periodicals delivered by Mail

Advertising spending translates into advertising revenue, and the key determinant of periodicals' profitability is advertising revenue.

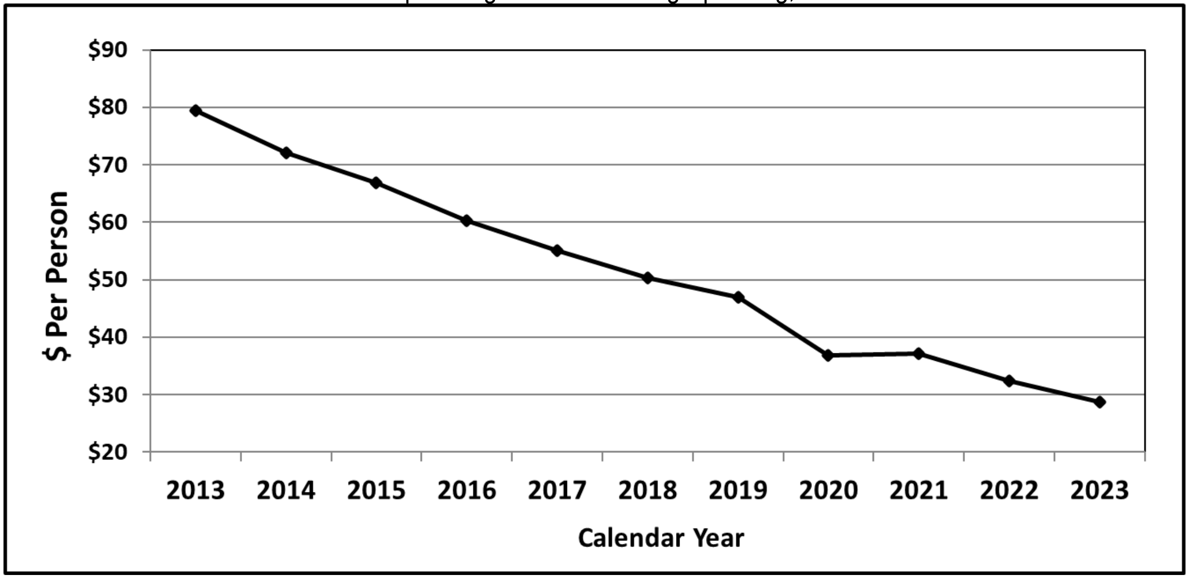
Advertising is a form of business investment and, as is the case with other investm1ents, when the economy takes a turn for the worse, advertising tends to slow. After reaching historically high levels in 2000, total ad spending as a percentage of Gross Domestic Product (GDP) quickly plummeted as the economy fell into recession in 2001. In 2002, however, growth in ad spending resumed, although at a slower pace than prior to the recession. After six years of economic expansion, spending reached new record highs in 2007. However, with the arrival of the 2008 recession, advertising spending suffered its sharpest decline in history, drastically shrinking the revenues and profitability of the magazine industry. After 2009, total advertising spending returned to growth (aided by a recovering economy) but spending on magazine advertising continued to decline, reaching its lowest point in 2023. As seen in Figure 6.2, just over the last ten years, annual magazine ad spending plummeted 64%, from $79 per person in 2013 to $29 in 2023.

Magazines represented 62% of all periodical mail delivered to households in 2023. Newspapers accounted for an additional 31%

GroupM projects that as the economy grows, so will total advertising spending. Growth in advertising spending would typically bode well for magazines. In the past, more magazines in circulation translated into higher volumes for the Postal Service since the mail continues to be their primary distribution channel. The Internet, however, has absorbed most advertising budgets, leaving fewer dollars for all other types of media, including periodicals. Furthermore, the variety of content available online, often free of charge, is likely to continue weakening periodical sales in the future and further reduce postal volumes.

Figure 6.2:

Real Per-Capita Magazine Advertising Spending, 2012-2023



Source: GroupM, U.S. Census Bureau

Household Periodicals

As shown in Table E.2 of the Executive Summary, in 2023 periodicals accounted for 3.5% of total household mail, with volumes declining 8% over the preceding two years, from 3.7 billion pieces in 2021 to 3.4 billion in 2023.

Table 6.1 provides a more detailed breakdown of periodicals, showing the number of pieces received annually, by type of publication. In 2023, households received an average of 16 magazines, unchanged from 2022 but 20% less than in 2021, accounting for 62% of periodicals. Monthly subscriptions accounted for 63% of total magazine, remaining the most popular type of publication received by households. Looking at historical trends, the number of magazines received in 2023 was 69% lower than in 1987, when they accounted for 59% of periodicals.

The number of newspapers received by mail increased 33%, from 6 pieces per household per year in 2022 (and in 2021) to 8 pieces in 2023, representing 31% of household periodicals (a significant increase from 21% in 2021). Compared to 1987, when they were still the predominant source of daily news, the number of newspapers received in 2023 fell 74%, from 10 pieces annually per household to 3 pieces per household, respectively.

While the long-term decline in newspapers mirrors reports of a similar trend in newspaper circulation, the latter was not the only reason for shrinking volumes. Current technology and the availability of alternate delivery systems have encouraged national newspapers to move a large portion of their publications online. They also continue delivering some of their papers directly to prime urban and suburban household customers. Such delivery alternatives mean that those publications no longer move through the mail.

Table 6.1:
Periodical Type by Year
(Pieces per Household per year)

Periodical Type	1987	2021	2022	2023
Magazines	**52**	**20**	**16**	**16**
Weekly	16	3	2	2
Monthly	31	13	11	10
Other	5	4	4	4
Newspapers	**31**	**6**	**6**	**8**
Daily	10	2	2	3
Weekly	16	3	3	3
Other	5	1	1	1
Newsletters	**5**	**2**	**1**	**2**
Unclassified		**1**	**1**	**.4**
Total Periodicals	**88**	**28**	**25**	**26**

Source: HDS Diary Sample, FY 1987, 2021, 2022 and 2023.
Note: Totals may not sum due to rounding.

Periodicals Mail and Household Characteristics

Income, Education, and Age

Tables 6.2 and 6.3 examine the relationship between the number of periodicals received and household income and its relationship with educational attainment.

In Table 6.2 we see that the number of periodicals received was strongly correlated to household income, with the volume increasing with income at each level of educational attainment and on average. As illustrated, household earnings over $100K received twice as many periodicals as households earning less than $35K. They also received 50% more periodicals than households earning $35K-$65K, and 20% more than households those earning $65K-$100K.

Unlike income, education appeared to have no clear relationship with periodicals received. The number of periodicals increased only when education rose from less than high school to high school-graduate. After high school, higher levels of education the number of periodicals received remained virtually unchanged. The lack of correlation at higher levels of education may be due to greater reliance on online content as information and research needs increase with higher education.

In Table 6.3 we see that age had a very strong influence on the number of periodicals received by households. As shown, heads of household younger than 35 received an average of just 0.1 periodical per week. By comparison, heads of household between the ages of 35 and 54 and those over 55 received an average of 0.3 and 0.8 pieces per week, respectively.

Heads of household over 55 received eight times more periodicals than heads under 35

Table 6.2:
Periodicals by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	Pieces than Educational Attainment of Head of Household				Average
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	0.1	0.3	0.4	0.3	**0.3**
$35 to $65	0.1	0.5	0.4	0.4	**0.4**
$65 to $100	0.3	0.5	0.6	0.5	**0.5**
Over $100	1.5	0.7	0.6	0.6	**0.6**
Average	**0.3**	**0.5**	**0.5**	**0.5**	**0.5**

Source: HDS Diary Sample, FY 2023.

Table 6.3:
Periodicals by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	**Age of Head of Household**			**Average**
	Under 35	**35 to 54**	**Over 55**	
Under $35	0.2	0.2	0.5	**0.3**
$35 to $65	0.1	0.2	0.7	**0.4**
$65 to $100	0.1	0.2	0.8	**0.5**
Over $100	0.2	0.4	1.0	**0.6**
Average	**0.1**	**0.3**	**0.8**	**0.5**

Source: HDS Diary Sample, FY 2023.

Household Size

Table 6.4 and Table 6.5 show that when households increased in size from one to two persons, the number of periodicals increased 20%. However, when household size was greater than two persons, there was no clear relationship with the number of periodicals received.

Similarly, when the number of adults in households increased from one to two adults, the number of periodicals increased 25%, but the presence of additional adults beyond two had no distinguishable impact on the number of periodicals received.

Table 6.4:
Periodicals by Size of Household
(Pieces per Household per Week)

Household Size	
One person	.5
Two	.6
Three	.4
Four	.3
Five or more	.4
Average	**.5**

Source: HDS Diary Sample, FY 2023.

Table 6.5:
Periodicals by Number of Adults in Household
(Pieces per Household per Week)

Number of Adults	
One	.4
Two	.5
Three or more	.5
Average	**.5**

Source: HDS Diary Sample, FY 2023.

Internet Access

In Table 6.6 we see that households with Internet access received 25% more periodicals in the mail than those without access. While it is true that Internet access provides several alternative ways to obtain a wide selection of content online, it is also true that households with access have significantly higher incomes and (as shown in Table 6.2) higher income households tend to receive more periodicals.

Table 6.6:
Periodicals by Type of Internet Access
(Pieces per Household per Week)

Type of Internet Access	
Broadband	.5
None	.4
Others	.6
Average	**.5**

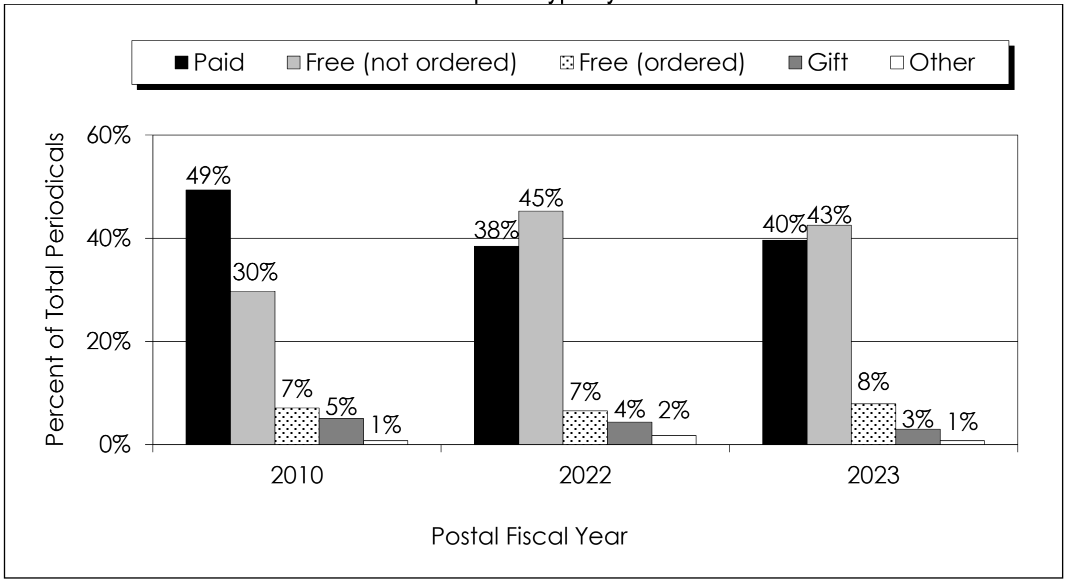
Source: HDS Diary Sample, FY 2023.

Subscription Type

Figure 6.3 shows the shares of periodicals received by households in 2023, 2022, and (for historical perspective) 2010, categorized by whether subscriptions were purchased or free. As illustrated, in 2023, households received a greater percentage of free not-ordered subscriptions (43%) than paid subscriptions (40%). Free subscriptions not-ordered typically include publication mailed as a benefit of membership in a professional, fraternal, or religious organization. An additional 8% of subscriptions were free but not ordered. When combined, total free subscriptions (whether they were ordered or not) accounted for the majority of periodicals received by mail.

In 2010, paid subscriptions accounted for almost half (49%) of periodical mail. However, as the Internet diverted many paid publications to the Internet where content is less expensive and often free, the paid share declined to 40% by 2023.

Figure 6.3:
Subscription Type by Year



Source: HDS Diary Sample, FY 2010, 2022, and 2023.
Note: Percentages do not add to 100 due to the exclusion of periodicals not classified by subscription type.

Examining the volume of periodicals by sender type, Table 6.7 shows that households received most periodicals (75%) from commercial organizations. The remining volume was received from several member organizations, which include professional affiliations, charitable, religious, and veteran organizations, educational groups, and unions. When combined, member organizations accounted for the remaining 25% of periodical mail.

Table 6.7:
Periodicals by Sender Type

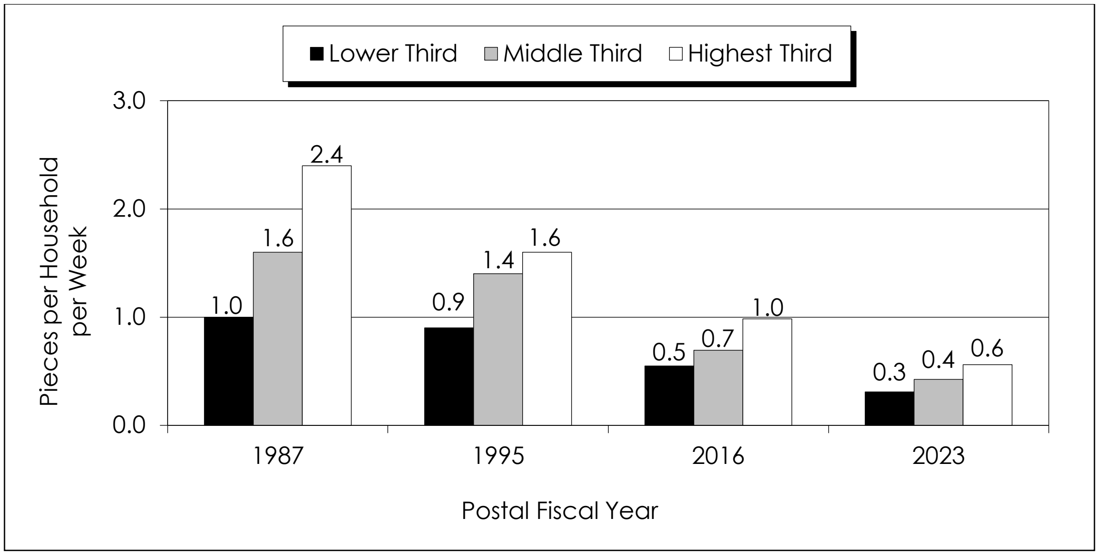
Sender Type	Annual Pieces per Household	Percent of Periodicals Received by HH
Commercial Organization	19.5	75%
Professional Organization	2.4	9%
Religious Organization	1.1	4%
Educational Organization	1.3	5%
Union	.4	1%
Charitable Organization	.3	1%
Veterans' Organization	.3	1%
Unclassified	.7	3%
Total	**26.0**	**100%**

Source: HDS Diary Sample, FY 2023.

Volume Drivers

Several factors may influence the volume of periodicals received by households. Some are demographic and others are more behavioral in nature. Given that a large share of periodicals received via paid subscriptions (see Fig 6.3), it would be reasonable to expect household income to have a strong influence on the volume of periodicals received (see Table 6.2). Figure 6.4 illustrates that, even though volumes have decreased dramatically since 1987, the relationship between periodicals and household income has remained relatively constant over time. In the chart, each time period shows a very similar positive relationship between income and periodicals, with the highest income group consistently receiving about twice as many periodicals as the lowest income group.

Figure 6.4:
Number of Periodicals Received per Week by Households by Income Group



Source: HDS Diary Sample, FY 1987, 1995, 2016, and 2023.

Chapter 7: Packages

Introduction

This chapter discusses packages sent and received by households. Packages can be mailed via the U.S. Postal Service using a variety of products and at rates that are based on weight, shape, and other relevant factors.

The Package Market

The package delivery market is an important segment of the economy. There are three major components of the package market:

- Overnight,
- Two- and three-day, and
- Ground.

The U.S. Postal Service provides services in all segments: Priority Mail Express in the overnight segment, Priority in the two- and three-day segment, and mainly Advantage Ground and Parcel Services in the ground segment.

Postal Service Package Volume

Compared to other types of mail, such as letters, bills, or advertising materials, most households do not receive packages regularly on a daily or weekly basis as they do with the mail. Consequently, the number of packages captured in the Household Diary Study survey is usually relatively small and the interpretation of the results should be conducted with this in mind.

Packages and Household Characteristics

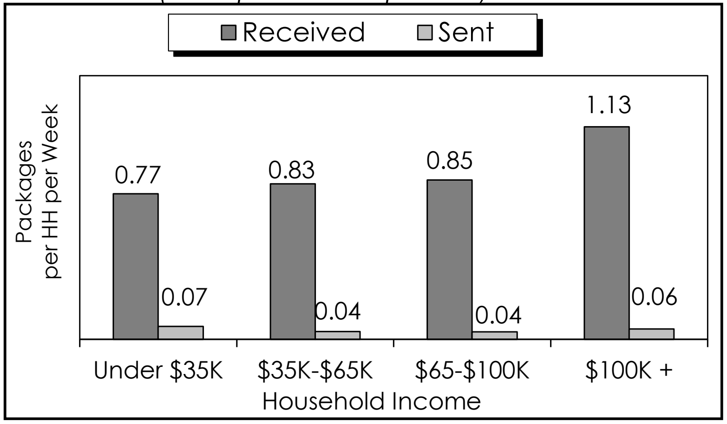
Income, Education, and Age

Survey results indicate that high-income households received consistently more packages than lower income earners. As Fig. 7.1 illustrates, households in the highest income bracket received close to 50% more packages than households earning less than $35,000. Packages sent, however, did not have a discernable relationship with income.

Figure 7.1:

Packages Sent and Received by Household Income

(Pieces per Household per Week)



Source: HDS Diary Sample, FY 2023.
Base: Packages Sent and Received by Households and Delivered by U.S. Postal Service.

The age of heads of household was another factor influencing the number of packages received, though the relationship was somewhat more complex. As shown in Table 7.1, heads of household between 35 and 54 years of age received more packages than either the younger or older counterparts. This outcome was likely influenced by the higher level of income earned by those in the 35-54 age group, as they typically earn more income than the other two groups.

In general, heads of household over 55 earn less income than 35–54-year-olds because a significant percentage of the cohort is either retired or approaching retirement. Since retirees tend to be less active buyers of good, they typically receive fewer packages. In Table 7.2 we also see that head of households over 55 sent fewer packages than younger cohorts.

Table 7.1:
Postal Service Packages Received by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	Age of Head of Household			Average
	Under 35	35 to 54	Over 55	
Under $35	0.75	0.80	0.77	**0.77**
$35 to $65	1.02	0.81	0.72	**0.83**
$65 to $100	0.61	0.93	0.91	**0.85**
Over $100	0.82	1.21	1.16	**1.13**
Average	**0.82**	**1.07**	**0.90**	**0.94**

Source: HDS Diary Sample, FY 2023.

Table 7.2:
Postal Service Packages Sent by Income and Age
(Pieces per Household per Week)

Household Income *(Thousands)*	Age of head of Household			Average
	Under 35	35 to 54	Over 55	
Under $35	0.01	0.17	0.04	**0.07**
$35 to $65	0.01	0.04	0.06	**0.04**
$65 to $100	0.08	0.04	0.03	**0.04**
Over $100	0.06	0.06	0.05	**0.06**
Average	**0.05**	**0.07**	**0.04**	**0.06**

Source: HDS Diary Sample, FY 2023.

Tables 7.3 and 7.4 illustrate the impact of education on the number of packages received and sent by households. Table 7.3 shows that, except for households with less than high school education, the weekly number of packages received per household increased as the level of education increased. For example, college graduates on average received 18% more packages than high-school graduates and 13% more than heads of household with some college education. In Table 7.4 we see that education had a stronger influence on the number of packages sent than on packages received. College graduates sent almost three times more packages than those with some high school and 60% more than high school graduates.

Table 7.3:
Postal Service Packages Received by Income and Education
(Pieces per Household per Week)

Household Income *(Thousands)*	Education of Head of Household				Average
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	1.19	0.56	0.78	0.77	**0.77**
$35 to $65	0.89	1.05	0.61	0.75	**0.83**
$65 to $100	1.06	0.65	0.88	0.91	**0.85**
Over $100	0.81	1.28	1.18	1.10	**1.13**
Average	**1.29**	**0.84**	**0.86**	**0.99**	**0.94**

Source: HDS Diary Sample, FY 2023.

Table 7.4:

Postal Service Packages Sent by Income and Education

(Pieces per Household per Week)

Household Income *(Thousands)*	Education of Head of Household				Average
	Less than High School	High School Graduate	Some College or Technical School	College Graduate	
Under $35	0.03	0.04	0.16	0.04	**0.07**
$35 to $65	0.15	0.03	0.03	0.05	**0.04**
$65 to $100	0.00	0.03	0.02	0.06	**0.04**
Over $100	0.00	0.05	0.06	0.06	**0.06**
Average	**0.03**	**0.05**	**0.07**	**0.08**	**0.06**

Source: HDS Diary Sample, FY 2023.

Household Size

Unsurprisingly, Table 7.5 shows that two-person households received 54% more packages than one-person households. It also shows that, in most cases, families with more than two members, received increasingly more packages as the family size increased. However, the number of packages sent did not show any correlation with family size.

Table 7.5:

Postal Service Packages Received and Sent by Size of Household

(Pieces per Household per Week)

Household Size	Received	Sent
One person	0.62	0.09
Two	0.97	0.06
Three	1.09	0.05
Four	1.07	0.08
Five or more	1.28	0.04
Average	**0.94**	**0.06**

Source: HDS Diary Sample, FY 2023.

In Table 7.6 we also see that households with two or more adults received significantly more packages than those with only one adult. For example, when the number of adults increased from one to two adults, packages received increased 35%. When the number of adults increased from two to three or more, the number of packages received increased an additional 16%. However, the table also shows that there was no clear relationship between packages sent and the number of adults in the household.

Table 7.6:

Postal Service Packages Received and Sent by Number of Adults in Household

(Pieces per Household per Week)

Number of Adults	Received	Sent
One	0.66	0.08
Two	1.01	0.06
Three or more	1.17	0.05
Average	**0.94**	**0.06**

Source: HDS Diary Sample, FY 2023.

Internet Access

The number of packages received by households in 2023 was strongly influenced by whether the household had access to the Internet. This was to be expected, given that the surge in eCommerce activity generated significant growth in package deliveries. As seen in Table 7.7, households with Broadband access received 96% more packages than those without access. However, the table also shows that households with access sent 40% fewer pieces than those without access. This result can be attributed in part to situations where online purchases (such as gifts) are mailed directly to the recipient by the seller.

Households with Internet access received 96% more packages than households without a connection

Relatedly, Table 7.8 shows that online-shoppers received more than twice as many packages as households who said they were not online shoppers. However, both groups sent approximately the same number of packages.

Table 7.7:

Packages Received and Sent
by Household Internet Access
(Pieces per Household per Week)

Type of Internet Access	Received	Sent
Broadband	0.96	0.06
None	0.49	0.10
Others	0.91	0.20
Average	**0.94**	**0.06**

Source: HDS Diary Sample, FY 2023.

Table 7.8:

Packages Received and Sent
by Household Online Shopping Behavior
(Pieces per Household per Week)

Shopping Online?	Received	Sent
Yes	0.98	0.06
No	0.46	0.07

Source: HDS Diary Sample, FY 2023.

Household Package Contents

As shown in Table 7.9, one-quarter (25%) of packages received by households contained clothing items. Health and medical products were the second most common content, with a 16% share of packages received.

Clothing items were also the most common content for packages sent by households, with 26% of the volume. Other common types of contents sent included electronic and household items, which are typically sent in large numbers during the holiday season.

Table 7.9:

Contents of Postal Service Sent and Received Packages

Contents	Volume (Millions of Pieces)					
	2021		2022		2023	
	Sent	Received	Sent	Received	Sent	Received
Clothing/Footwear/Shoes	154	1,806	171	1,947	116	1,581
Health/Medical/Dental/Vision products	9	806	18	1,058	16	1,020
Household/Kitchen/Lawn and garden products	60	868	29	803	32	847
Cosmetics/Beauty products/Toiletries	2	472	32	397	11	501
Toys	10	360	72	308	10	339
Book(s)	5	407	45	427	36	304
Electronic equipment	23	275	25	270	52	257
Food Products	3	333	2	356	6	250
Computer hardware, software, or accessories	17	180	10	175	11	208
Photos/Film	0	53	6	45	0	182
Music/Video	64	235	98	173	51	165
Travel products or information	2	53	0	49	7	97
Checkbooks	1	14	2	19	1	34
Other Contents	149	1,113	72	1,154	112	1,067
Total Packages	**491**	**6,503**	**576**	**6,770**	**440**	**6,432**

Table 7.9:

Contents of Postal Service Sent and Received Packages (cont.)

Contents	Percent of Pieces					
	2021		2022		2023	
	Sent	Received	Sent	Received	Sent	Received
Clothing/Footwear/Shoes	31%	28%	30%	29%	26%	25%
Health/Medical/Dental/Vision products	2%	12%	3%	16%	4%	16%
Household/Kitchen/Lawn and garden products	12%	13%	5%	12%	7%	13%
Cosmetics/Beauty products/Toiletries	0%	7%	6%	6%	3%	8%
Toys	2%	6%	12%	5%	2%	5%
Book(s)	1%	6%	8%	6%	8%	5%
Electronic equipment	5%	4%	4%	4%	12%	4%
Food Products	1%	5%	0%	5%	1%	4%
Computer hardware, software, or accessories	4%	3%	2%	3%	2%	3%
Photos/Film	0%	1%	1%	1%	0%	3%
Music/Video	13%	4%	17%	3%	11%	3%
Travel products or information	0%	1%	0%	1%	2%	2%
Checkbooks	0%	0%	0%	0%	0%	1%
Other Contents	30%	17%	12%	17%	26%	17%
Total Packages	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**

Source: HDS Diary Sample, FY 2021, 2022, and 2023.
Notes:
Contents questions are multiple response; as such, total packages do not equal the sum for each column.
Does not include contents for which no answer was given (DK/RF).
Music/Video packages include 0.1 billion pieces of CD/DVDs sent and received, reported in
First-Class Mail letters in Tables E.1, 1.5, and 1.6

Appendix A: Annual Trends

Table A8-1
First Class Mail Received by Type
Pieces in Millions
Years 2010 - 2023 (Diary Data)

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Correspondence														
Personal	4,855	4,301	4,266	3,895	3,603	3,249	3,157	3,136	3,061	2,803	2,690	2,894	2,728	2,476
Greeting Cards	3,278	3,083	3,069	2,767	2,562	2,251	2,243	2,265	2,320	1,967	1,904	2,085	1,994	1,863
Letter from Friend or Relative	837	701	693	652	592	588	493	515	436	464	499	470	427	323
Other Personal	740	517	505	477	449	410	421	356	305	372	288	339	306	290
Business/Government	9,234	10,586	10,691	10,243	10,644	10,650	10,889	10,866	10,735	10,297	10,301	10,236	10,156	9,649
Notice/ Announcement/ Business Invitation	4,297	4,403	4,748	4,518	4,905	5,116	5,051	5,049	4,767	4,546	4,492	4,309	4,259	4,498
Tax-Related (Docs, information, forms)	526	1,001	991	998	1,062	983	1,174	1,093	1,156	1,159	1,221	1,314	1,247	1,301
Insurance	2,001	2,602	2,443	2,426	2,571	2,401	2,353	2,442	2,509	2,565	2,222	2,286	2,204	1,841
Other Bus/Gov	2,410	2,580	2,509	2,301	2,106	2,150	2,310	2,282	2,302	2,026	2,366	2,328	2,445	2,010
Social	1,905	1,394	1,355	1,359	1,161	1,105	1,098	1,012	979	1,003	1,014	839	758	738
Notice/ Announcement/ Business Invitation	1,351	752	773	723	645	608	535	486	512	511	436	363	364	313
Other Social	554	642	582	635	516	497	563	526	467	492	579	475	394	425
Total	15,995	16,281	16,311	15,497	15,408	15,004	15,144	15,015	14,775	14,103	14,005	13,969	13,641	12,864
Transactions														
Bills	11,274	11,347	10,824	10,815	10,412	10,219	9,852	9,116	8,761	8,033	7,540	6,929	6,650	6,534
Financial Statements	5,418	4,655	4,744	4,285	4,514	4,204	3,994	4,051	3,931	3,741	3,871	3,577	3,445	3,477
Credit Card Statement/Bill	4,091	3,834	4,023	3,454	3,690	3,623	3,742	3,514	3,765	3,334	3,053	2,770	2,503	2,629
Request for donation	660	755	687	652	609	596	524	500	547	482	515	536	632	463
Payment/Check/Credit	1,194	1,009	993	1,055	998	1,039	895	894	740	745	807	881	725	778
Other	626	1,307	1,230	1,252	1,131	1,200	1,276	1,139	1,152	896	942	1,014	950	998
Total	23,263	22,906	22,501	21,512	21,355	20,881	20,283	19,214	18,896	17,231	16,728	15,708	14,906	14,879
Advertising (Ads Only)	6,115	5,448	5,021	4,240	3,925	3,593	3,826	3,712	3,504	3,973	3,470	3,642	3,947	3,236
CD/DVD/Video Games[1]	942	1,040	543	466	374	312	217	212	170	134	102	100	77	60
DK/RF[2]	3,748	2,598	2,828	2,426	1,768	1,610	1,811	1,641	1,613	1,559	1,423	1,281	976	1,189
Total First-Class Received	50,063	48,272	47,204	44,142	42,830	41,400	41,280	39,794	38,958	36,999	35,729	34,700	33,547	32,228

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

Note: Prior to 2011, historical data were reclassified to reflect movement of some categories between Transaction and Correspondence mail.

[1] CD/DVD/Video Games not collected as a separate category prior to 2007.

[2] Purpose of Correspondence and Transaction mail was not reported

Table A8-2
Shares of First Class Mail Received by Type
Years 2010 - 2023 (Diary Data)

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Correspondence														
Personal	10%	9%	9%	9%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%
Greeting Cards	7%	6%	7%	6%	6%	5%	5%	6%	6%	5%	5%	6%	6%	6%
Letter from Friend or Relative	2%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%
Other Personal	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%
Business/Government	18%	22%	23%	23%	25%	26%	26%	27%	28%	28%	29%	29%	30%	30%
Notice/ Announcement/ Business Invitation	9%	9%	10%	10%	11%	12%	12%	13%	12%	12%	13%	12%	13%	14%
Tax-Related (Docs, information, forms)	1%	2%	2%	2%	2%	2%	3%	3%	3%	3%	3%	4%	4%	4%
Insurance	4%	5%	5%	5%	6%	6%	6%	6%	6%	7%	6%	7%	7%	6%
Other Bus/Gov	5%	5%	5%	5%	5%	5%	6%	6%	6%	5%	7%	7%	7%	6%
Social	4%	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%	2%	2%	2%
Notice/ Announcement/ Business Invitation	3%	2%	2%	2%	2%	1%	1%	1%	1%	1%	1%	1%	1%	1%
Other Social	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	2%	1%	1%	1%
Total	32%	34%	35%	35%	36%	36%	37%	38%	38%	38%	39%	40%	41%	40%
Transactions														
Bills	23%	24%	23%	24%	24%	25%	24%	23%	22%	22%	21%	20%	20%	20%
Financial Statements	11%	10%	10%	10%	11%	10%	10%	10%	10%	10%	11%	10%	10%	11%
Credit Card Statement/Bill	8%	8%	9%	8%	9%	9%	9%	9%	10%	9%	9%	8%	7%	8%
Request for donation	1%	2%	1%	1%	1%	1%	1%	1%	1%	1%	1%	2%	2%	1%
Payment/Check/Credit	2%	2%	2%	2%	2%	3%	2%	2%	2%	2%	2%	3%	2%	2%
Other	1%	3%	3%	3%	3%	3%	3%	3%	3%	2%	3%	3%	3%	3%
Total	46%	47%	48%	49%	50%	50%	49%	48%	49%	47%	47%	45%	44%	46%
Advertising (Ads Only)	12%	11%	11%	10%	9%	9%	9%	9%	9%	11%	10%	10%	12%	10%
CD/DVD/Video Games[1]	2%	2%	1%	1%	1%	1%	1%	1%	0%	0%	0%	0%	0%	0%
DK/RF[2]	7%	5%	6%	5%	4%	4%	4%	4%	4%	4%	4%	4%	3%	4%
Total First-Class Received	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.
Note: Prior to 2011, historical data were reclassified to reflect movement of some categories between Transaction and Correspondence mail.

[1] CD/DVD/Video Games not collected as a separate category prior to 2007.

Table A8-3
First Class Mail Sent by type
Pieces in Millions
Years 2010 - 2023 (Diary Data)

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Correspondence														
Personal	4,234	3,547	3,631	3,647	3,160	3,274	2,810	2,471	2,540	2,346	2,400	2,460	2,314	1,585
Greeting Cards	2,834	2,572	2,705	2,652	2,261	2,263	2,138	1,816	1,850	1,718	1,700	1,784	1,640	1,125
Letter to Friend or Relative	798	636	667	675	561	678	461	409	468	423	575	408	493	214
Other Personal	602	340	259	320	338	332	211	246	221	206	125	267	181	247
Business/Government	1,094	1,209	1,195	1,106	1,067	1,185	1,080	963	771	816	782	784	785	660
Social	263	209	183	163	176	148	157	121	130	158	165	196	115	60
Total	5,591	4,965	5,009	4,916	4,403	4,607	4,047	3,555	3,441	3,319	3,346	3,439	3,215	2,305
Transactions														
Bill Payment[2]	5,632	5,517	5,098	4,513	4,470	4,383	3,753	3,341	3,100	2,784	2,467	2,415	1,946	1,750
Orders/Rebate request	279	359	321	247	249	224	189	190	157	143	169	134	145	112
Donations	345	355	391	330	332	288	286	260	251	247	304	272	267	236
Total	6,257	6,231	5,810	5,089	5,051	4,894	4,228	3,791	3,508	3,174	2,940	2,822	2,357	2,097
CD/DVD/Video Games[1]	700	758	380	360	318	218	174	168	130	80	63	56	51	33
DK/RF[3]	208	112	134	126	119	139	143	128	189	95	121	106	55	202
Total First-Class Sent	12,755	12,066	11,333	10,492	9,891	9,859	8,592	7,641	7,268	6,668	6,470	6,423	5,679	4,637

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

[1] CD/DVD/Video Games not collected as a separate category prior to 2007.

[2] Payments were restated 2000-2009

[3] Combination of Correspondence and Transactions (Purpose is unknown)

Table A8-4
Shares of First Class Mail Sent by type
Years 2010 - 2023 (Diary Data)

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Correspondence														
Personal	33%	29%	32%	35%	32%	33%	33%	32%	35%	35%	37%	38%	41%	34%
Greeting Cards	22%	21%	24%	25%	23%	23%	25%	24%	25%	26%	26%	28%	29%	24%
Letter to Friend or Relative	6%	5%	6%	6%	6%	7%	5%	5%	6%	6%	9%	6%	9%	5%
Other Personal	5%	3%	2%	3%	3%	3%	2%	3%	3%	3%	2%	4%	3%	5%
Business/Government	9%	10%	11%	11%	11%	12%	13%	13%	11%	12%	12%	12%	14%	14%
Social	2%	2%	2%	2%	2%	2%	2%	2%	2%	2%	3%	3%	2%	1%
Total	44%	41%	44%	47%	45%	47%	47%	47%	47%	50%	52%	54%	57%	50%
Transactions														
Bill Payment[2]	44%	46%	45%	43%	45%	44%	44%	44%	43%	42%	38%	38%	34%	38%
Orders	2%	3%	3%	2%	3%	2%	2%	2%	2%	2%	3%	2%	3%	2%
Donations	3%	3%	3%	3%	3%	3%	3%	3%	3%	4%	5%	4%	5%	5%
Total	49%	52%	51%	49%	51%	50%	49%	50%	48%	48%	45%	44%	42%	45%
CD/DVD/Video Games[1]	5%	6%	3%	3%	3%	2%	2%	2%	2%	1%	1%	1%	1%	1%
DK/RF	2%	1%	1%	1%	1%	1%	2%	2%	3%	1%	2%	2%	1%	4%
Total First-Class Sent	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.
[1] CD/DVD/Video Games not collected as a separate category prior to 2007.

[2] Payments were restated 2000-2009

Table A8-5a
Bills and Statements Received
Pieces in Millions by Sender Type
Years 2010 - 2023 (Diary Data)

Payee	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Financial														
Credit Card	4,091	3,834	4,023	3,454	3,690	3,623	3,742	3,532	3,760	3,349	3,051	2,749	2,552	2,629
Bank, S&L, Credit Union	4,216	3,703	3,517	3,474	3,531	3,264	3,015	2,960	2,821	2,596	2,537	2,323	2,188	2,171
Insurance Company[1]	2,459	1,668	1,661	1,725	1,719	1,740	1,667	1,704	1,498	1,362	1,437	1,294	1,322	1,238
Real Estate/Mortgage	376	317	313	390	374	318	397	356	342	306	382	302	273	216
Other Financial	1,578	1,543	1,654	1,331	1,401	1,426	1,353	1,280	1,253	1,273	1,282	1,139	1,112	1,306
Total Financial	12,719	11,064	11,168	10,376	10,716	10,370	10,175	9,832	9,674	8,886	8,689	7,806	7,447	7,560
Merchants														
Department Store	129	32	63	59	54	49	38	62	34	18	9	38	31	18
Publisher	300	327	290	275	202	210	158	146	163	107	119	95	66	52
Mail Order Company	139	101	88	71	77	94	65	49	34	16	10	46	22	9
Other Merchants	221	163	171	164	148	136	164	141	164	145	89	105	70	60
Total Merchants	790	623	612	569	482	488	424	399	395	287	227	284	189	138
Services														
Telephone /Cable Company	2,287	2,436	2,367	2,134	2,050	1,821	1,730	1,634	1,493	1,353	1,232	983	821	682
Utility Company	2,444	2,494	2,329	2,416	2,304	2,433	2,335	2,384	2,268	2,141	2,082	1,795	1,722	1,853
Medical and Other Professional	2,299	2,069	1,936	1,947	2,026	1,846	1,953	1,575	1,692	1,561	1,318	1,448	1,455	1,529
Other Service	393	404	402	387	352	369	376	313	372	326	403	319	259	278
Total Service	7,423	7,403	7,035	6,884	6,731	6,469	6,395	5,906	5,826	5,381	5,035	4,545	4,257	4,342
Manufacturers	59	51	26	21	47	46	27	26	18	24	28	60	27	70
Government	671	501	530	520	470	493	386	389	401	407	385	471	554	460
Social	173	159	191	159	127	136	149	100	113	87	80	76	84	57
Other/Don't Know/Refused	38	35	29	25	45	43	32	29	31	38	21	35	41	12
Total – All Industries	21,874	19,836	19,591	18,554	18,617	18,046	17,589	16,681	16,457	15,109	14,464	13,277	12,598	12,640

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

Table A8-6
Shares of Bills Paid by Method
Years 2010 - 2023 (Recruitment Data)

Method	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Mail	47%	44%	40%	37%	34%	31%	27%	30%	27%	23%	21%	19%	17%	15%
Internet using a Computer	29%	31%	31%	33%	35%	34%	34%	31%	31%	32%	30%	31%	30%	28%
Internet using Mobile Phone	0%	0%	2%	2%	3%	4%	7%	8%	10%	12%	14%	14%	17%	19%
Internet using Other Device	0%	0%	1%	2%	3%	6%	5%	3%	3%	3%	3%	3%	4%	4%
Auto Deduction from Bank	13%	13%	14%	14%	14%	14%	15%	15%	15%	17%	18%	19%	19%	19%
In Person	5%	5%	5%	5%	4%	4%	4%	4%	4%	3%	3%	2%	2%	2%
Credit Card	3%	3%	4%	4%	4%	4%	5%	5%	6%	6%	7%	8%	9%	10%
Telephone	2%	3%	3%	3%	3%	3%	3%	4%	4%	4%	3%	3%	3%	3%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Total Electronic[1]	48%	51%	55%	59%	62%	65%	69%	66%	69%	73%	76%	79%	81%	82%

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

[1] Includes bills paid by Internet, Auto Deduction from Bank Account, Credit Card, and Telephone

Table A8-7
Shares of Households using Method of Paying Bills
Years 2010 - 2023 (Recruitment Data)

Method	2000	2001	2002	2003	2004	2005	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Mail	95%	94%	95%	95%	95%	94%	84%	79%	78%	74%	71%	69%	68%	63%	59%	56%	55%	51%	50%	47%
Internet using a Computer	4%	8%	12%	16%	23%	28%	51%	54%	55%	59%	61%	61%	63%	56%	55%	58%	58%	59%	63%	60%
Internet using Mobile Phone	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	7%	11%	11%	15%	23%	22%	26%	31%	35%	37%	44%	49%
Internet using Other Device	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2%	5%	7%	12%	12%	8%	8%	8%	9%	9%	12%	11%
Auto Deduction from Bank Account	34%	35%	43%	45%	51%	54%	52%	50%	55%	55%	54%	54%	61%	52%	54%	57%	60%	62%	67%	67%
In Person	33%	29%	33%	34%	32%	31%	29%	28%	31%	27%	24%	24%	25%	22%	23%	19%	18%	14%	16%	17%
Credit Card	N/A	N/A	15%	17%	19%	22%	19%	18%	21%	22%	20%	21%	27%	23%	25%	27%	30%	32%	40%	42%
Telephone	4%	7%	9%	10%	11%	14%	13%	16%	18%	18%	16%	17%	19%	18%	18%	16%	16%	14%	15%	16%

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

Table A8-8
Type of Payments made by Mail
Pieces in Millions by Payee Type
Years 2010 - 2023 (Diary Data)

Payee	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Financial														
Credit Card	1,185	1,111	1,085	861	889	871	791	718	674	632	526	510	431	393
Bank, S&L, Credit Union	399	376	333	236	271	319	257	180	198	170	114	130	90	62
Insurance Company	486	478	407	389	410	383	343	304	291	259	252	237	161	116
Real Estate/Mortgage	228	219	168	182	157	175	129	153	98	81	113	67	37	41
Other Financial	37	26	27	36	31	35	43	26	18	21	13	24	16	28
Total Financial	2,336	2,210	2,020	1,704	1,758	1,783	1,562	1,380	1,280	1,163	1,018	967	734	640
Merchants														
Department Store	105	66	67	57	59	59	55	50	12	22	13	21	10	22
Publisher	151	157	159	179	112	104	83	72	68	77	87	47	38	21
Mail Order Company	82	52	50	52	52	47	53	47	35	12	10	19	11	8
Other Merchants	91	95	115	75	90	56	62	60	55	53	53	65	45	25
Total Merchants	429	369	391	362	313	266	253	229	170	164	162	152	103	76
Services														
Telephone /Cable Company	880	884	777	653	593	544	470	401	393	316	322	240	158	171
Utility Company	952	920	912	818	780	785	633	627	539	520	421	449	365	372
Medical and Other Professional	543	537	520	476	542	520	421	339	360	226	228	254	250	223
Other Service	203	247	212	192	189	190	171	156	163	166	140	145	153	121
Total Service	2,577	2,588	2,421	2,139	2,104	2,039	1,695	1,523	1,456	1,228	1,110	1,088	925	888
Manufacturers	7	9	10	6	8	9	12	10	5	8	8	10	3	5
Government	238	140	141	122	136	127	116	110	92	100	80	111	94	69
Social	0	106	71	97	63	65	54	60	55	68	39	46	39	34
Other/Don't Know/Refused	46	96	45	84	90	93	60	28	43	53	49	59	47	37
Total – All Industries	5,632	5,517	5,098	4,513	4,470	4,383	3,753	3,341	3,100	2,784	2,467	2,432	1,946	1,750

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.
Note: Payments were restated 2000-2009.

Table A8-9
Type of Payments made by Mail
Percent of Bill Payments by Payee Type
Years 2010 - 2023 (Diary Data)

Payee	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Financial														
Credit Card	21%	20%	21%	19%	20%	20%	21%	21%	22%	23%	21%	21%	22%	22%
Bank, S&L, Credit Union	7%	7%	7%	5%	6%	7%	7%	5%	6%	6%	5%	5%	5%	4%
Insurance Company	9%	9%	8%	9%	9%	9%	9%	9%	9%	9%	10%	10%	8%	7%
Real Estate/Mortgage	4%	4%	3%	4%	4%	4%	3%	5%	3%	3%	5%	3%	2%	2%
Other Financial	1%	0%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	2%
Total Financial	41%	40%	40%	38%	39%	41%	42%	41%	41%	42%	41%	40%	38%	37%
Merchants														
Department Store	2%	1%	1%	1%	1%	1%	1%	2%	0%	1%	1%	1%	1%	1%
Publisher	3%	3%	3%	4%	3%	2%	2%	2%	2%	3%	4%	2%	2%	1%
Mail Order Company	1%	1%	1%	1%	1%	1%	1%	1%	1%	0%	0%	1%	1%	0%
Other Merchants	2%	2%	2%	2%	2%	1%	2%	2%	2%	2%	2%	3%	2%	1%
Total Merchants	8%	7%	8%	8%	7%	6%	7%	7%	5%	6%	7%	6%	5%	4%
Services														
Telephone /Cable Company	16%	16%	15%	14%	13%	12%	13%	12%	13%	11%	13%	10%	8%	10%
Utility Company	17%	17%	18%	18%	17%	18%	17%	19%	17%	19%	17%	18%	19%	21%
Medical and Other Professional	10%	10%	10%	11%	12%	12%	11%	10%	12%	8%	9%	10%	13%	13%
Other Service	4%	4%	4%	4%	4%	4%	5%	5%	5%	6%	6%	6%	8%	7%
Total Service	46%	47%	47%	47%	47%	47%	45%	46%	47%	44%	45%	45%	48%	51%
Manufacturers	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Government	4%	3%	3%	3%	3%	3%	3%	3%	3%	4%	3%	5%	5%	4%
Social	0%	2%	1%	2%	1%	1%	1%	2%	2%	2%	2%	2%	2%	2%
Other/Don't Know/Refused	1%	2%	1%	2%	2%	2%	2%	1%	1%	2%	2%	2%	2%	2%
Total – All Industries	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

Table A8-10
Share of Households by Internet Access type
Years 2010 - 2023 (Recruitment Data)

Type of Access	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
None	21%	18%	17%	13%	13%	11%	10%	10%	8%	7%	6%	5%	2%	3%
Total Internet Access	78%	81%	83%	86%	87%	88%	89%	89%	90%	93%	94%	95%	98%	97%
Dial-up	6%	3%	3%	2%	1%	1%	1%	1%	1%	0%	1%	0%	0%	0%
Cable Modem	34%	38%	41%	44%	85%	86%	88%	88%	89%	91%	92%	93%	96%	93%
Other Broadband	10%	14%	14%	17%										
DSL	25%	23%	23%	20%										
Other/ Did Not Specify Connection	4%	3%	2%	3%	1%	1%	1%	1%	1%	1%	2%	1%	2%	5%
DK/RF/Missing	0%	1%	0%	1%	1%	1%	1%	1%	1%	0%	0%	0%	0%	0%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

Note: All types of Broadband services are combined beginning in 2014

Table A8-11
Number of Purchases Made over the Internet over the past month
Percent of Households
Years 2010 - 2023 (Recruitment Data)

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
All Households														
None	52%	46%	45%	40%	37%	36%	34%	29%	24%	20%	17%	14%	10%	11%
1	8%	8%	9%	9%	8%	8%	8%	6%	6%	6%	5%	3%	3%	3%
2	10%	12%	10%	11%	11%	11%	11%	10%	10%	10%	8%	6%	7%	7%
3-5	17%	20%	20%	22%	25%	25%	24%	30%	27%	27%	27%	24%	26%	29%
6-10	8%	9%	10%	11%	12%	14%	14%	16%	20%	20%	23%	25%	27%	27%
More than 10	5%	5%	6%	7%	7%	8%	9%	9%	14%	16%	21%	27%	27%	22%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Households that Made 1+ purchases														
1	16%	15%	16%	15%	13%	12%	12%	9%	8%	8%	6%	3%	4%	4%
2	21%	22%	18%	18%	18%	17%	16%	14%	13%	12%	9%	7%	8%	8%
3-5	36%	37%	36%	37%	39%	38%	37%	42%	35%	34%	32%	28%	29%	33%
6-10	16%	16%	18%	18%	19%	21%	21%	23%	26%	25%	27%	29%	30%	31%
More than 10	10%	10%	12%	12%	11%	12%	14%	13%	18%	20%	26%	32%	30%	25%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

Table A8-12
First Class and Marketing Mail Advertising
Pieces in Millions
Years 2010 - 2023 (Diary Data)

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
First-Class Advertising	12,793	12,710	11,437	9,709	8,987	8,370	8,292	7,795	7,594	7,475	6,284	6,642	6,748	5,344
Advertising Only	6,115	5,448	5,021	4,240	3,925	3,593	3,826	3,712	3,504	3,973	3,470	3,642	3,947	3,236
Secondary Advertising	6,678	7,262	6,416	5,469	5,062	4,776	4,466	4,083	4,090	3,503	2,813	3,000	2,801	2,108
Marketing Ads[1]	72,934	77,747	73,874	74,365	71,788	71,631	71,754	69,075	67,681	65,686	54,867	58,528	58,571	51,942
Total Ads	85,727	90,457	85,311	84,074	80,775	80,001	80,046	76,871	75,275	73,161	61,150	65,170	65,319	57,286
First-Class Ads Share of Total Ads	15%	14%	13%	12%	11%	10%	10%	10%	10%	10%	10%	10%	10%	9%

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

Table A8-13A
First Class and Marketing Mail Advertising By Sender Type
Pieces in Millions
Years 2010 - 2023 (Diary Data)

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
First-Class Ads [1]														
Financial	4,632	4,651	4,231	3,366	3,394	3,378	3,214	3,146	3,174	3,086	2,675	2,827	3,028	2,043
Merchants	2,422	2,172	2,186	1,832	1,817	1,594	1,517	1,385	1,391	1,403	1,142	1,093	1,117	833
Services	4,472	4,767	4,068	3,496	2,931	2,668	2,712	2,579	2,271	2,196	1,723	1,806	1,896	1,692
Manufacturers	262	212	165	190	148	153	202	174	115	178	142	228	164	175
Government	296	289	219	264	210	138	169	145	161	157	195	210	168	186
Social	611	587	549	538	458	421	438	342	450	434	373	384	335	331
Other	98	31	20	24	28	17	41	25	32	22	34	93	40	84
Total	12,793	12,710	11,437	9,709	8,987	8,370	8,292	7,795	7,594	7,475	6,284	6,642	6,748	5,344
Standard Ads														
Financial	14,809	18,472	15,404	15,775	15,437	15,687	17,177	16,567	16,269	16,658	12,651	14,174	15,742	12,813
Merchants	24,990	25,427	24,241	23,092	23,281	22,034	21,687	20,819	19,701	17,137	13,672	13,443	12,459	10,821
Services	10,798	11,324	10,813	11,641	11,365	11,874	11,902	11,391	11,107	11,011	8,899	10,035	10,178	10,587
Manufacturers	1,560	1,638	1,579	1,642	971	1,250	1,333	1,260	1,032	1,173	858	1,263	1,036	840
Government	1,046	976	1,068	992	892	921	951	908	1,018	1,089	1,428	1,336	1,277	1,005
Social	12,372	12,739	13,503	13,984	12,799	12,689	11,988	11,665	12,232	12,534	11,980	12,797	12,723	11,257
From Multiple Organizations	6,484	6,678	6,704	6,591	6,590	6,815	6,183	6,100	5,839	5,638	4,997	4,919	4,842	4,144
Other	882	494	563	647	453	361	534	366	483	447	383	561	313	474
Total	72,941	77,747	73,874	74,365	71,788	71,631	71,754	69,075	67,681	65,686	54,867	58,528	58,571	51,942
Total Ads														
Financial	19,441	23,123	19,634	19,141	18,831	19,065	20,391	19,713	19,442	19,743	15,326	17,001	18,769	14,857
Merchants	27,411	27,599	26,427	24,924	25,098	23,628	23,203	22,204	21,093	18,539	14,814	14,536	13,577	11,653
Services	15,270	16,092	14,881	15,137	14,296	14,542	14,613	13,970	13,378	13,208	10,622	11,841	12,074	12,279
Manufacturers	1,822	1,850	1,744	1,832	1,119	1,403	1,535	1,434	1,146	1,351	1,000	1,492	1,200	1,015
Government	1,342	1,265	1,287	1,257	1,102	1,059	1,120	1,053	1,179	1,245	1,623	1,547	1,445	1,191
Social	12,983	13,325	14,051	14,522	13,258	13,110	12,426	12,007	12,682	12,968	12,353	13,181	13,057	11,588
From Multiple Organizations	6,484	6,678	6,704	6,591	6,590	6,815	6,183	6,100	5,839	5,638	4,997	4,919	4,842	4,144
Other	980	525	583	671	481	378	574	391	514	470	416	654	353	558
Total	85,734	90,457	85,311	84,074	80,775	80,001	80,046	76,871	75,275	73,161	61,150	65,170	65,319	57,286

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

[1] Includes Secondary Advertising

Table A8-13A2
First Class Advertising ONLY (no secondary) and Marketing Mail By Sender Type
Pieces in Millions
Years 2010 - 2023 (Diary Data)

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
First-Class Ads[1]														
Financial	1,655	1,517	1,411	1,160	1,121	1,163	1,195	1,311	1,222	1,534	1,394	1,555	1,713	1,165
Merchants	2,028	1,777	1,703	1,411	1,493	1,222	1,153	1,063	1,099	1,148	957	902	923	697
Services	1,966	1,601	1,503	1,204	988	951	1,078	1,068	915	996	825	759	978	993
Manufacturers	245	183	133	150	123	106	183	141	95	149	128	212	138	158
Government	172	138	64	150	56	26	51	36	39	33	55	46	66	52
Social	0	215	191	157	134	119	150	86	129	108	87	119	100	116
Other	48	17	15	8	9	7	17	7	4	5	23	48	29	55
Total	6,115	5,448	5,021	4,240	3,925	3,593	3,826	3,712	3,504	3,973	3,470	3,642	3,947	3,236
Standard Ads														
Financial	14,809	18,472	15,404	15,775	15,437	15,687	17,177	16,567	16,269	16,658	12,651	14,174	15,742	12,813
Merchants	24,990	25,427	24,241	23,092	23,281	22,034	21,687	20,819	19,701	17,137	13,672	13,443	12,459	10,821
Services	10,798	11,324	10,813	11,641	11,365	11,874	11,902	11,391	11,107	11,011	8,899	10,035	10,178	10,587
Manufacturers	1,560	1,638	1,579	1,642	971	1,250	1,333	1,260	1,032	1,173	858	1,263	1,036	840
Government	1,046	976	1,068	992	892	921	951	908	1,018	1,089	1,428	1,336	1,277	1,005
Social	12,372	12,739	13,503	13,984	12,799	12,689	11,988	11,665	12,232	12,534	11,980	12,797	12,723	11,257
From Multiple Organizations	6,484	6,678	6,704	6,591	6,590	6,815	6,183	6,100	5,839	5,638	4,997	4,919	4,842	4,144
Other	882	494	563	647	453	361	534	366	483	447	383	561	313	474
Total	72,941	77,747	73,874	74,365	71,788	71,631	71,754	69,075	67,681	65,686	54,867	58,528	58,571	51,942
Total Ads														
Financial	16,464	19,989	16,815	16,935	16,558	16,849	18,371	17,877	17,491	18,192	14,045	15,730	17,455	13,978
Merchants	27,018	27,204	25,944	24,503	24,774	23,256	22,840	21,883	20,801	18,285	14,629	14,345	13,382	11,518
Services	12,765	12,925	12,316	12,845	12,353	12,825	12,980	12,459	12,022	12,007	9,724	10,795	11,157	11,580
Manufacturers	1,805	1,821	1,713	1,792	1,094	1,356	1,516	1,401	1,127	1,321	986	1,475	1,174	998
Government	1,219	1,114	1,132	1,143	948	948	1,002	944	1,057	1,122	1,483	1,382	1,343	1,057
Social	12,372	12,954	13,693	14,140	12,933	12,807	12,138	11,752	12,361	12,642	12,066	12,916	12,822	11,373
From Multiple Organizations	6,484	6,678	6,704	6,591	6,590	6,815	6,183	6,100	5,839	5,638	4,997	4,919	4,842	4,144
Other	930	511	578	655	462	368	550	372	487	452	406	609	342	529
Total	79,056	83,195	78,895	78,605	75,713	75,225	75,580	72,788	71,185	69,659	58,337	62,170	62,518	55,178

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

[1] Excludes Secondary Advertising

Table A8-14
First Class and Marketing Mail Advertising By Sender Type
Percent of Pieces
Years 2010 - 2023 (Diary Data)

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
First-Class Ads[1]														
Financial	36%	37%	37%	35%	38%	40%	39%	40%	42%	41%	43%	43%	45%	38%
Merchants	19%	17%	19%	19%	20%	19%	18%	18%	18%	19%	18%	16%	17%	16%
Services	35%	38%	36%	36%	33%	32%	33%	33%	30%	29%	27%	27%	28%	32%
Manufacturers	2%	2%	1%	2%	2%	2%	2%	2%	2%	2%	2%	3%	2%	3%
Government	2%	2%	2%	3%	2%	2%	2%	2%	2%	2%	3%	3%	2%	3%
Social	5%	5%	5%	6%	5%	5%	5%	4%	6%	6%	6%	6%	5%	6%
Other	1%	0%	0%	0%	0%	0%	0%	0%	0%	0%	1%	1%	1%	2%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Standard Ads														
Financial	20%	24%	21%	21%	22%	22%	24%	24%	24%	25%	23%	24%	27%	25%
Merchants	34%	33%	33%	31%	32%	31%	30%	30%	29%	26%	25%	23%	21%	21%
Services	15%	15%	15%	16%	16%	17%	17%	16%	16%	17%	16%	17%	17%	20%
Manufacturers	2%	2%	2%	2%	1%	2%	2%	2%	2%	2%	2%	2%	2%	2%
Government	1%	1%	1%	1%	1%	1%	1%	1%	2%	2%	3%	2%	2%	2%
Social	17%	16%	18%	19%	18%	18%	17%	17%	18%	19%	22%	22%	22%	22%
From Multiple Organizations	9%	9%	9%	9%	9%	10%	9%	9%	9%	9%	9%	8%	8%	8%
Other	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%	1%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Total Ads														
Financial	23%	26%	23%	23%	23%	24%	25%	26%	26%	27%	25%	26%	29%	26%
Merchants	32%	31%	31%	30%	31%	30%	29%	29%	28%	25%	24%	22%	21%	20%
Services	18%	18%	17%	18%	18%	18%	18%	18%	18%	18%	17%	18%	18%	21%
Manufacturers	2%	2%	2%	2%	1%	2%	2%	2%	2%	2%	2%	2%	2%	2%
Government	2%	1%	2%	1%	1%	1%	1%	1%	2%	2%	3%	2%	2%	2%
Social	15%	15%	16%	17%	16%	16%	16%	16%	17%	18%	20%	20%	20%	20%
From Multiple Organizations	8%	7%	8%	8%	8%	9%	8%	8%	8%	8%	8%	8%	7%	7%
Other	1%	1%	1%	1%	1%	0%	1%	1%	1%	1%	1%	1%	1%	1%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

[1] Includes Secondary Advertising

Table A8-15
Treatment Of Marketing Mail Advertising Material By Household Income
Percent of Households
Years 2010 - 2023 (Recruitment Data)

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Usually Read - Total	20%	22%	22%	20%	20%	19%	18%	16%	16%	14%	15%	14%	12%	41%
Under $25K	44%	44%	42%	40%	42%	41%	41%	37%	32%	30%	24%	26%	26%	5%
$25 - $ 49.9	26%	22%	23%	24%	26%	26%	25%	31%	30%	31%	33%	31%	28%	18%
$50 - $64.9	11%	10%	12%	13%	10%	12%	10%	10%	11%	11%	11%	11%	10%	11%
$65 +	19%	24%	24%	23%	22%	21%	24%	22%	27%	27%	32%	32%	37%	66%
Usually Scan - Total	27%	26%	25%	21%	21%	22%	22%	22%	24%	22%	23%	22%	17%	22%
Under 25K	26%	24%	22%	23%	22%	25%	23%	21%	18%	17%	14%	14%	11%	4%
$25 - $ 49.9	26%	21%	21%	24%	28%	23%	23%	22%	26%	24%	24%	25%	23%	14%
$50 - $64.9	14%	12%	13%	13%	14%	12%	13%	13%	11%	14%	11%	11%	11%	12%
$65 +	35%	43%	44%	40%	37%	40%	41%	44%	45%	44%	51%	50%	55%	70%
Read Some - Total	35%	32%	34%	36%	37%	36%	35%	36%	34%	37%	36%	35%	38%	32%
Under 25K	24%	22%	22%	21%	17%	19%	18%	18%	14%	15%	14%	11%	12%	5%
$25 - $ 49.9	23%	21%	22%	23%	25%	23%	23%	22%	25%	22%	21%	22%	21%	12%
$50 - $64.9	14%	12%	14%	13%	14%	13%	13%	12%	13%	14%	14%	12%	12%	12%
$65 +	40%	46%	42%	43%	45%	45%	46%	48%	49%	49%	51%	55%	55%	71%
Usually Don't Read - Total	18%	21%	20%	23%	21%	23%	25%	25%	27%	27%	26%	29%	33%	4%
Under 25K	28%	25%	29%	25%	19%	21%	18%	19%	15%	14%	12%	12%	11%	9%
$25 - $ 49.9	20%	22%	19%	19%	20%	23%	20%	21%	23%	19%	19%	17%	19%	17%
$50 - $64.9	13%	13%	12%	10%	12%	12%	14%	10%	12%	13%	9%	11%	10%	7%
$65 +	39%	41%	41%	46%	48%	44%	49%	51%	49%	54%	60%	61%	60%	67%

Note: Beginning in 2010 data was restated due to changes in weighing methodologies.

Appendix B: Methodology

Study Design and Methodology

The U.S. Postal Service Household Diary Study (HDS), conducted by NuStats on behalf of the Finance and Strategy group, is a continuously fielded study that measures household mail volumes, mail usage, and attitudes about the mail and advertising. The HDS uses a two-stage survey design: Stage 1 is an online and interviewer-mediated household recruitment interview. Stage 2 is a self-completion mail diary. The diary can be completed on paper or online. The HDS uses a multi-mode approach to minimize response bias, to improve data accuracy through efficient data checking and household re-contacts, and to provide immediate telephone assistance to participants during their diary week.

Household Recruitment Interview
The **household recruitment interview** collects information on head of households' demographics, recall of mail sent and received, adoption and use of communications technologies, bill payment behavior, and attitudes towards advertising.

Mail Diary
The **mail diary** covers a seven-day period (Monday to Sunday) and collects information on the number of mail pieces received and sent, industry source, mail characteristics, and attitudes regarding mail received.

Sample Design

This section describes the household selection process for participation in the HDS. A sample is the representative subset of the survey population used to gain information about the entire population. The population of inference for the HDS is all U.S. households. The probability design ensures each household has an equal chance of selection.

The sample design allows projections of results to all U.S. households. The Postal Service provided an address sample that NuStats matched for known telephone listings. Invitation letters are mailed to all sampled households with information about how to participate. Generally, the study was conducted using telephone and online sampling for household selection and screening, followed by diaries and/or instructional materials mailed to eligible households and completed by each household unit either by paper, or by internet. Households without telephones were contacted via U.S. Mail. The sample design involves a systematic sample stratified by strata (or urban/rural location) and Census regions, ensuring even coverage across the United States.

A master national sample was specified and drawn by in-house sampling statisticians. The Postal Service drew the household probability sample from the national address database following NuStats specifications. The master list, sorted by ZIP code, was used to draw a systematic stratified sample, which was then tagged with variables indicating each housing unit's geographic location in terms of Census region and stratum.

Sample was drawn for each of the four quarters based on known proportions of households within a Census region and urban or rural location. Census regions are defined by state. Urban and rural location is defined by county and metropolitan status as defined by the U.S. Census Bureau. The strata are defined by county as follows:

- **Stratum 1**: Counties that are part of the 30 largest metropolitan areas in the United States, as defined by the 2020 US Census.
- **Stratum 2:** Counties that are part of metropolitan areas but are not in Stratum 1.
- **Stratum 3**: Counties that are not part of a metropolitan area.

Quarterly sample frames were then derived based on the amount of sample needed for each quarter, and sample was allocated to region and strata cells based on known proportions as indicated by Census 2020 count of households.

The sample was continuously "fielded" throughout all 52 weeks of the year. Sample was released in a manner designed to recruit equal sample sizes for each diary week. For this year, NuStats encountered challenges in the recruitment and diary completion of recruited households, resulting in only 4,009 completes vs. the target of 5,200 households. Table B.1 shows the distribution of recruited and completed households.

Table B.1:
Sample by Postal Quarter

Quarter	Required Sample	Recruited Households	Completed Households
Quarter 1	1,300	1,444	572
Quarter 2	1,300	2,443	1,267
Quarter 3	1,300	2,477	1,367
Quarter 4	1,300	1,776	803
Total	**5,200**	**8,140**	**4,009**

Data Collection Method

The study uses a two-stage design in which households are recruited to participate in the diary study by choosing to use a telephone interview or an online survey (Stage 1) and recruited households complete a seven-day diary of mail received and sent (Stage 2) either by paper or online.

Stage 1: Household Recruitment Interview

The main function of the household recruitment interview is to recruit households to participate in the diary study. In addition, the interview collects information on household and person demographics, recall of mail sent and received, adoption and use of communication technologies, bill payment behavior, and attitudes towards advertising.

Households completed the recruitment interview via computer-assisted telephone interviewing (CATI) technology or by using an online survey. The FY 2023 household interview consisted of 7,731 online surveys and 409 phone interviews completed with an adult member (age 18 or older) in the household. Table B.2 below shows the distribution of recruited households by recruitment type.

Table B.2:
Sample by Recruitment Type

Recruitment Interview	Completed	Sample Percent
Phone	409	5.0%
Web	7,731	95.0%
Total	**8,140**	**100%**

These respondents represented a cross-section of U.S. households by geography. The household interview contained 130 data items and took an average of 24.04 minutes to administer. The flow of the interview included the following elements:

- **Introduction.** Each interview began with an introduction and purpose of the interview. The interviewer also verified the respondent's address.
- **Technology adoption and use.** Questions were asked about ownership and use of personal computers, Internet, and other electronic communication.
- **Mail volume recall.** The respondent was asked to summarize how many bills, statements, and packages all members of the household have sent in a particular time period.
- **Bills and Statements.** Volumes, methods, and timing were explored in depth.
- **Bill payments.** Bill payment volumes, methods, and timing were explored in depth.
- **Advertising.** Descriptions of advertising received by the household as well as attitudes about the advertising, and orders placed because of it, were elicited.
- **Periodicals.** A summary of magazine and newspaper volumes received by the household were collected.
- **Use of postal services.** The use of post offices, post office boxes, and private mailing services was explored.
- **Online shopping.** Respondents were asked about their online shopping habits, including questions about shipping methods.
- **Household and person demographics.** Demographic items included gender, age, marital status, employment status, educational attainment, race/ethnicity, household income, household wage earners, home ownership, residence tenure, and dwelling type.

The completion rate for the FY 2023 study (defined as the proportion of respondents who completed the diary portion relative to all recruited respondents) was 49.3 percent compared to 57.5 percent in FY 2022. Most recruitment refusals took place prior to hearing who NuStats was and why the firm was calling. Refusal households that were later re-contacted cited time constraints and privacy concerns as reasons for not participating.

Stage 2: Mail Diary Package

Recruited households were mailed the website for the online diary, instructions, and a toll-free "help" telephone number. If they chose the diary on paper, they were sent seven mail diaries.

The diary package contains a Certificate of Appreciation, Instruction Booklet, and a photo-based "Quick Start" sheet. The Instruction Booklet provided information about the study, answers to frequently asked questions, instructions for filling out the diary, guidelines for sorting mail, and examples of mail markings. The paper diary instrument was composed of two parts:

- **The Question sheets.** The Question sheets are color-coded by mail classification (First-Class Mail received, First-Class Mail sent, Marketing Mail, Nonprofit, etc.). Information collected about each mail classification included: type of mail piece (i.e., envelope, postcard, catalog, etc.), mail classification, mail type, sender type, information about advertising enclosed, and receiver reaction or responses to the mail piece.
- **Seven answer booklets,** each specific to a day of the week. Each booklet was arranged by mail classification and color-coded to correspond to the question sheets.

The online diary guided Respondents through the questions and easily recorded each day's mail all in one website. Respondents participating by mail were instructed to enclose pertinent information from the envelopes of each mail piece received to enable NuStats editors to verify or clarify quantity and classes of mail recorded in the diaries. Respondents participating online were instructed to upload photographs of the envelopes. NuStats used a three-stage editing process to check the accuracy of information recorded on paper diaries. First, returned diary packages were culled for those that represented a reasonable attempt to complete the diary. Second, the diary information recorded for each day was checked to ensure that answers were complete and logical, as well as to verify recorded information against the mail markings returned in the package. The diaries were then scanned using Optical Character Recognition (OCR) software. In stage three, a verifier re-checked the diary information recorded in the OCR software for each day. This second edit functions as a quality control check to ensure data accuracy.

Overall, about one percent of returned diaries did not pass the edit checking process.

Of the 8,140 households recruited to receive a diary package 4,009 actually returned acceptable completed diaries (defined as containing data suitable for analysis) to NuStats, for a completion rate of 49.3 percent.

Data Management

Data management entails processing the information resulting from the Household Interview and Mail Diaries, making it available for analysis, storing it, and documenting it. Household interviews were conducted using both web and CATI technology. Questionnaire and relevant data checks were programmed into a master questionnaire that was used in the web and phone survey. Recorded data was extracted from the web and CATI software into a database management file.

Diary information was recorded through the web and paper. Returned paper diary information was recorded (entered) through optical scanning technology. The diary data, once scanned using Teleform software, was captured in a database management file. Similarly, web diary information was stored in a web server and later captured in a database management file.

After completion of data collection, editing and entry tasks, the survey data were contained in nine data files. One data file contained the Household Interview data and another eight contained Mail Diary—one for each mail classification (First-Class Mail received, First-Class Mail sent, etc.). These files were all developed in SAS-PC.

The file variables were identified by variable name. For each file variable, the File Information contains:

- Label, which is a brief description of the variable.
- Measurement level, which specifies the level of measurement as scale (numeric data on an interval or ratio scale), ordinal, or nominal. Nominal and ordinal data can be either string (alphanumeric) or numeric.
- Value formats, which identify the response codes.
- Column width and alignment.

Several SAS programming operations were necessary to put the Mail Diary data in the desired form for analysis. The structure for these programs was contained in a separate File Information document that accompanied the data delivery.

Various edit routines were used to check the consistency of the reported data and to identify reporting or entry errors. Routine edit checks were conducted to examine questionnaire responses for reasonableness and consistency across items. Routine checks included such items as:

- Response code range checks.
- Checks for proper data skips and patterns of answering questions consistent with prior answers.
- Checks for realistic responses (e.g., number of online purchases possible in one month).
- Checks for high frequency of item non-response (missing data from question refusals).

When conducting these checks, data were compared against the actual survey forms. NuStats identified extreme values that were impossible or unlikely, and corrected inconsistent data when possible.

Some extreme/inconsistent data values unable to be corrected or verified were edited to missing values.

In addition, NuStats performed in-depth customized data checks to ensure data within each record of the Household Interview were logically consistent. For example, a respondent should have reported paying bills online only if he/she also reported having Internet access.

Raw variables, derived variables, and analytical programs were documented in a data documentation binder that accompanied the data delivery. Any information that could be directly or indirectly used to identify individual respondents, such as respondent names, addresses, or telephone numbers, were removed to protect respondent confidentiality and privacy. Such information is stored in a locked archival file.

Sample Demographic Profile (All Counts Unweighted), Government Fiscal Year 2023

Table B.3:

Annual Household Income by Recruitment/Retrieval Status

Annual Household Income	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
$0 - $24,999	229	372	601	5.7%	15.8%
$25,000 - $34,999	251	349	600	6.3%	7.6%
$35,000 - $49,999	301	351	652	7.5%	10.6%
$50,000 - $64,999	363	331	694	9.1%	10.3%
$65,000 - $79,999	380	336	716	9.5%	8.7%
$80,000 - $99,999	356	355	711	8.9%	9.5%
$100,000 or more	1,382	1,107	2,489	34.5%	37.5%
Don't Know	65	133	198	1.6%	N/A
Refused	682	797	1,479	17.0%	N/A
Total	**4,009**	**4,131**	**8,140**	**100.0%**	**100.0%**

Notes:
Sample Percent based only on retrieved households that provided a response to the Household Income question.
Population percent based on U.S. Census Bureau, Current Population Survey Annual Demographic File (March 2023).

Table B.4:

Number of Adults in Household by Recruitment/Retrieval Status

Number of Adults in Household	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
One	957	1,185	2,142	23.9%	33.2%
Two	1,839	1,519	3,358	45.9%	50.7%
Three	531	621	1,152	13.2%	10.8%
Four	436	474	910	10.9%	3.9%
Five or More	246	332	578	6.1%	1.3%
Total	**4,009**	**4,131**	**8,140**	**100.0%**	**100.0%**

Notes:
Sample Percent based only on retrieved households.
Population percent based on U.S. Census Bureau, Current Population Survey Annual Demographic File (March 2023).

Table B.5:

Geographic Region by Recruitment/Retrieval Status

Geographic Region	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
Northeast	679	718	1,397	16.9%	17.6%
Midwest	947	768	1,715	23.6%	21.7%
South	1,472	1,714	3,186	36.7%	38.3%
West	911	931	1,842	22.7%	22.3%
Total	**4,009**	**4,131**	**8,140**	**100.0%**	**100.0%**

Notes:
Sample Percent based only on retrieved households.
Population percent based on 2020 Decennial Census P.L. 94-171 Redistricting Data Summary Files

Table B.6:

Urban/Rural Location by Recruitment/Retrieval Status

Urban/Rural Location	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
30 Largest Metro Areas	1,789	1,811	3,600	44.6%	44.3%
Other Metro Areas	1,646	1,743	3,389	41.1%	41.4%
Non-Metropolitan Areas	574	577	1,151	14.3%	14.3%
Total	**4,009**	**4,131**	**8,140**	**100.0%**	**100.0%**

Notes:
Sample Percent based only on retrieved households.
Population percent based on 2020 Decennial Census P.L. 94-171 Redistricting Data Summary Files

Table B.7:

Age of Head of Household by Recruitment/Retrieval Status

Age of Head of Household	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
18 - 24	50	101	151	1.2%	4.6%
25 - 44	907	1,192	2,099	22.6%	32.9%
45 - 64	1,419	1,376	2,795	35.4%	34.2%
65+	1,561	1,341	2,902	38.9%	28.3%
Refused	72	121	193	1.8%	N/A
Total	**4,009**	**4,131**	**8,140**	**100.0%**	**100.0%**

Notes:
Sample Percent based only on retrieved households that provided a valid response.
Population percent based on U.S. Census Bureau, Current Population Survey Annual Demographic File (March 2023).

Table B.8:

Educational Attainment of Head of Household by Recruitment/Retrieval Status

Educational Attainment of Head of Household	Recruited Households		Total	Sample Percent	Population Percent
	Retrieved	Not Retrieved			
8th grade or less	17	42	59	0.4%	2.8%
Some high school	45	96	141	1.1%	4.9%
High school graduate	494	667	1,161	12.3%	25.9%
Some college	701	810	1,511	17.5%	16.4%
Technical school graduate	264	304	568	6.6%	4.4%
College graduate	1,284	1,260	2,544	32.0%	30.2%
Postgraduate work	1,111	764	1,875	27.7%	15.4%
Refused	93	188	281	2.3%	N/A
Total	**4,009**	**4,131**	**8,140**	**100.0%**	**100.0%**

Notes:
Sample Percent based only on retrieved households that provided a valid response.
Population percent based on U.S. Census Bureau, Current Population Survey Annual Demographic File (March 2023).

Data Weighting and Expansion

This section explains the methodology used for creating sampling and expansion weights for the FY 2023 Household Diary Study.

The FY 2023 HDS uses both weighting and expansion factors to 1) adjust the sample data to match population parameters and 2) expand mail volumes exhibited in the diary sample to all U.S. households.

Weighting Procedures, FY 2023 Recruitment Data

Sampling weights were produced separately for the households that participated in the recruitment phase of the FY 2023 HDS, and those that completed and returned a diary. There were four main weighting variables: Geography, Education, Age, and Homeownership. FY 2023 recruitment geographic weights were derived from sample households' strata and region:

Strata: As mentioned previously, there are three strata. A household was classified within strata as residing in the top 30 metropolitan areas nationwide, any other metropolitan area, or a non-metropolitan area.[1] Table B.9 provides unweighted sample counts from FY 2023 recruitment data for strata:

Table B.9:
HDS 2023 Recruitment Data: Urban/Rural Location

Urban/ Rural Location	Household	Percent	Cumulative Percent
30 Largest Metro Areas	3,600	44.2%	44.2%
Other Metro Areas	3,389	41.6%	85.9%
Non-Metro Counties	1,151	14.1%	100.0%
Total	**8,140**	**100.0%**	

Regions: Table B.10 provides unweighted sample counts from FY 2023 recruitment data for region. Households were classified by state. There are four mutually exclusive regions as defined by the U.S. Census Bureau (along with respective states):

Four Census Regions:

Northeast: Connecticut, Maine, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, and Vermont.

Midwest: Illinois, Indiana, Iowa, Kansas, Michigan, Minnesota, Missouri, Nebraska, North Dakota, Ohio, South Dakota, and Wisconsin.

South: Alabama, Arkansas, Delaware, District of Columbia, Florida, Georgia, Kentucky, Louisiana, Maryland, Mississippi, North Carolina, Oklahoma, South Carolina, Tennessee, Texas, Virginia, and West Virginia.

West: Arizona, Alaska, California, Colorado, Hawaii, Idaho, Montana, Nevada, New Mexico, Oregon, Utah, Washington, and Wyoming.

Table B.10:
HDS 2023 Recruitment Data: Geographic Region

Geographic Region	Households	Percent	Cumulative Percent
Northeast	1,397	17.2%	17.2%
Midwest	1,715	21.1%	38.2%
South	3,186	39.1%	77.4%
West	1,842	22.6%	100.0%
Total	**8,140**	**100.0%**	

Strata/Regions: Table B.11 indicates the distribution of households from the FY 2023 recruitment sample within strata and regions.

Population parameters for the intersection of the three strata and four regions were based on 2020 Census counts of households by county. As Table B.12 shows, each county was grouped according to its location within these 12 mutually exclusive and collectively exhaustive geographic categories.

To calculate the weight for each strata/region interval, the population percentage was divided by the sample percentage. Geography weights appear in the last column to the right in Table B.12.

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[1] Core-Based Statistical Area (CBSA) is defined within the sample according to the official definition used by the U.S. Census Bureau. CBSAs refer collectively to both metropolitan and micropolitan statistical areas. Non-metropolitan areas are counties that do not belong to a metropolitan area. Each sample county was assigned to a stratum according to its metropolitan status.

Table B.11:
Distribution of Households within Strata and Region

Geographic Region	Stratum *(Urban/Rural Location)*			Total
	30 Largest Metro Areas	Other Metro Areas	Non-Metro Areas	
Northeast	817	474	106	**1,397**
Midwest	543	809	363	**1,715**
South	1,240	1,434	512	**3,186**
West	1,000	672	170	**1,842**
Total	**3,600**	**3,389**	**1,151**	**8,140**

Table B.12:
HDS 2023 Recruitment Data: Construction of Geographic Weight

Stratum	Geographic Region	Households *(Population)*	Percent	Households *(Sample)*	Percent	Weight
30 Largest Metro Areas	Northeast	12,951,999	9.85%	817	10.0%	.98
	Midwest	9,049,808	6.89%	543	6.7%	1.03
	South	19,583,112	14.90%	1,240	15.2%	.98
	West	16,628,406	12.65%	1,000	12.3%	1.03
Other Metro Areas	Northeast	8,299,775	6.31%	474	5.8%	1.08
	Midwest	13,264,647	10.09%	809	9.9%	1.02
	South	22,980,499	17.48%	1,434	17.6%	.99
	West	9,923,962	7.55%	672	8.3%	.91
Non-Metro Areas	Northeast	1,933,626	1.47%	106	1.3%	1.13
	Midwest	6,253,157	4.76%	363	4.5%	1.07
	South	7,819,183	5.95%	512	6.3%	.95
	West	2,745,379	2.09%	170	2.1%	1.00
Totals		**131,433,555**	**100.0%**	**8,140**	**100.0%**	1.00

Source: Household Population Estimates based on U.S. Census Bureau, 2023 Census.

Education: In addition to weighting for differences in geography between the sample and the population, an additional weight was created based on differences in the educational attainment of the head of household. For those households in which either more than one person was identified as the head of household or no individual was identified as the head of household, one was chosen based on the following sequence of criteria: 1) oldest male or 2) oldest female (if no male exists). For cases in which two candidates for the head of the household were of the same age, the respondent on the phone was chosen.

Known population parameters were based on weighted proportions derived from the U.S. Census Bureau's Current Population Survey annual demographic file for March 2023. For cases in which the head of household refused to provide his/her education level, an educational level was imputed based on the average educational level of like cases. There were 281 such cases in 2023; mean levels of educational attainment were based on geography (strata and regions), as well as age and income level, if provided.

Table B.13:

HDS 2023 Recruitment Data: Construction of Educational Attainment Weight

Educational Attainment	Households *(Population)*	Percent	Households *(Sample)*	Percent	Weight
Less than high school diploma	9,279,467	7.1%	200	2.5%	2.87
High school graduate	33,991,340	25.9%	1,161	14.3%	1.81
Some college	21,764,730	16.6%	1,530	18.8%	0.88
Technical school graduate	5,883,910	4.5%	763	9.4%	0.48
College graduate	39,960,953	30.4%	2,611	32.1%	0.95
Postgraduate work	20,553,154	15.6%	1,875	23.0%	0.68
Totals	**131,433,555**	**100.0%**	**8,140**	**100.0%**	**1.00**

Note: Education responses include imputed Don't Know/Refused answers.

Age: Additional weight was also created based on differences in the age of the head of household. Known population parameters were based on weighted proportions derived from the U.S. Census Bureau's Current Population Survey annual demographic file for March 2023. For cases in which the head of household refused to provide his/her age, an age level was imputed based on the average age level of like cases. There were 193 such cases in 2023; mean levels of age were based on geography (strata and regions), as well as education level, if provided.

Table B.14:

HDS 2023 Recruitment Data: Construction of Age Weight

Educational Attainment	Households *(Population)*	Percent	Households *(Sample)*	Percent	Weight
18-24	5,949,992	4.5%	151	1.9%	2.44
25-34	20,570,571	15.7%	870	10.7%	1.46
35-44	22,290,906	17.0%	1,231	15.1%	1.12
45-54	21,451,290	16.3%	1,233	15.1%	1.08
55-64	23,620,516	18.0%	1,717	21.1%	0.85
65-74	21,690,564	16.5%	1,814	22.3%	0.74
75+	15,859,716	12.1%	1,124	13.8%	0.87
Total	**131,433,555**	**100.0%**	**8,140**	**100.0%**	**1.00**

Note: Age responses include imputed Don't Know/Refused answers.

Homeownership: In addition to weighing for differences in income, education and geography between the sample and the population, data were weighed to account for homeownership rates for U.S. households. For those households in which homeownership was unknown, one was imputed based on the average income level and geography of like cases. There were 272 such cases in 2023; mean levels of income attainment were based on geography (strata and regions). Known population parameters were based on weighted proportions derived from the U.S. Census Bureau's Current Population Survey Quarterly Housing Vacancies and Homeownership file for November 2023. The average Homeownership rate in 2023 is 66 percent.

Weighting Procedures, FY 2023 Diary Data

As mentioned above, 8,140 households participated in the recruitment phase of the FY 2023 HDS, and 4,009 households completed usable diaries. Balancing weights for the FY 2023 HDS diary data were developed by applying the same approach used for recruitment data. An additional age weight was derived based on the age of the head of household using the following categories: 18–24, 25–34, 35–44, 45–54, 55–64, 65–74, and over 75 years old.

All component weights were multiplied together and normalized to ensure that the total number of weighted cases equals the number of unweighted cases.

A final adjustment in the form of expansion factors was made to expand the sample to the level of total households in the United States at the time of data collection, which was 131.4 million. The number of households in the United States was divided into the number of households that participated in the diary portion of the survey. The resultant factor was applied to each household in the survey. The expansion factor was multiplied by the sampling weight and then multiplied by 52 (the number of calendar weeks in one year) to derive nationwide annual volume estimates from the sample data.

Expansion Factor
131433555.02/ 4,009 = 32784.6

Component Weight:

$$\omega = \frac{Ps / Pt}{Ss / St},$$

Where Ps = population count in cohort and
Pt = total population count
Ss = sample count in cohort
St = total sample count